Algebraic Geometry



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Foreword

Chapter 1

Classical Algebraic Geometry

Chapter 2

Spectra

In this chapter the notion of spectra is introduced. This is the set of prime ideals of a ring which naturally can be equipped with a topology, called the Zariski topology. We discuss spectra of familiar rings and the connection of this definition to classical algebraic geometry. Further topological properties of spectra are discussed; the most important one is the fact that the spectrum of a ring is quasi-compact. Finally we prove that mapping a ring to its spectrum defines a functor from the category of rings to the category of topological spaces.

Recall our convention that rings are required to be commutative and unital, and every ring homomorphism is unital.

2.1 Spectrum of a Ring

Definition 2.1. Given a ring R, its **(prime) spectrum** $\operatorname{Spec} R$ is the set of its prime ideals.

It turns out that the spectrum $\operatorname{Spec} R$ comes with a natural topology. To define this topology, we introduce the following notation. Given a subset $S \subseteq R$, define

$$V(S) := \{\mathfrak{p} \in \operatorname{Spec} R : \mathfrak{p} \supseteq S\}.$$

It is straightforward to check that if $\mathfrak{a}$ is the ideal generated by S, then $V(S) = V(\mathfrak{a})$. Hence it suffices to consider only the sets $V(\mathfrak{a})$ whenever $\mathfrak{a}$ is an ideal of R. We shall also write $V(f_1, \dots, f_n)$ for $V(\langle f_1, \dots, f_n \rangle)$, where, as per usual, $\langle f_1, \dots, f_n \rangle$ denotes the ideal generated by $f_1, \dots, f_n$.

▸ **Exercise 2.2.** Show that for any ideal $\mathfrak{a}$ of R, $V(\sqrt{\mathfrak{a}}) = V(\mathfrak{a})$, where $\sqrt{\mathfrak{a}}$ denotes the radical of $\mathfrak{a}$.

The sets $V(\mathfrak{a})$ satisfy the following properties.

Proposition 2.3. *Let R be a ring, and let $\mathfrak{a}$, $\mathfrak{b}$ be ideals of R. Further, for each $k \in S$ let $\mathfrak{a}_k$ be ideals of R where S is some index set. Then*

(1.) If $\mathfrak{a} \subseteq \mathfrak{b}$, then $V(\mathfrak{b}) \subseteq V(\mathfrak{a})$.

(2.) $V(\emptyset) = \operatorname{Spec} R$ and $V(1) = \emptyset$.

(3.) $V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b})$.

(4.) $V(\sum_{k\in S} \mathfrak{a}_k) = \bigcap_{k\in S} V(\mathfrak{a}_k)$.

▸ **Exercise 2.4.** Prove the above proposition. *[Hint: the fact that $\sqrt{\mathfrak{a}\mathfrak{b}} = \sqrt{\mathfrak{a} \cap \mathfrak{b}}$ might be useful.]*

The above proposition shows that the sets of the form $V(\mathfrak{a})$, where $\mathfrak{a}$ runs through all ideals of R, are closed sets of a topology on $\operatorname{Spec} R$. This is the **Zariski topology** on $\operatorname{Spec} R$. With the topology in mind, we will refer to elements of $\operatorname{Spec} R$ as points. Given $x \in \operatorname{Spec} R$, we shall write $\mathfrak{p}_x$ if we want to view x as a prime ideal of R.

There is a reverse correspondence of the map $V(-)$. If Y is a subset of $\operatorname{Spec} R$, we define the ideal $I(Y)$ associated to Y by

$$I(Y) := \bigcap_{\mathfrak{p}\in Y} \mathfrak{p}.$$

Note that $I(\emptyset) = R$.

Proposition 2.5. *Let R be a ring, $\mathfrak{a} \subseteq R$ an ideal, and Y a subset of $\operatorname{Spec} R$.*

(1.) If $X \subseteq Y \subseteq \operatorname{Spec} R$, then $I(Y) \subseteq I(X)$.

(2.) $\sqrt{I(Y)} = I(Y)$, i.e. $I(Y)$ is a radical ideal.

(3.) $I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}}$ and $V(I(Y)) = \overline{Y}$ where $\overline{Y}$ is the topological closure of Y in $\operatorname{Spec} R$.

Proof. (1.) Trivial.

(2.) We only need to check the non-trivial inclusion $\sqrt{I(Y)} \subseteq I(Y)$. Suppose $a \in R$ and $n \geq 1$ satisfy $a^n \in I(Y)$. Then for each prime ideal $\mathfrak{p} \in Y$, $a^n \in \mathfrak{p}$. Then $a \in \mathfrak{p}$, and so $a \in I(Y)$.

(3.) The assertion that $I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}}$ follows from a theorem in commutative algebra which states that, for any ideal $\mathfrak{a}$ of R,

$$\sqrt{\mathfrak{a}} = \bigcap_{\substack{\mathfrak{p} \text{ prime} \\ \mathfrak{p}\supseteq\mathfrak{a}}} \mathfrak{p}.$$

For the second assertion, note that $V(I(Y))$ contains Y. Now if $V(\mathfrak{b})$ is a closed set containing Y, then $I(Y) \supseteq I(V(\mathfrak{b})) = \sqrt{\mathfrak{b}} \supseteq \mathfrak{b}$ by the first assertion. It follows that

$$V(I(Y)) \subseteq V(\mathfrak{b}),$$

proving that $V(I(Y))$ is the closure of Y. □

The following corollaries are immediate.

Corollary 2.6. *The maps $Y \mapsto I(Y)$ and $\mathfrak{a} \mapsto V(\mathfrak{a})$ give rise to mutually inverse and inclusion-reversing bijections between the sets*

$$\{\textit{radical ideals of } R\} \rightleftarrows \{\textit{closed subsets of } \operatorname{Spec} R\}.$$

Corollary 2.7. *Let R be a ring and $\mathfrak{a}, \mathfrak{b}$ ideals of R. Then $V(\mathfrak{a}) = V(\mathfrak{b})$ if and only if $\sqrt{\mathfrak{a}} = \sqrt{\mathfrak{b}}$.*

There is a connection between the sets $V(I)$ with the new definition in this chapter and the vanishing sets in the sense of classical algebraic geometry. For this, elements of R need to be interpreted as "functions" on $\operatorname{Spec} R$. Given $f \in R$, we define **the value of f at a prime** $\mathfrak{p} \in \operatorname{Spec} R$ to be the image of f under the composition

$$R \to R/\mathfrak{p} \hookrightarrow \operatorname{Frac}(R/\mathfrak{p}).$$

Here we used the fact that $R/\mathfrak{p}$ is an integral domain so that its field of fractions may be constructed. Via this map, we see that $f(\mathfrak{p}) = 0$ if and only if $f \in \mathfrak{p}$. Hence $V(f)$ consists of prime ideals $\mathfrak{p}$ such that $f(\mathfrak{p}) = 0$. Therefore $V(f)$ is the *vanishing set* of f. Compare this situation to the classical case when R is the polynomial ring $k[x_1, \dots, x_n]$ and f is a polynomial in R.

Here we discuss spectra of familiar rings.

Example 2.8 (Spectrum of a Field)**.** Every field k has only one prime ideal $\{0\}$. Thus $\operatorname{Spec} k$ consists of a single point.

Example 2.9 (Spectrum of $\mathbb{Z}$)**.** The prime ideal of $\mathbb{Z}$ are the zero ideal (0) and the ideal generated by prime numbers (p). Hence $\mathbb{Z}$ has infinitely many points.

More generally, the spectrum of an integral domain R is non-empty. It contains the point corresponding to the zero ideal. Let us denote this point by η. The fact that every prime ideal of R contains the zero ideal means that $\overline{\{\eta\}} = \operatorname{Spec} R$: the closure of the singleton $\{\eta\}$ is equal to the whole spectrum. Such a point is called a *generic point* of $\operatorname{Spec} R$ (Proposition 2.26). As strange as it is, generic points will turn out to be important to the theory of integral schemes.

For the last example in this section, we relate the coordinate ring $k[V]$ of an affine variety $V \subseteq \mathbb{A}^n$ to the spectrum $\operatorname{Spec} k[V]$ when k is an algebraically closed field.

Example 2.10. Let $V \subseteq \mathbb{A}^n$ be an affine variety given as the vanishing set $Z(f_1, \dots, f_i)$ of polynomials in $k[x_1, \dots, x_n]$. Let I denote the ideal consisting of polynomials vanishing on V. Recall that there is a one-to-one correspondence

$$\{\text{maximal ideals of } k[x_1, \dots, x_n]/I\} \iff \{\text{points of } V\}.$$

Let us denote the coordinate ring of V by $k[V] = k[x_1, \dots, x_n]/I$. If we write $\operatorname{Spec}_M(A)$ for the set of maximal ideals of the ring A, then the points of V are in bijection with points of $\operatorname{Spec}_M(k[V])$ via Hilbert Nullstellensatz. The set $\operatorname{Spec}_M(k[V])$ is a subspace of $\operatorname{Spec}(k[V])$. It is easy to check that the subspace topology on $\operatorname{Spec}_M(k[V])$ is the same one as the Zariski topology on V considered as a subspace of $k[x_1, \dots, x_n]$.

In topological terms maximal ideals are the closed points of the spectrum (see Proposition 2.26 below). Hence we "recover" classical algebraic geometry when we consider the closed points of the spectrum of a coordinate ring.

2.2 Topological Properties of Spectra

Basis of Spec R

We now turn to the topological properties of $\operatorname{Spec} R$ under the Zariski topology. Firstly, there is a basis for this topology which is derived from elements of R.

Definition 2.11. Let R be a ring and $f \in R$. We define

$$D(f) = \operatorname{Spec} R \setminus V(f).$$

Plainly, $D(f)$ is open in $\operatorname{Spec} R$. An open set of this form is called a **principal open subset** of $\operatorname{Spec} R$.

The set $D(f)$ can also be regarded as the set of points where f does not vanish, that is, $D(f) = \{\mathfrak{p} \in \operatorname{Spec} R : f(\mathfrak{p}) \neq 0\}$.

Proposition 2.12. *Let R be a ring. Principal open subsets of* $\operatorname{Spec} R$ *form a basis of the Zariski topology.*

Proof. Let $U \subseteq \operatorname{Spec} R$ be an open set, so that $U = \operatorname{Spec} R \setminus V(\mathfrak{b})$ for some ideal $\mathfrak{b}$ of R. It is easy to see that

$$V(\mathfrak{b}) = \bigcap_{f \in \mathfrak{b}} V(f),$$

and consequently $U = \bigcup_{f \in \mathfrak{b}} D(f)$. □

The basis $\mathcal{B}$ consisting of principal open subsets is *stable under finite intersections*. This means that if $B_1, B_2 \in \mathcal{B}$, then $B_1 \cap B_2 \in \mathcal{B}$. This follows from the proposition below, and the fact that $D(0) = \emptyset$.

Proposition 2.13. *Let R be a ring. Then $D(f) \cap D(g) = D(fg)$ for all $f, g \in R$. Consequently the basis consisting of principal open subsets is stable under finite intersections.*

▸ **Exercise 2.14.** Prove the above proposition.

The following lemma is useful for turning a topological statement about a covering by principal open subsets into an algebraic statement.

Lemma 2.15 ("Covering Trick")**.** *Let $\{f_i\}_{i\in I}$ be a family of elements of a ring R and let $g \in R$. Then $D(g) \subseteq \bigcup_{i\in I} D(f_i)$ if and only if there is an integer $n > 0$ such that g^n is contained in the ideal $\mathfrak{a}$ generated by f_i.*

Proof. The condition $D(g) \subseteq \bigcup_{i\in I} D(f_i)$ is equivalent to $V(g) \supseteq \bigcap_{i\in I} V(f_i) = V(\mathfrak{a})$, which is equivalent to $g \in \sqrt{\mathfrak{a}}$. □

Two special cases of the above theorem are often used. Firstly, $D(g) \subseteq D(f)$ if and only if $g \in \sqrt{(f)}$, if and only if $g^n = fa$ for some $a \in R$. This is important when we define the structure sheaf of $\operatorname{Spec} R$.

Secondly, setting $g = 1$ shows that the collection $\{D(f_i)\}_{i\in I}$ covers $\operatorname{Spec} R$ if and only if $1 \in \langle f_i : i \in I\rangle$. Hence there is a finite subset $J \subseteq I$ such that $1 = \sum_{j\in J} r_j f_j$ where $r_j \in R$. This implies that $\operatorname{Spec} R = \bigcup_{j\in J} D(f_j)$. Thus we have found a finite subcover of $\{D(f_i)\}_{i\in I}$, implying that $\operatorname{Spec} R$ is quasi-compact. More generally, every principal open subset of $\operatorname{Spec} R$ is quasi-compact by the same argument.

Proposition 2.16. *Let R be a ring. Then every principal open subset $D(g)$, $g \in R$, is quasi-compact.*

▸ **Exercise 2.17.** Prove this proposition by adapting the proof that $\operatorname{Spec} R$ is quasi-compact.

Irreducibility

A non-empty topological space is **irreducible** if it cannot be written as a union of two proper closed subsets. In other words, a topological space $X \neq \emptyset$ is irreducible if and only if whenever $X = C_1 \cup C_2$, where C_1 and C_2 are closed subsets, then either $C_1 = X$ or $C_2 = X$. A topological space is **reducible** if it is not irreducible. It is useful to consider the empty topological space as being reducible.

A subset of a topological space is irreducible if it is irreducible under the subspace topology. Similar to connected components, we define an **irreducible component** to

be a subspace $Z \subseteq X$ such that Z is maximally irreducible subset of X, that is, Z is irreducible and whenever Y is irreducible and $Z \subseteq Y$, then $Z = Y$.

▸ **Exercise 2.18.** Show that if X is a topological space and $x \in X$, then the closure $\overline{\{x\}}$ is irreducible.

We collect some properties of irreducible spaces here, whose proofs are mechanical.

Proposition 2.19. *The following are equivalent for a non-empty topological space X.*

(1.) X is irreducible.

(2.) Every non-empty open subset of X is dense in X.

(3.) Every two non-empty open subsets of X has non-empty intersection.

(4.) Every non-empty open subset of X is irreducible.

▸ **Exercise 2.20.** Prove the above proposition.

Proposition 2.21. *Let $f\colon X \to Y$ be a continuous map between topological spaces. If $Z \subseteq X$ is an irreducible subset, then its image $f(Z)$ is irreducible.*

Proof. Clearly $f(Z)$ is non-empty because Z is. Suppose $f(Z) = C_1 \cup C_2$ where C_1 and C_2 are closed subsets of $f(Z)$ under the subspace topology. Hence $C_1 = D_1 \cap f(Z)$ and $C_2 = D_2 \cap f(Z)$ for some closed sets D_1 and D_2 of Y. We then have

$$Z = (f^{-1}(D_1) \cap Z) \cup (f^{-1}(D_2) \cap Z).$$

This is a decomposition of Z into two closed subsets. Since Z is irreducible, either $f^{-1}(D_1) \cap Z = Z$ or $f^{-1}(D_2) \cap Z = Z$. Hence either $C_1 = f(Z)$ or $C_2 = f(Z)$. □

Proposition 2.22. *Let X be a topological space. Then $Y \subseteq X$ is irreducible if and only if its closure $\overline{Y}$ is irreducible.*

Proof. First assume that $Y \subseteq X$ is irreducible. Then $\overline{Y}$ is clearly non-empty. If $\overline{Y} = C_1 \cup C_2$ for some closed subsets C_1, C_2 of X, then $Y = (C_1 \cap Y) \cup (C_2 \cap Y)$. Since Y is irreducible, one of the subsets must be equal to Y, say $C_1 \cap Y = Y$. Hence C_1 is a closed set containing Y, and so $\overline{Y} \subseteq C_1$. Plainly, $C_1 \subseteq \overline{Y}$ and therefore $\overline{Y} = C_1$.

For the converse, assume that Y is reducible. If Y is empty, then $\overline{Y} = \emptyset$ is also reducible. Hence we may assume that Y is non-empty and write $Y = C_1 \cup C_2$, where C_1 and C_2 are proper closed subsets of Y under the subspace topology. Then $\overline{Y} = \overline{C_1} \cup \overline{C_2}$ by the property of closure. If $\overline{Y} = \overline{C_1}$, then

$$C_1 = \overline{C_1} \cap Y = \overline{Y} \cap Y = Y,$$

a contradiction. Similarly we conclude that $\overline{Y} \neq \overline{C_2}$. Therefore we have found a decomposition of $\overline{Y}$ into two proper closed subsets. □

We now turn to irreducible subsets of $\operatorname{Spec} R$, when R is a ring. The starting point is the following proposition which says that irreducible subsets are related to prime ideals.

Proposition 2.23. *A subset $Y \subseteq \operatorname{Spec} R$ is irreducible if and only if $I(Y)$ is a prime ideal of R.*

Proof. Assume that Y is irreducible. Then $\overline{Y}$ is also irreducible. Let $fg \in I(Y) \subseteq R$. Using Proposition 2.3 and Proposition 2.5, we have the identities

$$V(f) \cup V(g) = V(fg) \supseteq V(I(Y)) = \overline{Y}.$$

Hence $\overline{Y} = (V(f) \cap \overline{Y}) \cup (V(g) \cap \overline{Y})$. Since $\overline{Y}$ is irreducible, it must be equal to one of the subsets, say $\overline{Y} = V(f) \cap \overline{Y}$. Therefore $\overline{Y} \subseteq V(f)$. Again, by Proposition 2.5, we have $I(Y) \supseteq \sqrt{(f)}$. This implies that $f \in I(Y)$. This proves that $I(Y)$ is prime.

Now assume that $\mathfrak{p} = I(Y)$ is prime. By Proposition 2.5, $\overline{Y} = V(\mathfrak{p})$. Let $V(\mathfrak{a}), V(\mathfrak{b})$ be closed sets such that

$$V(\mathfrak{p}) = V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{a} \cap \mathfrak{b}).$$

Recall that $\mathfrak{p} = \sqrt{\mathfrak{a} \cap \mathfrak{b}} = \sqrt{\mathfrak{a}} \cap \sqrt{\mathfrak{b}}$ by commutative algebra. Since $\mathfrak{p}$ is prime, either $\mathfrak{p} = \sqrt{\mathfrak{a}}$ or $\mathfrak{p} = \sqrt{\mathfrak{b}}$. Hence $V(\mathfrak{p}) = V(\mathfrak{a})$ or $V(\mathfrak{p}) = V(\mathfrak{b})$. Thus $V(\mathfrak{p}) = \overline{Y}$ is irreducible, and so does Y. □

Proposition 2.24. *There is a one-to-one correspondence between $\operatorname{Spec} R$ and the set of closed irreducible subsets of $\operatorname{Spec} R$ given in one direction by $\mathfrak{p} \mapsto \overline{\{\mathfrak{p}\}}$. Under this correspondence, minimal prime ideals of R correspond bijectively to irreducible components of $\operatorname{Spec} R$.*

Proof. If $\mathfrak{p}$ is prime, then $I(\{\mathfrak{p}\}) = \mathfrak{p}$ and thus $\overline{\{\mathfrak{p}\}} = V(I(\{\mathfrak{p}\})) = V(\mathfrak{p})$ is irreducible by Proposition 2.23. Now if Y is a closed irreducible subset of $\operatorname{Spec} R$, then by Proposition 2.23, $I(Y)$ is a prime ideal. Hence the correspondence is given by the functions $V(-)$ and $I(-)$ which are proved to be mutually inverse bijections in Corollary 2.6.

The second statement follows immediately from the observation that $\overline{\{\mathfrak{p}\}} \subseteq \overline{\{\mathfrak{q}\}}$ if and only if $\mathfrak{p} \supseteq \mathfrak{q}$ for all prime ideals $\mathfrak{p}, \mathfrak{q}$ of R. □

Special Points

We introduce more definitions for points of a topological space which has interesting properties.

Definition 2.25. Let X be a topological space.

(1.) A point $x \in X$ is a **closed point** if $\{x\}$ is closed.

(2.) A point $\eta \in X$ is a **generic point** if $\overline{\{\eta\}} = X$.

(3.) Let x and x' be points of X. We say that x is a **generization** of x' and that x' is a **specialisation** of x if $x' \in \overline{\{x\}}$.

(4.) A point $x \in X$ is a **maximal point** of X if $\overline{\{x\}}$ is an irreducible component of X.

For spectra of rings, the topological notions above correspond to the following algebraic statements.

Proposition 2.26. *Let R be a ring.*

(1.) A point $x \in \operatorname{Spec} R$ is closed if and only if the ideal $\mathfrak{p}_x$ is maximal.

(2.) A point $\eta \in \operatorname{Spec} R$ is generic if and only if the ideal $\mathfrak{p}_\eta$ is the unique minimal prime ideal.

(3.) A point x is a generization of x' if and only if $\mathfrak{p}_x \subseteq \mathfrak{p}_{x'}$.

(4.) A point $x \in \operatorname{Spec} R$ is maximal if and only if $\mathfrak{p}_x$ is a minimal prime ideal.

▸ **Exercise 2.27.** Prove the proposition.

Corollary 2.28. *Let R be a ring. Each closed irreducible subset $Z \subseteq \operatorname{Spec} R$ contains a unique generic point η such that $Z = \overline{\{\eta\}}$.*

Proof. Combine Proposition 2.24 and Proposition 2.26. □

Example 2.29 (Spectrum of a PID)**.** Let R be a principal ideal domain. The zero ideal (0) is prime and furthermore is a generic point of $\operatorname{Spec} R$. Clearly this is the unique generic point of $\operatorname{Spec} R$.

The other prime ideals of R are ideals generated by prime elements. Since prime elements and irreducible elements coincide in a PID, it follows that every point of $\operatorname{Spec} R$ distinct from the generic point is closed.

2.3 Spec as a Functor

We have seen that a ring R gives rise to the topological space $\operatorname{Spec} R$. In addition to this, a ring homomorphism $\varphi\colon R \to S$ gives rise to a continuous map between the underlying topological space $\operatorname{Spec} S \to \operatorname{Spec} R$, as we shall see in the following lemma.

Lemma 2.30. *Let $\varphi\colon R \to S$ be a ring homomorphism. Then there is a continuous map $\operatorname{Spec}\varphi\colon \operatorname{Spec} S \to \operatorname{Spec} R$ given by $\mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q})$.*
In particular, the identity $(\operatorname{Spec}\varphi)^{-1}(V(\mathfrak{b})) = V(\varphi(\mathfrak{b}))$ holds for all ideals $\mathfrak{b}$ of R.

Proof. It is an easy exercise in commutative algebra to show that $\varphi^{-1}(\mathfrak{q})$ is a prime ideal of R if $\mathfrak{q}$ is a prime ideal of S. To prove that $f := \operatorname{Spec}\varphi$ is continuous, let $V(\mathfrak{b})$ be a closed subset of $\operatorname{Spec} R$. Then

$$f^{-1}(V(\mathfrak{b})) = \{\mathfrak{p} \in \operatorname{Spec} S : \mathfrak{b} \subseteq \varphi^{-1}(\mathfrak{p})\},$$

from the definition, as one can check. But $\mathfrak{b} \subseteq \varphi^{-1}(\mathfrak{p})$ if and only if $\varphi(\mathfrak{b}) \subseteq \mathfrak{p}$, and thus $f^{-1}(V(\mathfrak{b})) = V(\varphi(\mathfrak{b}))$ as claimed. Since every preimage under f of a closed set is closed, this shows that f is continuous. □

We left the reader to check that, for any ring R and the identity map $\mathrm{id}_R\colon R \to R$, the map $\operatorname{Spec}(\mathrm{id}_R)$ is the identity function of $\operatorname{Spec} R$. Moreover, we have $\operatorname{Spec}(\psi \circ \varphi) = \operatorname{Spec}\varphi \circ \operatorname{Spec}\psi$ for any ring homomorphisms $\varphi\colon R \to S$ and $\psi\colon S \to T$. Hence Spec is a contravariant functor from the category of rings to the category of topological spaces.

A continuous map between topological spaces $f\colon X \to Y$ is **dominant** if $f(X)$ is dense in Y. For the morphism $\operatorname{Spec}\varphi\colon \operatorname{Spec} S \to \operatorname{Spec} R$ between spectra, this happens precisely when $\ker\varphi$ is contained in the nilradical of R.

Corollary 2.31. *Let $\varphi\colon R \to S$ be a ring homomorphism. Then $\operatorname{Spec}\varphi$ is dominant if and only if $\ker\varphi \subseteq \operatorname{Nil}(R)$.*

Proof. See Problem 2.8. □

In particular, an injective ring map $\varphi\colon R \to S$ induces a dominant continuous map $\operatorname{Spec}\varphi$ on spectra.

Proposition 2.32. *Let $\varphi\colon R \to S$ be a surjective homomorphism of rings with kernel $\mathfrak{a}$. Then $\operatorname{Spec}\varphi$ defines a homeomorphism from $\operatorname{Spec} S$ to the set $V(\mathfrak{a})$ of $\operatorname{Spec} R$.*

Proof. By the first isomorphism theorem, $S \cong R/\mathfrak{a}$. It is then clear that $\operatorname{Spec}\varphi$ is a bijection between prime ideals of S and prime ideals of R containing $\mathfrak{a}$. Hence $\operatorname{Spec}\varphi\colon \operatorname{Spec} S \to V(\mathfrak{a})$ is a bijective continuous map. Now let $\mathfrak{b}$ be an ideal of S. Then $\varphi^{-1}(\mathfrak{b})$ is an ideal of R. The identity in Lemma 2.30 takes the form

$$(\operatorname{Spec}\varphi)^{-1}(V(\varphi^{-1}(\mathfrak{b}))) = V(\varphi(\varphi^{-1}(\mathfrak{b}))).$$

Since φ is surjective, $\varphi(\varphi^{-1}(A)) = A$ for all $A \subseteq S$. Thus we obtain

$$(\operatorname{Spec}\varphi)^{-1}(V(\varphi^{-1}(\mathfrak{b}))) = V(\mathfrak{b}).$$

Using the fact that $\mathfrak{a} \subseteq \varphi^{-1}(\mathfrak{b})$, we get $V(\varphi^{-1}(\mathfrak{b})) \subseteq V(\mathfrak{a})$. Together with $\operatorname{Spec}\varphi\colon \operatorname{Spec} S \to V(\mathfrak{a})$ being bijective, we conclude that $V(\varphi^{-1}(\mathfrak{b})) = (\operatorname{Spec}\varphi)(V(\mathfrak{b}))$. Thus $\operatorname{Spec}\varphi$ is a closed map, and so it is a homeomorphism. □

Since every quotient homomorphism is surjective, we obtain the following corollary.

Corollary 2.33 (Homeomorphism between $\operatorname{Spec} R/\mathfrak{a}$ and $V(\mathfrak{a})$)**.** *Let R be a ring, $\mathfrak{a}$ its ideal, and $\pi\colon R \to R/\mathfrak{a}$ the quotient homomorphism. Then $\operatorname{Spec}\pi$ is a homeomorphism from $\operatorname{Spec} R/\mathfrak{a}$ onto the set $V(\mathfrak{a})$.*

Proposition 2.34. *Let R be a ring, $S \subseteq R$ a multiplicative subset, and $\psi\colon R \to S^{-1}R$ the canonical homomorphism $r \mapsto r/1$. Then $\operatorname{Spec}\psi$ is a homeomorphism from $\operatorname{Spec}(S^{-1}R)$ onto the set of prime ideals $\mathfrak{p}$ of R with $\mathfrak{p} \cap S = \emptyset$.*

Proof. Let $D(S)$ denote the set of prime ideals $\mathfrak{p}$ of R with $\mathfrak{p} \cap S = \emptyset$. It is a result in commutative algebra which states that $\operatorname{Spec}\psi$ is a bijection from $\operatorname{Spec}(S^{-1}R)$ to $D(S)$. We show that it is closed. Let J be an ideal of $S^{-1}R$. Then it is easily checked that

$$(\operatorname{Spec}\psi)(V(J)) = V(\psi^{-1}(J)) \cap D(S),$$

proving that $\operatorname{Spec}\psi$ is a closed map. □

2.4 Problems

2.1 Prove that, for any ring R and any ideal I of R, $V(I) = \emptyset$ if and only if $I = R$.

2.2 Prove the following theorem from commutative algebra: if R is a ring and $\mathfrak{a}$ is an ideal of R, then

$$\sqrt{\mathfrak{a}} = \bigcap_{\substack{\mathfrak{p} \text{ prime} \\ \mathfrak{p} \supseteq \mathfrak{a}}} \mathfrak{p}.$$

2.3 Show that if $f \in R$ vanishes at every point of $\operatorname{Spec} R$ (that is, $f(\mathfrak{p}) = 0$ for all $\mathfrak{p} \in \operatorname{Spec} R$), then f is nilpotent.

2.4 Prove that $\operatorname{Spec} R$ is irreducible if and only if the nilradical of R is prime.

2.5 Show that if $\mathfrak{p} \subseteq R$ is a prime ideal, then $\{\mathfrak{p}\}$ is the only generic point of $V(\mathfrak{p})$.

2.6 Show that if R is an integral domain, then $\operatorname{Spec} R$ is irreducible.

2.7 Given a ring homomorphism $\varphi\colon R \to S$, let f denote the corresponding continuous map $\operatorname{Spec}\varphi\colon \operatorname{Spec} S \to \operatorname{Spec} R$. Show that

(a) $f^{-1}(D(r)) = D(\varphi(r))$ for all $r \in R$.

(b) $V(\varphi^{-1}(\mathfrak{b})) = \overline{f(V(\mathfrak{b}))}$ for all ideal $\mathfrak{b} \subseteq S$.

2.8 Prove Corollary 2.31. *[Hint: Use Problem 2.7].*

2.9 Let R be a ring and $\mathfrak{p}$ a prime ideal of R.

(a) Prove that $\operatorname{Spec} R_{\mathfrak{p}}$ has a unique closed point.

(b) Prove that if U is an open subset which contains the unique closed point of $\operatorname{Spec} R_{\mathfrak{p}}$, then $U = \operatorname{Spec} R_{\mathfrak{p}}$.

2.10 Let $\varphi\colon R \to S$ be a ring homomorphism. If $\operatorname{Spec} S$ is irreducible with a generic point η, prove that $(\operatorname{Spec}\varphi)(\eta)$ is the generic point of the closure $\overline{\operatorname{im}\operatorname{Spec}}$.

2.11 Show that if R is a ring, then $\operatorname{Spec} R$ and $\operatorname{Spec}(R/\operatorname{Nil} R)$ are homeomorphic.

Chapter 3

Sheaves and Affine Schemes

3.1 Presheaves and Sheaves

Presheaves

Definition 3.1. Let X be a topological space. A **presheaf of Abelian groups** $\mathcal{F}$ on X consists of the following data:

(1.) an assignment of an Abelian group $\mathcal{F}(U)$ to each open subset $U \subseteq X$;

(2.) for each inclusion of open subsets $U \subseteq V$, there is a group homomorphism $\mathrm{res}_U^V \colon \mathcal{F}(V) \to \mathcal{F}(U)$ such that

(a) $\mathrm{res}_U^U = \mathrm{id}_{\mathcal{F}(U)}$ for all open subset U;

(b) if $U \subseteq V \subseteq W$ are open subsets, then $\mathrm{res}_U^W = \mathrm{res}_U^V \circ \mathrm{res}_V^W$.

An element of $\mathcal{F}(U)$ is called a **section** of $\mathcal{F}$ over U. The morphism res_U^V is called the **restriction map** from V to U. If $s \in \mathcal{F}(V)$, the more suggestive notation for $\mathrm{res}_U^V(s)$ is $s|_U$. The condition for restriction maps is then $(s|_V)|_U = s|_U$ for all open subsets $U \subseteq V \subseteq W$ and $s \in \mathcal{F}(W)$. Another notation for $\mathcal{F}(U)$ is $\Gamma(U, \mathcal{F})$ which will be useful when we define sheaf cohomology.

Example 3.2. An example of presheaves on a topological space is the presheaf of real-valued smooth function on a smooth manifold M. If $U \subseteq M$ is open, we define $\mathcal{C}^\infty(U)$ to be the set of real-valued smooth functions on U. The restriction maps are simply restriction of functions, which is the motivation for the notation $s|_U$.

We now turn to the definition of morphisms between presheaves.

Definition 3.3. Given two presheaves $\mathcal{F}$ and $\mathcal{G}$, a **morphism of presheaves** $f\colon \mathcal{F} \to \mathcal{G}$ is the collection of maps $f_U\colon \mathcal{F}(U) \to \mathcal{G}(U)$ for each open subset $U \subseteq X$ such that whenever $U \subseteq V$ are open subsets, the diagram

$$\begin{array}{ccc} \mathcal{F}(V) & \xrightarrow{f_V} & \mathcal{G}(V) \\ \Big\downarrow{\scriptstyle \mathrm{res}_U^V} & & \Big\downarrow{\scriptstyle \mathrm{res}_U^V} \\ \mathcal{F}(U) & \xrightarrow[f_U]{} & \mathcal{G}(U) \end{array}$$

commutes.

We can alternatively give a categorical perspective on presheaves. Let X be a topological space. The open sets of X form a category $(\mathrm{Ouv}(X))$ of open subsets of X. We declare that there is exactly one morphism $U \to V$ in $(\mathrm{Ouv}(X))$ if and only if $U \subseteq V$. Put simply, this is the order category of open subsets of X with the ordering given by inclusion of sets. Then a presheaf on X is simply a contravariant functor $\mathcal{F}\colon (\mathrm{Ouv}(X))^{\mathrm{op}} \to (\mathrm{Ab})$. A morphism of presheaves $f\colon \mathcal{F} \to \mathcal{G}$ is then a natural transformation between the two functors.

By changing the target so that $\mathcal{F}\colon (\mathrm{Ouv}(X))^{\mathrm{op}} \to \mathcal{C}$ maps to another category $\mathcal{C}$, we obtain a presheaf $\mathcal{F}$ on X with value in $\mathcal{C}$, not just Abelian groups. In the following we mostly consider the following types of presheaves:

(1.) Presheaves of sets $(\mathrm{Ouv}(X))^{\mathrm{op}} \to (\mathrm{Set})$.

(2.) Presheaves of Abelian groups $(\mathrm{Ouv}(X))^{\mathrm{op}} \to (\mathrm{Ab})$.

(3.) Presheaves of rings $(\mathrm{Ouv}(X))^{\mathrm{op}} \to (\mathrm{Ring})$.

(4.) Given a ring R, presheaves of R-modules $(\mathrm{Ouv}(X))^{\mathrm{op}} \to (R\text{-Mod})$ and presheaves of R-algebras $(\mathrm{Ouv}(X))^{\mathrm{op}} \to (R\text{-Alg})$.

Example 3.4 (Constant Presheaf)**.** Let A be a fixed Abelian group and let X be a topological space. We can define a presheaf $\mathcal{A}$ on X by setting $\mathcal{A}(U) = A$ for all open subsets $U \subseteq X$ and all restriction maps are the identity homomorphism. This is called the **constant presheaf** with value A on X.

Example 3.5 (Restriction of Presheaf)**.** Let X be a topological space and $\mathcal{F}$ a presheaf on X. If U is an open subset of X, then the assignment

$$\mathcal{F}|_U(V) = \mathcal{F}(V) \quad \text{for each open subset } V \subseteq U$$

define a presheaf on U. The presheaf $\mathcal{F}|_U$ is called the **restriction** of the presheaf $\mathcal{F}$ onto U.

Example 3.6 (Sub-presheaf)**.** Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on a topological space X with values in algebraic objects such as Abelian groups or rings. If $\mathcal{F}(U)$ is a subgroup of $\mathcal{G}(U)$ for all open $U \subseteq X$, then $\mathcal{F}$ is said to be a **sub-presheaf** of $\mathcal{G}$.

By abuse of language we also say that $\mathcal{F}$ is a sub-presheaf of $\mathcal{G}$ if $\mathcal{F}$ is isomorphic to a sub-presheaf of $\mathcal{G}$.

Sheaves

The fundamental property of functions on a topological space X is that we can glue functions $f_i : U_i \to X$ defined on open subsets $U_i \subseteq X$ together into a function $f : U \to X$ defined on the union $U = \bigcup_i U_i$ of the open sets. Since we like to think of sections on U as "functions" on U, the above property should hold for the type of presheaves which are desirable for our study. Hence we are led to the following definition.

Definition 3.7. Let $\mathcal{F}$ be a presheaf of Abelian groups on a topological space X. We say that $\mathcal{F}$ is a **sheaf** if it satisfies the following two properties: for all open subsets $U \subseteq X$ and for all open coverings $\{U_i\}_{i\in I}$ of U,

(1.) (Uniqueness) Let $s, t \in \mathcal{F}(U)$ be sections such that $s|_{U_i} = t|_{U_i}$ for all $i \in I$. Then $s = t$.

(2.) (Gluing) Given a collection of sections $s_i \in \mathcal{F}(U_i)$ such that $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$ for all $i, j \in I$, there exists $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all $i \in I$.

The above two properties are called the **sheaf axioms**. Given a collection of sections $s_i \in \mathcal{F}(U_i)$ and $s \in \mathcal{F}(U)$ as in the second part, the section s is said to be obtained by **gluing the sections** s_i. It is unique by the first part.

Note that we may replace the target category from the category of Abelian groups to other categories, as noted in the previous section. For sheaves with values in a category of algebraic structures, the sheaf axioms can be restated in terms of exact sequences as follows.

Lemma 3.8. *Let $\mathcal{F}$ be a presheaf of Abelian groups (or rings, R-modules, R-algebras with R a ring) on a topological space X. Then $\mathcal{F}$ is a sheaf if and only if for every open subset $U \subseteq X$ and open cover $\{U_i\}_{i\in I}$ of U, the sequence*

$$0 \longrightarrow \mathcal{F}(U) \xrightarrow{\varphi} \prod_{i\in I} \mathcal{F}(U_i) \xrightarrow{\sigma} \prod_{(i,j)\in I\times I} \mathcal{F}(U_i \cap U_j),$$

is exact, where $\varphi : s \mapsto (s|_{U_i})_{i\in I}$ and $\sigma : (s_i)_{i\in I} \mapsto (s_i|_{U_i\cap U_j} - s_j|_{U_i\cap U_j})_{i,j}$.

Proof. The uniqueness condition is equivalent to the sequence being exact at $\mathcal{F}(U)$. Similarly, the gluing condition is equivalent to the sequence being exact at $\prod_i \mathcal{F}(U_i)$. □

A **morphism of sheaves** is a morphism of presheaves. We then obtain the category of sheaves on a topological space X, denoted by $(\mathrm{Sh}(X))$. Note that we really should have chosen a notation such as $(\mathrm{Ab}\,\mathrm{Sh}(X))$ to unambiguously clarify that it is the category of sheaves of *Abelian groups* on X that we are considering. Still, in this section and wherever appropriate, the proof works for any target category of sheaves. With this in mind, we shall prove most statement for sheaves of Abelian groups and left the other cases for the reader.

The category of presheaves on X is denoted by $(\mathrm{PSh}(X))$. Clearly $(\mathrm{Sh}(X))$ is a full subcategory of $(\mathrm{PSh}(X))$.

Remark 3.9. For a sheaf $\mathcal{F}$ on a topological space X, $\mathcal{F}(\emptyset)$ is precisely the final object in the respective category. This can be seen from Lemma 3.8 by considering the empty covering of the empty set. Since the empty product is the terminal object, we obtain the exact sequence $0 \to \mathcal{F}(\emptyset) \to 0 \to 0$, showing that $\mathcal{F}(\emptyset) \cong 0$.

When $\mathcal{F}$ is a sheaf of sets, we can argue directly from the definition of sheaves that $\mathcal{F}(\emptyset)$ is a set with one element. Note that Lemma 3.8 does not apply to this case.

Sheaves can be restricted onto any open subset.

Example 3.10 (Restriction Sheaf)**.** Suppose we have a sheaf $\mathcal{F}$ on a topological space X. If U is an open subset of X, then the restriction $\mathcal{F}|_U$ is a sheaf.

Example 3.11 (Subsheaf)**.** Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on a topological space X with values in algebraic objects such as Abelian groups or rings. If $\mathcal{F}(U)$ is a subgroup of $\mathcal{G}(U)$ for all open $U \subseteq X$, then $\mathcal{F}$ is said to be a **subsheaf** of $\mathcal{G}$.

It is not true in general that, if $\mathcal{G}$ is a sheaf, and $\mathcal{F}$ is a presheaf such that $\mathcal{F}(U)$ is a subgroup of $\mathcal{G}(U)$ for all $U \subseteq X$, then $\mathcal{F}$ is a subsheaf of $\mathcal{G}$. Hence a sub-presheaf of a sheaf is not necessarily a sheaf.

Example 3.12 (Sheaves of Continuous and Smooth Functions)**.** Let X and Y be topological spaces. Define $\mathcal{C}$ to be the presheaf of continuous functions, that is, for every open subset $U \subseteq X$,

$$\mathcal{C}(U) := \{f : U \to Y : \ f \text{ is continuous}\}.$$

The restriction maps are restrictions of functions.

In a similar manner, let M be a smooth manifold. Then the presheaf of real-valued smooth functions on M (Example 3.2) is a sheaf.

There are further examples of sheaves coming from differential geometry such as the sheaf of holomorphic function on $\mathbb{C}$.

Example 3.13 (Hom-set as a Sheaf)**.** Let X be a topological space and Y a set. Then the assignment $U \mapsto \{f : U \to Y : f \text{ is a function}\}$ define a sheaf on X.

Example 3.14 (Skyscraper Sheaves)**.** Let X be a topological space and A an Abelian group. The **skyscraper sheaf in the point $p \in X$ with value A** is defined to be the sheaf

$$\mathcal{F}(U) = \begin{cases} A, & p \in U; \\ 0, & p \notin U. \end{cases}$$

Example 3.15. Ring of regular functions on a variety

Not every presheaf is a sheaf, as the following non-examples show.

Example 3.16. Constant presheaves are in general not a sheaf. As an example, consider the two-point topological space $X = \{x, y\}$ with discrete topology. The constant presheaf $\mathcal{F}$ with value $\mathbb{Z}$ is not a sheaf, as the reader may check.

Note that, even if we modify $\mathcal{F}(\emptyset)$ to be equal to $\{0\}$, it is still not a sheaf because the gluing axiom cannot be satisfied.

Example 3.17 (Presheaf of Bounded Functions)**.** Let X be a topological space. For each open set $U \subseteq X$, define

$$\mathcal{F}(U) := \{f : U \to \mathbb{R} : \ f \text{ is bounded}\}.$$

Then $\mathcal{F}$ is a presheaf on X. In general it will not be a sheaf.

▸ **Exercise 3.18.** Verify all the examples.

Sheaves on a Basis

Let X be a topological space and $\mathcal{B}$ a basis of X. It will be useful, when we define affine schemes, to consider sheaves defined explicitly only on the basis $\mathcal{B}$. The main result of this section is that there is a unique sheaf on the whole topological space X extending the sheaf defined on a basis.

Definition 3.19. Suppose X is a topological space and $\mathcal{B}$ a basis of X.

(1.) A **presheaf on $\mathcal{B}$** is a contravariant functor $\mathcal{F}\colon \mathcal{B} \to (\mathrm{Ab})$ where $\mathcal{B}$ is regarded as a full subcategory of (Ouv_X).

(2.) A **sheaf on $\mathcal{B}$** is a presheaf $\mathcal{F}$ on $\mathcal{B}$ which satisfies the analogous conditions about gluing as for sheaves: for all $U \in \mathcal{B}$ and an open covering $U = \bigcup_{i\in I} U_i$ where $U_i \in \mathcal{B}$,

(a) (Uniqueness) Let $s, t \in \mathcal{F}(U)$ be such that $s|_{U_i} = t|_{U_i}$ for all i. Then $s = t$.

(b) (Gluing) A collection of sections $s_i \in \mathcal{F}(U_i)$ is said to be **compatible** if for all $i, j \in I$, and all open set $U_k \in \mathcal{B}$ with $U_k \subseteq U_i \cap U_j$, we have

$$s_i|_{U_k} = s_j|_{U_k}.$$

It is required that, whenever the sections $s_i \in \mathcal{F}(U_i)$ are compatible, then there is a section $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$.

Note that the gluing condition has to be modified from the gluing condition for sheaves. This is because the intersection of two basic open sets $U_i, U_j \in \mathcal{B}$ might not belong in $\mathcal{B}$. It is still open, however, and so we may cover the intersection $U_i \cap U_j$ by open sets from $\mathcal{B}$. So we need to check compatibility of sections in those smaller open sets instead.

Remark 3.20. If the basis $\mathcal{B}$ is stable under finite intersection (as is the case for the basis of spectra consisting of principal open subsets, Proposition 2.13), then the usual gluing condition can be used instead. For such a basis $\mathcal{B}$, a presheaf $\mathcal{F}$ on $\mathcal{B}$ is a sheaf if for all $U \in \mathcal{B}$ and an open covering $U = \bigcup_{i\in I} U_i$ where $U_i \in \mathcal{B}$,

(1.) it satisfy uniqueness, and,

(2.) for all sections $s_i \in \mathcal{F}(U_i)$, if $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$, then there is a section $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$.

Proposition 3.21. *Let X be a topological space, $\mathcal{B}$ a basis of X, and $\mathcal{F}$ a sheaf on $\mathcal{B}$. There is a sheaf $\mathcal{F}'$ on X such that $\mathcal{F}'(U) \cong \mathcal{F}(U)$ for all $U \in \mathcal{B}$. The sheaf $\mathcal{F}'$ is unique up to isomorphism.*

Proof. content... □

Morphisms between sheaves on a basis can be extended to the morphisms between the extended sheaves.

3.2 Stalks

Let X be a topological space and let $x \in X$ be a point. The collection of Abelian groups $\mathcal{F}(U)$, where U is an open neighbourhood of x, with restriction maps is a direct system. The direct limit of this system leads to the notion of stalks.

Definition 3.22. Let $\mathcal{F}$ be a presheaf on X. The **stalk** at $x \in X$ of $\mathcal{F}$ is the Abelian group

$$\mathcal{F}_x := \varinjlim_{\substack{U \ni x, \\ U \text{ open}}} \mathcal{F}(U).$$

More explicitly, $\mathcal{F}_x$ is the set of equivalence classes of pairs (U, s) where U is an open neighbourhood of x and $s \in \mathcal{F}(U)$. Two pairs (U, s) and (V, t) are definitionally equivalent if there is an open neighbourhood W of x such that $W \subseteq U \cap V$ and $s|_W = t|_W$. Let us denote the equivalence class of (U, s) by $[U, s]$. The group operation on $\mathcal{F}_x$ is defined by

$$[U, s] + [V, t] = [U \cap V, s|_{U\cap V} + t|_{U\cap V}].$$

▸ **Exercise 3.23.** Prove that the above relation is indeed an equivalence relation on the set of pairs (U, x), and that the multiplication is well-defined which makes $\mathcal{F}_x$ into an Abelian group.

By the definition of direct limits, for each open neighbourhood U of x there is a natural homomorphism $\mathcal{F}(U) \to \mathcal{F}_x$ given by $s \mapsto [U, s]$. If U is understood, we shall write s_x for the equivalent class $[U, s]$. The element s_x is called the **germ** of s in x.

If $\mathcal{F}$ and $\mathcal{G}$ are presheaves on X with a morphism $\varphi \colon \mathcal{F} \to \mathcal{G}$, then there is an induced homomorphism of Abelian group between stalks

$$\begin{aligned} \varphi_x \colon \mathcal{F}_x &\longrightarrow \mathcal{G}_x \\ [U, s] &\longmapsto [U, \varphi_U(s)]. \end{aligned}$$

This is the same map from the universal property of direct limits.

▸ **Exercise 3.24.** Show that φ_x is a group homomorphism.

We thus see that taking stalks gives a functor from the category of presheaves on X to the category of Abelian groups. Further, if the presheaves take values in sets or rings, then stalks will be sets or rings, respectively.

Lemma 3.25. *Let $\mathcal{F}$ be a sheaf on a topological space X and let $s, t \in \mathcal{F}(X)$ be sections such that $s_x = t_x$ for all $x \in X$. Then $s = t$.*

Proof. For each $x \in X$, there is an open neighbourhood U_x such that $s|_{U_x} = t|_{U_x}$. Since the open sets of the form U_x covers X, this shows that $s = t$. □

Injectivity and bijectivity of morphisms in the data of a morphism of sheaves can be checked on stalks, as the following proposition shows.

Proposition 3.26. *Let X be a topological space, $\mathcal{F}$ and $\mathcal{G}$ sheaves on X, and $\varphi, \psi \colon \mathcal{F} \to \mathcal{G}$ morphisms of sheaves. Then*

(1.) The homomorphisms $\varphi_U \colon \mathcal{F}(U) \to \mathcal{G}(U)$ are injective for all open $U \subseteq X$ if and only if the induced morphisms $\varphi_x \colon \mathcal{F}_x \to \mathcal{G}_x$ are injective for all $x \in X$.

(2.) The homomorphisms $\varphi_U \colon \mathcal{F}(U) \to \mathcal{G}(U)$ are bijective for all open $U \subseteq X$ if and only if the induced morphisms $\varphi_x \colon \mathcal{F}_x \to \mathcal{G}_x$ are bijective for all $x \in X$.

(3.) $\varphi_x = \psi_x$ for all $x \in X$ if and only if $\varphi = \psi$.

Proof. We shall prove the statements (2.) and (3.), leaving (1.) as an exercise.

If the homomorphisms φ_U are bijective for all open subset U, then the stalks are isomorphic under φ_x by the universal property of direct limits. (Abstract nonsense: every functor preserves isomorphisms.) So we only have to prove the converse. Assume that $\varphi_x \colon \mathcal{F}_x \to \mathcal{G}_x$ are bijective for all $x \in X$.

(Injectivity.) Let $s, t \in \mathcal{F}(U)$ be sections and suppose that $\varphi_U(s) = \varphi_U(t)$. Then for every $x \in U$, we have $\varphi_x(s_x) = [U, \varphi_U(s)] = [U, \varphi_U(t)] = \varphi_x(t_x)$. Hence $s_x = t_x$ for all $x \in U$, and so $s = t$ by Lemma 3.25.

(Surjectivity.) Let $b \in \mathcal{G}(U)$. For each $x \in X$, there is a germ $s_x \in \mathcal{F}_x$ such that $\varphi_x(s_x) = b_x$. Let s be a section defined on an open neighbourhood V of x such that $[V, s] = s_x$. Then by assumption $[V, \varphi_V(s)] = b_x$. By definition, there is an open neighbourhood W_x of x and a section $w(x) := s|_{W_x} \in \mathcal{F}(W_x)$ such that $\varphi_{W_x}(w(x)) = b|_{W_x}$.

The open sets $\{W_x\}$, $x \in U$, form an open cover of U. Furthermore, if $x, y \in U$, then $w(x)|_{W_x \cap W_y} = w(y)|_{W_x \cap W_y}$. Indeed, we have

$$\begin{aligned}
\varphi_{W_x \cap W_y}(w(x)|_{W_x \cap W_y}) &= \varphi_{W_x}(w(x))|_{W_x \cap W_y} && \text{(Definition of morphism of sheaves)} \\
&= b|_{W_x}|_{W_x \cap W_y} && \text{(Definition of } w(x)\text{)} \\
&= b|_{W_y}|_{W_x \cap W_y} && \text{(Definition of morphism of sheaves)} \\
&= \varphi_{W_y}(w(y))|_{W_x \cap W_y} && \text{(Definition of } w(y)\text{)} \\
&= \varphi_{W_x \cap W_y}(w(y)|_{W_x \cap W_y}) && \text{(Definition of morphism of sheaves).}
\end{aligned}$$

(From now on we will drop justification for routine computations like these.) Since we have just proved that the morphism $\varphi_{W_x \cap W_y}$ is injective, and thus $w(x)|_{W_x \cap W_y} = w(y)|_{W_x \cap W_y}$ as claimed. In particular, the sections $w(x)$ can be glued together. Let $w \in \mathcal{F}(U)$ denote the resulting section so that $w|_{W_x} = w(x)$ for all $x \in U$. Then

$$\varphi_U(w)|_{W_x} = \varphi_{W_x}(w|_{W_x}) = \varphi_{W_x}(w(x)) = b|_{W_x}$$

for all open set W_x in the cover. By the sheaf condition, $\varphi_U(w) = b$.

Finally we prove that if $\varphi_x = \psi_x$ for all $x \in X$, then $\varphi = \psi$. Note that the converse is trivial. Let $s \in \mathcal{F}(U)$ be a section. Then

$$(\varphi_U(s))_x = \varphi_x(s_x) = \psi_x(s_x) = (\psi_U(s))_x$$

for all $x \in X$. Hence $\varphi_U(s) = \psi_U(s)$ by Lemma 3.25. □

▸ **Exercise 3.27.** Finish the proof of Proposition 3.26.

Remark 3.28. Keep the notation of Proposition 3.26.

(1.) Note that this proof does not use the fact that $\mathcal{F}$, $\mathcal{G}$ is a sheaf of Abelian groups, so the proposition holds for sheaves of sets.

(2.) The analogous statement for surjections is false. It is in general not true that $\varphi_U \colon \mathcal{F}(U) \to \mathcal{G}(U)$ are surjective for all open $U \subseteq X$ if and only if the induced morphisms $\varphi_x \colon \mathcal{F}_x \to \mathcal{G}_x$ are surjective for all $x \in X$.

The following proposition is much harder to prove.

Proposition 3.29. *Let X be a topological space, $\mathcal{F}$, $\mathcal{G}$ sheaves on X, and $\varphi \colon \mathcal{F} \to \mathcal{G}$ a morphism of sheaves. Then*

(1.) φ is a monomorphism in $(\mathrm{Ab}(X))$ if and only if φ_x is injective for all $x \in X$.

(2.) φ is an epimorphism in $(\mathrm{Ab}(X))$ if and only if φ_x is surjective for all $x \in X$.

(3.) φ is an isomorphism in $(\mathrm{Ab}(X))$ if and only if φ_x is bijective for all $x \in X$.

Proof. (1) First assume that φ is a monomorphism. In light of Proposition 3.26, it suffices to show that each morphism $\varphi_U \colon \mathcal{F} \to \mathcal{G}$ is injective. Let $\mathcal{K}$ be the kernel sheaf of φ, i.e. $\mathcal{K}(U) = \ker \varphi_U$ for all open $U \subseteq X$. This is a subsheaf of $\mathcal{F}$ (Problem 3.2).

Let $i \colon \mathcal{K} \to \mathcal{F}$ be the inclusion morphism, i.e. i_U is the inclusion $\mathcal{K}(U) \hookrightarrow \mathcal{F}(U)$ for all open subset $U \subseteq X$. There is also another morphism of sheaves $0 \colon \mathcal{K} \to \mathcal{F}$ such that $0_U \colon \mathcal{K}(U) \to \mathcal{F}(U)$ is the zero homomorphism for all U. Clearly $\varphi \circ i = 0 = \varphi \circ 0$. Since φ is a monomorphism, $i = 0$. Hence for each open $U \subseteq X$, we see that $a = i_U(a) = 0_U(a) = 0$ for all $a \in \ker(\varphi_U)$. In other words, φ_U is injective for all open subset U.

Conversely, assume that φ_x is injective for all $x \in X$. Suppose $\psi, \eta \colon \mathcal{H} \to \mathcal{F}$ are morphisms of sheaves such that $\varphi \circ \psi = \varphi \circ \eta$. Since taking stalks is a functor, $\varphi_x \circ \psi_x = \varphi_x \circ \eta_x$ for all $x \in X$. Using the fact that φ_x is injective, we have $\psi_x = \eta_x$ for all $x \in X$. By Proposition 3.26, $\psi = \eta$, and so φ is a monomorphism.

(2) and (3) □

Remark 3.30. Proposition 3.29 remains true if the sheaves considered are sheaves of sets instead of sheaves of Abelian groups, although the arguments must be modified.

Stalks behave well with basis.

Lemma 3.31 (Stalks Can Be Computed on a Basis)**.** *Let X be a topological space with a basis $\mathcal{B}$, $x \in X$ a point, and $\mathcal{F}$ a presheaf on X. The collection of Abelian groups of the form*

$$\mathcal{F}(B), \quad B \in \mathcal{B},$$

together with the restriction maps of $\mathcal{F}$ form a direct system. Furthermore, the direct limit of this system is equal to the stalk of $\mathcal{F}$ at x:

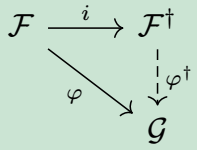
$$\mathcal{F}_x = \varinjlim_{\substack{B \ni x, \\ B \in \mathcal{B}}} \mathcal{F}(B).$$

Proof. arg1 □

3.3 Sheafification

As seen above, some presheaves fail to be sheaves because of the failure to glue sections. There is a canonical way to associate a sheaf to a presheaf.

Theorem 3.32 (Sheafification)**.** *Let X be a topological space and $\mathcal{F}$ a presheaf. There exists a sheaf $\mathcal{F}^\dagger$ (called the **sheafification** of $\mathcal{F}$) with a morphism of presheaves $i\colon \mathcal{F} \to \mathcal{F}^\dagger$ such that if $\mathcal{G}$ is a sheaf with a morphism of presheaves $\varphi\colon \mathcal{F} \to \mathcal{G}$, then there is a unique morphism of sheaves $\varphi^\dagger\colon \mathcal{F}^\dagger \to \mathcal{G}$ such that the following diagram commutes:*



The pair $(\mathcal{F}^\dagger, i)$ is unique up to unique isomorphism. Moreover, $i_x\colon \mathcal{F}_x \to \mathcal{F}_x^\dagger$ is an isomorphism for all $x \in X$.

The sheaf $\mathcal{F}^\dagger$ is also called the **sheaf associated to the presheaf $\mathcal{F}$**.

Proof. We divide the proof into several parts.

[**Definition.**] Let $U \subseteq X$ be an open subset. Define $\mathcal{F}^\dagger(U)$ to be the set of functions $f\colon U \to \coprod_{x \in U} \mathcal{F}_x$ satisfying

(∗) for all $x \in U$, there is an open neighbourhood $V \subseteq U$ of x and $s \in \mathcal{F}(V)$ such that for all $y \in V$, $f(y) = s_y$.

With pointwise addition this is an abelian group. The restriction map $\operatorname{res}_W^U\colon \mathcal{F}^\dagger(U) \to \mathcal{F}^\dagger(W)$ for $W \subseteq U$ is defined to be the restriction of functions, so $\operatorname{res}_W^U(f) = f|_W$ for

all $f \in \mathcal{F}^\dagger(U)$. It is easily checked that $f|_W \in \mathcal{F}^\dagger(W)$, and that restrictions are group homomorphisms. From this it follows that $\mathcal{F}^\dagger$ is a presheaf on X.

[$\mathcal{F}^\dagger$ **is a sheaf.**] To show that $\mathcal{F}^\dagger$ is a sheaf, we assume that $U \subseteq X$ is an open set with an open covering $\bigcup_{i\in I} U_i = U$.

(1.) (Uniqueness) Easy.

(2.) (Gluing.) For each $i \in I$, let $f_i \in \mathcal{F}^\dagger(U_i)$ be functions such that $f_i|_{U_i\cap U_j} = f_j|_{U_i\cap U_j}$ for all $i, j \in I$. Define $f\colon U \to \coprod_{x\in U} \mathcal{F}_x$ by $f(x) = f_i(x)$ if U_i contains x. This is well-defined because if U_i and U_j both contains x, then $x \in U_i \cap U_j$ and so $f_i(x) = f_i|_{U_i\cap U_j}(x) = f_j|_{U_i\cap U_j} = f_j(x)$. Further, the condition $(*)$ hold for f.

[**Definition of** i**.**] Next we define a morphism of presheaves $i\colon \mathcal{F} \to \mathcal{F}^\dagger$. For each $a \in \mathcal{F}(U)$, let $i_U(a)$ be the function $x \mapsto a_x$ for all $x \in U$. This is clearly seen to be a group homomorphism compatible with restriction maps of $\mathcal{F}$ and $\mathcal{F}^\dagger$. Thus i is really a morphism of presheaves.

[i_x **is surjective.**] Let $x \in X$ We now show that $\mathcal{F}_x^\dagger \cong \mathcal{F}_x$ under the map i_x. Let U be an open neighbourhood of x and let $f \in \mathcal{F}^\dagger(U)$. By definition there is an open neighbourhood V of x and $s \in \mathcal{F}(V)$ such that for all $y \in V$, we have $f(y) = s_y$. Then $[U, f] = [V, i_V(s)]$. In particular, the map $i_x\colon \mathcal{F}_x \to \mathcal{F}_x^\dagger$ is a surjection.

[i_x **is injective.**] To show that i_x is injective, let 0 denote the function $x \mapsto 0_x$ for all $x \in X$. Plainly, $[X, 0]$ is the additive identity of $\mathcal{F}_x$. Suppose that V is a neighbourhood of x and $s \in \mathcal{F}(V)$ is a section such that $i_x([V, s]) = [V, i_V(s)] = [X, 0]$. By definition, there is an open neighbourhood $W \subseteq V$ of x such that $i_W(s|W) = i_V(s)|_W = 0|_W$. But this implies that $s_y = 0_y$ for all $y \in W$. By Lemma 3.25, $s|_W = 0$. Hence $[V, s] = [W, s|_W] = [W, 0]$, and so i_x is an isomorphism.

[**Universal property of** $\mathcal{F}^\dagger$**.**] Now we show the universal property. Let $\mathcal{G}$ be a sheaf and $\varphi\colon \mathcal{F} \to \mathcal{G}$ a morphism of presheaves. Consider the diagram

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\ i\ } & \mathcal{F}^\dagger \\ {\scriptstyle\varphi}\big\downarrow & & \big\vdots \\ \mathcal{G} & \xrightarrow[\ j\ ]{} & \mathcal{G}^\dagger \end{array}$$

where $\mathcal{G}^\dagger$ is defined as set of functions similar to $\mathcal{F}^\dagger$, and j is morphism of preshaves $\mathcal{G} \to \mathcal{G}^\dagger$. We have shown above that j_x is an isomorphism for all $x \in X$, and so j is an isomorphism of sheaves (Proposition 3.29).

Since we are given a morphism $\varphi\colon \mathcal{F} \to \mathcal{G}$, we have the induced morphisms on stalks $\varphi_x\colon \mathcal{F}_x \to \mathcal{G}_x$ for all $x \in X$. Define a morphism $\varphi^\sharp\colon \mathcal{F}^\dagger \to \mathcal{G}^\dagger$ on an open set $U \subseteq X$ by

$$\varphi^\sharp_U(f) = \text{ the composition } U \xrightarrow{f} \coprod_{x\in U} \mathcal{F}_x \xrightarrow{\coprod_x \varphi_x} \coprod_{x\in U} \mathcal{G}_x,$$

that is, $\varphi^\sharp_U(f)(x) = \varphi_x(f(x))$ for all $x \in U$. It is easy to check that this is a group homomorphism. By composing with j^{-1} we get a morphism $\varphi^\dagger = j^{-1} \circ \varphi^\sharp\colon \mathcal{F}^\dagger \to \mathcal{G}$.

To verify that $\varphi^\dagger \circ i = \varphi$, let $U \subseteq X$ be an open set and $s \in \mathcal{F}(U)$. Then $\varphi^\sharp_U(i_U(s))$ is equal to the function given by $s \mapsto \varphi_U(s)_x$ for all $x \in X$. Because j_U is an isomorphism, we immediately have $j_U^{-1}(\varphi^\sharp_U(i_U(s))) = \varphi_U(s)$. This proves the claim.

The fact that $\mathcal{F}^\dagger$ and $\varphi^\dagger$ is unique follows from a standard uniqueness proof of universal objects. □

The set $\coprod_{x \in U} \mathcal{F}_x$ used in the construction of the sheafification is known as the **étalé space** of the presheaf $\mathcal{F}$. The word *étalé* comes from French meaning *spreaded out*.

The most important fact about sheafification can be phrased in the following form: it is a left adjoint to the inclusion functor of sheaves into the category of presheaves.

Proposition 3.33 (Sheafification is left adjoint to the inclusion functor)**.** *Let X be a topological space. Sheafification defines a functor $(-)^\dagger$ from the category of presheaves $(\mathrm{PSh}(X))$ to the category of sheaves $(\mathrm{Sh}(X))$. Furthermore, the sheafification functor is left adjoint to the inclusion functor of $(\mathrm{Sh}(X))$ into $(\mathrm{PSh}(X))$.*

Proof. Follows from Theorem 3.32 □

Example 3.34 (Constant Sheaf)**.** Constant Sheaves

3.4 Direct Image and Inverse Image

Suppose we have a continuous map $f : X \to Y$ between two topological spaces. There is a way to transfer a sheaf on X to a sheaf on Y, and vice versa. We first start with the notion of pushing a sheaf on X by f into a sheaf on Y.

Definition 3.35 (Pushforward, Direct Image Sheaf)**.** Let $f : X \to Y$ be a continuous map of topological spaces. If $\mathcal{F}$ is a sheaf on X, then the **pushforward of $\mathcal{F}$ under f** or the **direct image sheaf of $\mathcal{F}$** is the sheaf $f_*\mathcal{F}$ defined by

$$f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(V))$$

for all open subset $V \subseteq Y$. The restriction maps of $f_*\mathcal{F}$ is the restriction maps of $\mathcal{F}$.

▸ **Exercise 3.36.**

(1.) Prove that if $\mathcal{F}$ is a presheaf on X, then $f_*\mathcal{F}$ defined as above is a presheaf on Y.

(2.) Prove that if $\mathcal{F}$ is a sheaf, then $f_*\mathcal{F}$ is also a sheaf on Y.

▸ **Exercise 3.37.** Prove that f_* define a functor $(\mathrm{Sh}(X)) \to (\mathrm{Sh}(Y))$.

On the other hand, pulling back a sheaf on Y to a sheaf on X is harder. This is due to the fact that the image $f(U)$ of an open set $U \subseteq X$ under a continuous map need not be

open. We circumvent this obstruction by considering the filtered inductive system of open sets containing $f(U)$. If $\mathcal{G}$ is a presheaf on Y, then the assigment

$$U \mapsto \varinjlim_{\substack{V \supseteq f(U) \\ V \text{ open in } Y}} \mathcal{G}(V)$$

is a presheaf on X with the restriction maps induced by the restriction maps of $\mathcal{G}$. Let us write $f^+\mathcal{G}$ for this presheaf.

▸ **Exercise 3.38.** Show that $f^+\mathcal{G}$ is a presheaf.

The **inverse image sheaf** $f^{-1}\mathcal{G}$ of $\mathcal{G}$ is the sheafification of the presheaf $f^+\mathcal{G}$ defined above. Note that, even if $\mathcal{G}$ is a sheaf, $f^+\mathcal{G}$ might not be a sheaf, so the sheafification is required. If $i\colon Z \to X$ is the inclusion of a subspace Z into X, and $\mathcal{G}$ is a sheaf on X, then we also write $\mathcal{G}|_Z$ for the inverse image sheaf $i^{-1}\mathcal{G}$.

Remark 3.39. Let X be a topological space and $\mathcal{G}$ a sheaf on X. If $U \subseteq X$ is an open subset and $i\colon U \to X$ is the inclusion map, then the inverse image presheaf $f^+\mathcal{G}$ is already a sheaf because it is equal to $\mathcal{G}|_U$. It then follows that $i^{-1}\mathcal{G} = \mathcal{G}|_U$ in this case.

Proposition 3.40. *Let $f\colon X \to Y$ and $g\colon Y \to Z$ be continuous maps of topological spaces, and let $\mathcal{G}$ be a sheaf on Z. Then*

$$(g \circ f)^{-1}(\mathcal{G}) = f^{-1}(g^{-1}\mathcal{G}).$$

Proof. content... □

Lemma 3.41 (Stalk of Inverse Image Sheaf)**.** *Let $f\colon X \to Y$ be a continuous map of topological spaces and $\mathcal{G}$ a sheaf on Y. Then for all $x \in X$,*

$$(f^{-1}\mathcal{G})_x \cong \mathcal{G}_{f(x)}.$$

Proof. By Theorem 3.32, we can compute the stalk at $x \in X$ of the presheaf $f^+\mathcal{G}$ instead. We have

$$(f^+\mathcal{G})_x = \varinjlim_{U \ni x} f^{-1}\mathcal{G}(U) = \varinjlim_{U \ni x} \varinjlim_{\substack{V \supseteq f(U) \\ V \text{ open in } Y}} \mathcal{G}(V).$$

Write $\mathcal{I}$ for the category of pairs

$$\{(U, V) : U \text{ open in } X, U \ni x, V \text{ open in } Y, Y \supseteq f(U)\}.$$

There is a unique morphism $(U, V) \to (U', V')$ if and only if $U \supseteq U'$ and $V \supseteq V'$. Then the stalks at x of $f^+\mathcal{G}$ is equal to the colimit

$$(f^+\mathcal{G})_x = \varinjlim_{\mathcal{I}} \mathcal{G}(V).$$

Now define $\mathcal{J}$ to be the category of open neighbourhoods of $f(x)$ ordered by inverse inclusion. Let $p\colon \mathcal{I} \to \mathcal{J}$ be the projection functor defined by $p(U,V) = V$ for all $(U,V) \in \mathcal{I}$. Then p is a final functor. Indeed, if $W \in \mathcal{J}$, then the comma category $(W \downarrow p)$ is equal to the category of pairs $(U,V) \in \mathcal{I}$ such that $V \subseteq W$. There is a morphism $(U,V) \to (U',V')$ if and only if $V' \subseteq V$ and this morphism is unique. (Unravel the definition!) It is then clear that $(W \downarrow p)$ is not empty because $(f^{-1}(W), W) \in \mathcal{I}$. Further, given two pairs (U,V) and (U',V'), there is an object

$$(U \cap U', V \cap V') \in \mathcal{I}$$

such that $(U,V) \to (U \cap U', V \cap V') \leftarrow (U',V')$. Hence W is non-empty and connected, and so p is a final functor as needed.

By definition, it follows that restricting diagrams along p does not change their colimit. Consequently,

$$(f^+\mathcal{G})_x = \varinjlim_{\mathcal{I}} \mathcal{G} \circ p \cong \varinjlim_{W \in \mathcal{J}} \mathcal{G}(W) = \mathcal{G}_{f(x)}. \qquad \square$$

Remark 3.42. Let X be a topological space with $\mathcal{G}$ a sheaf on X. Suppose that $i\colon \{x\} \to X$ is an inclusion of a point $x \in X$ into X. Then

$$i^{-1}\mathcal{G} = \mathcal{G}_x.$$

The latter is to be interpreted as a sheaf on the one-point space $\{x\}$ which is equal to $\mathcal{G}_x$ on $\{x\}$. This gives another proof of Lemma 3.41.

Proposition 3.43 (Inverse Image is Left Adjoint to Direct Image)**.** *Let $f\colon X \to Y$ be a continuous map between two topological spaces. Let $\mathcal{F}$ be a sheaf on X, and let $\mathcal{G}$ be a presheaf on Y. Then there is a bijection*

$$\Phi_{\mathcal{F},\mathcal{G}}\colon \operatorname{Hom}_{(\mathrm{Sh}(X))}(f^{-1}\mathcal{G}, \mathcal{F}) \to \operatorname{Hom}_{(\mathrm{PSh}(Y))}(\mathcal{G}, f_*\mathcal{F})$$

which is functorial in $\mathcal{F}$ and $\mathcal{G}$.

Hence there is an adjunction $f^{-1} \dashv f_*$.

Proof. We divide the proof into several steps.

(1.) We first determine the unit $\eta\colon 1_{(\mathrm{PSh}(Y))} \Rightarrow f_*f^{-1}$ of this adjunction. Let $\mathcal{G}$ be a presheaf on Y. We need to find a morphism $\eta(\mathcal{G})\colon \mathcal{G} \to f_*f^{-1}\mathcal{G}$. To do this, let $V \subseteq Y$ be an open subset. Since $f(f^{-1}(V)) \subseteq V$, the open set V appears in the direct system in the definition of the presheaf $f^+\mathcal{G}$:

$$f^+\mathcal{G}(f^{-1}(V)) = \varinjlim_{\substack{O \supseteq f(f^{-1}(V)) \\ O \text{ open in } X}} \mathcal{G}(O).$$

Hence there is a canonical morphism $\mathcal{G}(V) \to f^+\mathcal{G}(f^{-1}(V))$. After passing to the sheafification, we obtain a morphism $\mathcal{G}(V) \to f^{-1}\mathcal{G}(f^{-1}(V)) = f_*f^{-1}\mathcal{G}(V)$. It is

easily checked that this is compatible with restriction, and thus define a genuine morphism of presheaves $\eta(\mathcal{G})\colon \mathcal{G} \to f_* f^{-1}\mathcal{G}$, which we take to be $\eta(\mathcal{G})$.

We also write $\kappa^V_{f^{-1}(V)}$ for the canonical morphism $\mathcal{G}(v) \to f^+\mathcal{G}(f^{-1}(V))$ into the direct limit. Let $i\colon f^+\mathcal{G} \to f^{-1}\mathcal{G}$ be the sheafification map, and so $\eta(\mathcal{G})$ can also be given by the formula

$$\eta(\mathcal{G})_V = i_{f^{-1}(V)} \circ \kappa^V_{f^{-1}(V)}.$$

(2.) The proof that η is a natural transformation essentially boils down to the fact that taking colimits and sheafification are natural. The proof is omitted.

(3.) The next step is to define a morphism

$$\Phi_{\mathcal{F},\mathcal{G}}\colon \operatorname{Hom}_{(\operatorname{Sh}(X))}(f^{-1}\mathcal{G}, \mathcal{F}) \to \operatorname{Hom}_{(\operatorname{PSh}(Y))}(\mathcal{G}, f_*\mathcal{F}).$$

Let $\varphi\colon f^{-1}\mathcal{G} \to \mathcal{F}$ be a morphism of sheaves. We need to define a morphism $\Phi_{\mathcal{F},\mathcal{G}}(\varphi)\colon \mathcal{G} \to f_*\mathcal{F}$. This is basically forced upon us since we know the unit η. Let $V \subseteq Y$ be an open set. Then define $\Phi_{\mathcal{F},\mathcal{G}}(\varphi)_V$ by

$$\mathcal{G}(V) \xrightarrow{\eta(\mathcal{G}_V)} f_* f^{-1}\mathcal{G}(V) = f^{-1}\mathcal{G}(f^{-1}(V)) \xrightarrow{\varphi_{f^{-1}(V)}} \mathcal{F}(f^{-1}(V)) = f_*\mathcal{F}(V).$$

In particular, $\Phi_{\mathcal{F},\mathcal{G}}(\varphi) := f_*\varphi \circ \eta(\mathcal{G})$. This shows that $\Phi_{\mathcal{F},\mathcal{G}}(\varphi)$ is indeed a morphism of presheaves.

(4.) We now construct the inverse $\Psi_{\mathcal{F},\mathcal{G}}$ of $\Phi_{\mathcal{F},\mathcal{G}}$. Let $\psi\colon \mathcal{G} \to f_*\mathcal{F}$ be a morphism of presheaves. We seek to define a morphism $\Psi(\psi)\colon f^{-1}\mathcal{G} \to \mathcal{F}$. By the universal property of sheafification (Theorem 3.32) it is enough for us to define a morphism of presheaves $a\colon f^+\mathcal{G} \to \mathcal{G}$. Let $U \subseteq X$ be an open subset. As a reminder, a_U is a morphism

$$\varinjlim_{\substack{V \supseteq f(U),\\ V \text{ open in } Y}} \mathcal{G}(U) \to \mathcal{F}(U).$$

A morphism out of a direct limit is uniquely determined by the morphisms out of each object in the direct system. With this in mind, let $V \supseteq f(U)$ be an open subset of Y and let κ^V_U be the canonical map $\mathcal{G}(V) \to f^+\mathcal{G}(U)$ into the direct limit. Then $f^{-1}(V) \supseteq f^{-1}(f(U)) \supseteq U$. Hence there exists a series of morphisms

$$\mathcal{G}(V) \xrightarrow{\psi_V} f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(V)) \xrightarrow{\operatorname{res}_U^{f^{-1}(V)}} \mathcal{F}(U)$$

which is compatible with restriction. Hence there is a unique morphism $a_U\colon f^+\mathcal{G}(U) \to \mathcal{F}(U)$ such that

$$\operatorname{res}_U^{f^{-1}(V)} \circ \psi_V = a_U \circ \kappa^V_U.$$

Together they give rise to a morphism $a\colon f^+\mathcal{G} \to \mathcal{F}$ of presheaves, as is easily checked. Define $\Psi(\psi)$ to be the unique morphism of sheaves $a^\dagger\colon f^{-1}\mathcal{G} \to \mathcal{F}$ associated to a. In particular, if we write $i\colon f^+\mathcal{G} \to f^{-1}\mathcal{G}$ for the sheafification map, then $\Psi(\psi) \circ i = a$ and $\Psi(\psi)$ is uniquely determined by this formula.

(5.) Finally we have to show that $\Phi_{\mathcal{F},\mathcal{G}}$ and $\Psi_{\mathcal{F},\mathcal{G}}$ are mutually inverse. This is simply algebra (or diagram chasing) from the properties we have already noted.

(a) Let $\varphi\colon f^{-1}\mathcal{G} \to \mathcal{F}$ be a morphism of sheaves. Then $\Phi(\varphi) = f_*\varphi \circ \eta(\mathcal{G})$. By definition $\Psi(\Phi(\varphi))$ is the unique morphism such that for any open subset $U \subseteq X$ and open subset $V \supseteq f(U)$, the equation

$$\Psi(\Phi(\varphi))_U \circ i_U \circ \kappa_U^V = \mathrm{res}_U^{f^{-1}V} \circ \Phi(\varphi)_V$$

holds. Hence we check that

$$\begin{aligned}
\varphi_U \circ i_U \circ \kappa_U^V &= \varphi_U \circ i_U \circ (\mathrm{res}_U^{f^{-1}V} \circ \kappa_{f^{-1}V}^V) \\
&= \varphi_U \circ (\mathrm{res}_U^{f^{-1}V} \circ i_{f^{-1}V}) \circ \kappa_{f^{-1}V}^V \\
&= (\mathrm{res}_U^{f^{-1}V} \circ \varphi_{f^{-1}V}) \circ i_{f^{-1}V} \circ \kappa_{f^{-1}V}^V \\
&= \mathrm{res}_U^{f^{-1}V} \circ (f_*\varphi_V \circ \eta(\mathcal{G})_V) \\
&= \mathrm{res}_U^{f^{-1}V} \circ \Phi(\varphi)_V
\end{aligned}$$

using the fact that all the maps are compatible with restrictions. By uniqueness, $\Psi(\Phi(\varphi)) = \varphi$.

(b) We now show that $\Phi(\Psi(\psi)) = \psi$. Let $V \subseteq Y$ be an open set. Then by definition

$$\begin{aligned}
\Phi(\Psi(\psi))_V = f_*\Psi(\psi)_V \circ \eta(\mathcal{G})_V &= \Psi(\psi)_{f^{-1}V} \circ (i_{f^{-1}V} \circ \kappa_{f^{-1}V}^V) \\
&= a_{f^{-1}V} \circ \kappa_{f^{-1}V}^V \\
&= \mathrm{res}_{f^{-1}V}^{f^{-1}V} \circ \psi_V = \psi_V.
\end{aligned}$$

Since this holds for every open set $V \subseteq Y$, $\Phi(\Psi(\psi)) = \Psi$.

(c) We finally prove that $\Phi_{\mathcal{F},\mathcal{G}}$ is natural in $\mathcal{F}$ and $\mathcal{G}$. First let $\rho\colon \mathcal{F} \to \mathcal{F}'$ and $\varphi\colon f^{-1}\mathcal{G} \to \mathcal{F}$ be morphisms of sheaves. Using the definition and expanding, we obtain

$$\Phi_{\mathcal{F},\mathcal{G}}(\rho \circ \varphi) = f_*(\rho \circ \varphi) \circ \eta(\mathcal{G}) = f_*\rho \circ f_*\varphi \circ \eta(\mathcal{G}) = f_*\rho \circ \Phi_{\mathcal{F},\mathcal{G}}(\varphi).$$

Therefore $\Phi_{\mathcal{F},\mathcal{G}}$ is natural in $\mathcal{F}$.
Finally let us check that $\Phi_{\mathcal{F},\mathcal{G}}$ is natural in $\mathcal{G}$. Let $\tau\colon \mathcal{G}' \to \mathcal{G}$ and $\psi\colon \mathcal{G} \to f_*\mathcal{F}$ be morphisms of sheaves. we have to show that

$$\Phi_{\mathcal{F},\mathcal{G}'}(\psi \circ f^{-1}\tau) = \Phi_{\mathcal{F},\mathcal{G}}(\psi) \circ \tau.$$

The left hand side is equal to

$$\Phi_{\mathcal{F},\mathcal{G}'}(\psi \circ f^{-1}\tau) = f_*(\psi \circ f^{-1}\tau) \circ \eta(\mathcal{G}') = f_*(\psi) \circ f_*f^{-1}\tau \circ \eta(\mathcal{G}'),$$

whereas the right hand side is $\Phi_{\mathcal{F},\mathcal{G}}(\psi) \circ \tau = f_*\psi \circ \eta(\mathcal{G}) \circ \tau$. We would be done if we can prove that

$$f_*f^{-1}\tau \circ \eta(\mathcal{G}') = \eta(\mathcal{G}) \circ \tau,$$

but this is simply the naturality of η with respect to the morphism τ. □

Remark 3.44. The proof might be simpler if we instead show that $f^+ \dashv f_*$ and use category theory to formally conclude that $f^{-1} \dashv f_*$.

Lemma 3.45 (Sheaf Gluing Lemma)**.** *Let X be a topological space with an open cover $\{U_i\}_{i\in I}$. Write U_{ij} for $U_i \cap U_j$. Suppose we are given sheaves $\mathcal{F}_i$ on U_i and an isomorphism of sheaves $\varphi_{ij} \colon \mathcal{F}_i|_{U_{ij}} \to \mathcal{F}_j|_{U_{ji}}$ for each $i, j \in I$ satisfying the following conditions:*

(1.) $\varphi_{ii} = \mathrm{id}_{\mathcal{F}_i}$.

(2.) $\varphi_{ji} = \varphi_{ij}^{-1}$.

(3.) $\varphi_{ik}|_{U_{ijk}} = \varphi_{jk} \circ \varphi_{ij}|_{U_{ijk}}$ *for all $i, j, k \in I$, where $U_{ijk} = U_i \cap U_j \cap U_k$. (This is called the* ***cocycle condition*** *of i, j, k.)*

Then there is a sheaf $\mathcal{F}$ on X and isomorphisms of sheaves $\psi_i \colon \mathcal{F}|_{U_i} \to \mathcal{F}_i$ such that $\phi_j^{-1} \circ \varphi_{ij} \circ \psi_i|_{U_{ij}}$ is the natural isomorphism $\mathcal{F}|_{U_i}|_{U_{ij}} \cong \mathcal{F}|_{U_j}|_{U_{ji}}$

Proof. TODO. □

3.5 Locally Ringed Spaces

Ringed Spaces

Definition 3.46. A **ringed space** is a pair $(X, \mathcal{O}_X)$ where X is a topological space and $\mathcal{O}_X$ is a sheaf of rings on X.

The sheaf $\mathcal{O}_X$ is called the **structure sheaf** of X. A **morphism of ringed spaces** $(f, f^\sharp) \colon (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a pair consisting of a continuous function $f \colon X \to Y$ and a morphism $f^\sharp \colon \mathcal{O}_Y \to f_*\mathcal{O}_X$ of sheaves of rings.

▸ **Exercise 3.47.** Define the composition of two morphisms of ringed spaces. Use this to show that ringed spaces and their morphisms form a category of ringed spaces.

Locally Ringed Spaces

Recall that a **local ring** is a ring A with a unique maximal ideal. We write $\mathfrak{m}_A$ for the maximal ideal of A, and define the **residue class field** of A to be $\kappa(A) = A/\mathfrak{m}_A$. A ring homomorphism $\varphi \colon A \to B$ between local rings is said to be **local** if $\varphi(\mathfrak{m}_A) \subseteq \mathfrak{m}_B$, or, equivalently, if $\varphi^{-1}(\mathfrak{m}_B) = \mathfrak{m}_A$.

A morphism of ringed spaces $f \colon X \to Y$ induces a morphism on stalks at each point of $x \in X$ as follows. Let $f^\sharp \colon \mathcal{O}_Y \to f_*\mathcal{O}_X$ be the morphism of sheaves. Then by the adjunction

between inverse image and direct image function, we have a morphism $f^\flat\colon f^{-1}\mathcal{O}_Y \to \mathcal{O}_X$ of sheaves on X. Let $x \in X$ and take the stalks at x to get

$$f_x^\sharp\colon \mathcal{O}_{Y,f(x)} = (f^{-1}\mathcal{O}_Y)_x \to \mathcal{O}_{X,x},$$

where the equality comes from Lemma 3.41. More concretely, $f_x^\sharp$ sends the class $[V,s]$ where V is an open neighbourhood of $f(x)$, and $s \in \mathcal{O}_Y(V)$ to the class $[f^{-1}V, \varphi_V(s)]$.

motivation for locally ringed condition required to make affine schemes equivalent to rings.

Definition 3.48.

(1.) A **locally ringed space** is a ringed space $(X,\mathcal{O}_X)$ such that for all $x \in X$ the stalk $\mathcal{O}_{X,x}$ is a local ring. The ring $\mathcal{O}_{X,x}$ is called the **local ring of X in x**.

(2.) A **morphism of locally ringed spaces** $(f,f^\sharp)\colon (X,\mathcal{O}_X) \to (Y,\mathcal{O}_Y)$ is a morphism of ringed spaces such that the induced map of stalks

$$f_x^\sharp\colon (f^{-1}\mathcal{O}_Y)_x = \mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$$

is a local ring homomorphism for all $x \in X$.

Let $(X,\mathcal{O}_X)$ be a locally ringed space, $x \in X$, and $\kappa(x)$ the residue field at x. If $f \in \mathcal{O}_X(U)$ is a section defined on an open neighbourhood U of x, we define $f(x)$ to be the image of f under the composition

$$\mathcal{O}_X(U) \to \mathcal{O}_{X,x} \to \mathcal{O}_{X,x}/\mathfrak{m}_x = \kappa(x).$$

In this manner we may think of a section on U as a "function" from U whose value at each point lies in the different fields $\kappa(x)$.

Lemma 3.49. *Let $(X,\mathcal{O}_X)$ be a locally ringed space, and let $f \in \mathcal{O}_X(X)$. Define*

$$X_f := \{x \in X : f(x) \neq 0\}.$$

Then X_f is an open subset of X.

Proof. Let $x \in X_f$ and thus $f_x \notin \mathfrak{m}_x$ by unravelling the definitions. Since $\mathcal{O}_{X,x}$ is a local ring, this means that f_x is invertible in $\mathcal{O}_{X,x}$. So there is $g_x \in \mathcal{O}_{X,x}$ such that $f_x g_x = 1$. Let $g \in \mathcal{O}_X(U)$ be a lift of g_x to an open neighbourhood U of x. Therefore there is an open neighbourhood $V \subseteq U$ such that

$$f|_V g|_V = 1.$$

This equation implies that f_y is invertible for all $y \in V$, and so $f(y) \neq 0$ for all $y \in V$. Hence $x \in V \subseteq X_f$ and thus the set X_f is open. □

Compare this to the case of affine varieties.

3.6 Affine Schemes

Let A be a ring. We are going to make $X = \operatorname{Spec} A$ into a locally ringed space by constructing a sheaf of rings on X. Recall that X has a basis consisting of principal open subsets. Define a sheaf $\mathcal{O}_X$ on this basis by setting

$$\mathcal{O}_X(D(f)) = A_f \quad \text{for each } f \in A,$$

where A_f is the localisation of A with respect to the set $\{f^i : i \geq 0\}$. Let us denote the canonical homomorphism $A \to A_f$ by i_f. To see that this is well-defined, let $D(f) = D(g)$. By Lemma 2.15, $D(g) \subseteq D(f)$ if and only $g^n = fa$ for some $a \in A$ and $n \geq 1$. In particular, $f/1$ is invertible in A_g because $(a/g^n)(f/1) = 1$. By the universal property of localisation, there exists a unique ring homomorphism $\rho_g^f \colon A_f \to A_g$ such that $\rho_g^f \circ i_f = i_g$. In fact, the homomorphism ρ_g^f is a homomorphism of R-algebras.

Suppose $D(h) \subseteq D(g) \subseteq D(f)$. Then by uniqueness, $\rho_h^g \circ \rho_g^f = \rho_h^f$. In particular, if $D(g) = D(f)$, then ρ_g^f is an isomorphism which allows us to identify A_f with A_g. The condition $\rho_h^g \circ \rho_g^f = \rho_h^f$ implies that $\mathcal{O}_X$ is a presheaf of rings on $\mathcal{B}$ with the restriction maps given by ρ_g^f.

Theorem 3.50. *The presheaf $\mathcal{O}_X$ is a sheaf on $\mathcal{B}$.*

Proof. Let $D(f)$ be a principal open subset and let $\{D(f_i) : i \in I\}$ be an open cover of $D(f)$. By restricting the presheaf $\mathcal{O}_X$ to $D(f)$, and replacing A_f with A, we may assume that $f = 1$, and hence $D(f) = X$, to ease the notation. Therefore the assumption translates to the statement that $X = \operatorname{Spec} A$ is covered by $\{D(f_i) : i \in I\}$.

We first prove the sheaf conditions when I is finite.

(1.) Let $s \in \mathcal{O}_X(X) = A$ satisfy $s|_{D(f_i)} = 0$ for all $i \in I$. For each i there is $n_i \in \mathbb{N}$ such that $f_i^{n_i} s = 0$. Set $n = \max_{i \in I} n_i$, so that $f_i^n s = 0$.

Since $D(f_i) = D(f_i^n)$ for all i, it follows that $\bigcup_{i \in I} D(f_i^n) = \bigcup_{i \in I} D(f_i) = X$ and so $\sum_{i \in I} f_i^n A = A$. Thus there are $b_i \in A$ such that $\sum_{i \in I} f_i^n b_i = 1$. This gives

$$s = \left(\sum_{i \in I} f_i^n b_i \right) s = \sum_{i \in I} f_i^n b_i s = 0.$$

(2.) Let $s_i \in \mathcal{O}_X(D(f_i))$ satisfy $s_i|_{D(f_i) \cap D(f_j)} = s_j|_{D(f_i) \cap D(f_j)}$ for all $i, j \in I$. Let us write $s_i = \frac{a_i}{f_i^{n_i}}$. We may adjust the fractions so that all n_i are equal, so that $s_i = \frac{a_i}{f_i^n}$, where $n \in \mathbb{N}$ is independent of i.

By hypothesis, for all $i, j \in I$,

$$\left. \frac{a_i}{f_i^n} \right|_{D(f_i f_j)} = \frac{a_i f_j^n}{(f_i f_j)^n} = \frac{a_j f_i^n}{(f_i f_j)^n} = \left. \frac{a_j}{f_j^n} \right|_{D(f_i f_j)}.$$

So that there is $m \geq 1$ such that $(f_i f_j)^m (a_i f_j^n - a_j f_i^n) = 0$. Again we may choose m to be independent of i and j. Thus by setting $k = n + m$, we obtain

$$(f_i^m a_i) f_j^k = (f_j^m a_j) f_i^k$$

for all $i, j \in I$.

Arguing as above, there are $b_i \in A$ such that $\sum_{j \in I} f_j^k b_j = 1$. Put $s := \sum_{j \in I} b_j f_j^m a_j$. Then

$$f_i^k s = f_i^k \left(\sum_{j \in I} b_j f_j^m a_j \right) = \sum_{j \in I} b_j (f_i^k f_j^m a_j) = \sum_{j \in I} b_j (f_i^m a_i f_j^k) = f_i^m a_i \left(\sum_{j \in I} b_j f_j^k \right) = f_i^m a_i.$$

Hence $s \in A = \mathcal{O}_X(X)$ is such that its image in A_{f_i} is equal to

$$\frac{s}{1} = \frac{f_i^k s}{f_i^k} = \frac{f_i^m a_i}{f_i^{n+m}} = \frac{a_i}{f_i^n} = s_i.$$

Finally we consider the case when I is infinite. Via Lemma 2.15, there is a finite subset $J \subseteq I$ such that $1 = \sum_{j \in J} f_j A$. From what was just proved, we see that gluing sections indexed by finite subsets work.

(1.) Let $s \in \mathcal{O}_X(X) = A$ satisfy $s|_{D(f_i)} = 0$ for all $i \in I$. Then $s|_{D(f_j)} = 0$ for all $j \in J$ and hence $s = 0$.

(2.) Let $s_i \in \mathcal{O}_X(D(f_i))$ satisfy $s_i|_{D(f_i) \cap D(f_k)} = s_k|_{D(f_i) \cap D(f_k)}$ for all $i, k \in I$. Consider the sections s_j with $j \in J$. From the finite case, they glue to a section $s \in \mathcal{O}_X(X)$ such that $s|_{D(f_j)} = s_j$ for all $j \in J$. We claim that the section s is the one we get from gluing s_i.

We only need to check that for all $i \in I$, $s|_{D(f_i)} = s_i$. Note that $\{D(f_i) \cap D(f_j)\}_{j \in J}$ is a finite covering of $D(f_i)$. Hence, for all $i \in I$ and $j \in J$,

$$s|_{D(f_i)}|_{D(f_i) \cap D(f_j)} = s|_{D(f_j)}|_{D(f_i) \cap D(f_j)} = s_j|_{D(f_i) \cap D(f_j)} = s_i|_{D(f_i) \cap D(f_j)}.$$

By (1), $s|_{D(f_i)} = s_i$. This finishes the proof. □

Corollary 3.51. *The sheaf $\mathcal{O}_X$ extends to a sheaf $\mathcal{O}_X$ on X, called the* ***structure sheaf*** *of* Spec A.

Lemma 3.52 (Stalks of Spec A)**.** *Let $X = \operatorname{Spec} A$, $\mathcal{O}_X$ its structure sheaf, and $x \in \operatorname{Spec} A$. Then*

$$\mathcal{O}_{X,x} = A_{\mathfrak{p}_x},$$

the localisation of A at the prime ideal $\mathfrak{p}_x$. In particular, $(X, \mathcal{O}_X)$ is a locally ringed space.

Proof. By Lemma 3.31, we can compute the stalk $\mathcal{O}_{X,x}$ on the basis of principal open subsets, and thus

$$\mathcal{O}_{X,x} = \varinjlim_{D(f)\ni x} \mathcal{O}_X(D(f)) = \varinjlim_{f\notin\mathfrak{p}} A_f = A_{\mathfrak{p}},$$

the last equality follows from commutative algebra (Problem 3.8). □

Definition 3.53. A locally ringed space $(X, \mathcal{O}_X)$ is said to be an **affine scheme** if it is isomorphic to a locally ringed space $(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$ for suitable ring A.

A **morphism of affine schemes** is simply a morphism of locally ringed spaces. We thus obtain the category of affine schemes which we shall denote by (Aff). Let $\varphi\colon A \to B$ be a ring homomorphism. We shall define a morphism of affine schemes $\operatorname{Spec}\varphi = (f, f^\sharp)\colon (\operatorname{Spec} B, \mathcal{O}_{\operatorname{Spec} B}) \to (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$. Here is the definition:

(1.) $f = \operatorname{Spec}\varphi$ is the induced continuous map on spectra, and,

(2.) $f^\sharp\colon \mathcal{O}_{\operatorname{Spec} A} \to f_*\mathcal{O}_{\operatorname{Spec} B}$ is given on principal open subsets by

$$f^\sharp_{D(s)} : \mathcal{O}_{\operatorname{Spec} A}(D(s)) \to \mathcal{O}_{\operatorname{Spec} B}(f^{-1}(D(s))).$$

Since $f^{-1}(D(s)) = D(\varphi(s))$ by Problem 2.7, we need to define a map $f^\sharp_{D(s)}\colon A_s \to B_{\varphi(s)}$. The natural map for this is the unique map $\varphi_s\colon A_s \to B_{\varphi(s)}$ induced by the canonical localisation maps:

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & B \\ \downarrow & & \downarrow \\ A_s & \xrightarrow[\varphi_s]{} & B_{\varphi(s)} \end{array}$$

Therefore we take $f^\sharp_{D(s)}$ to be φ_s.

The above data actually define a morphism of ringed spaces $\operatorname{Spec} B \to \operatorname{Spec} A$, as shown in the following proposition.

Proposition 3.54. *The maps $f^\sharp_{D(s)}$ is compatible with restrictions, and furthermore, $f^\sharp_x$ is a local ring homomorphism for all $x \in \operatorname{Spec} B$. Hence $(f, f^\sharp)$ defines a morphism of locally ringed spaces.*

Proof. Let $D(g) \subseteq D(s)$ be principal open subsets of $\operatorname{Spec} A$. Then

$$D(\varphi(g)) = f^{-1}(D(g)) \subseteq f^{-1}(D(s)) = D(\varphi(s)).$$

Therefore we have a commutative diagram of ring homomorphisms

$$\begin{array}{ccccccc} \mathcal{O}_{\operatorname{Spec} A}(D(s)) & \xrightarrow{\simeq} & A_s & \xrightarrow{\varphi_s} & B_{\varphi(s)} & \xrightarrow{\simeq} & \mathcal{O}_{\operatorname{Spec} B}(D(\varphi(s))) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \mathcal{O}_{\operatorname{Spec} A}(D(g)) & \xrightarrow[\simeq]{} & A_g & \xrightarrow[\varphi_g]{} & B_{\varphi(g)} & \xrightarrow{\simeq} & \mathcal{O}_{\operatorname{Spec} B}(D(\varphi(g))) \end{array}$$

Hence the homomorphisms φ_s assemble to a morphism of sheaves on the basis of principal open subsets, and so it extends to a unique morphism of sheaves $f^\sharp\colon \mathcal{O}_{\operatorname{Spec} A} \to f_*\mathcal{O}_{\operatorname{Spec} B}$.

Lastly we have to check that $f^\sharp_x$ is local for all $x \in \operatorname{Spec} B$. Unravelling the definition of $f^\sharp$, it is clear that $f^\sharp_x$ is the homomorphism

$$\mathcal{O}_{\operatorname{Spec} A, f(x)} = A_{\varphi^{-1}(\mathfrak{p}_x)} \xrightarrow{\varphi_{\mathfrak{p}_x}} B_{\mathfrak{p}_x} = \mathcal{O}_{\operatorname{Spec} B, x}$$

induced from $\varphi\colon A \to B$, which is indeed local. This proves the proposition. □

▸ **Exercise 3.55.** Verify that if $\varphi\colon A \to B$ and $\psi\colon B \to C$ are ring homomorphisms, then $\operatorname{Spec}(\psi \circ \varphi) = \operatorname{Spec}(\varphi) \circ \operatorname{Spec}(\psi)$ and for any ring A, $\operatorname{Spec}(\mathrm{id}_A)$ is the identity morphism of the affine scheme $\operatorname{Spec} A$. Hence $\operatorname{Spec}\colon (\mathrm{Ring}) \to (\mathrm{Aff})$ is a contravariant functor.

Anti-equivalence of the Categories (Aff) *and* (Ring)

In the last section, we constructed a morphism of affine schemes $(f, f^\sharp)\colon \operatorname{Spec} B \to \operatorname{Spec} A$ out of a ring homomorphism $\varphi\colon A \to B$. On the other hand, if $(f, f^\sharp)\colon X \to Y$ is a morphism of ringed spaces, then

$$f^\sharp_Y\colon \mathcal{O}_Y(Y) \to f_*\mathcal{O}_X(Y) = \mathcal{O}_X(X)$$

is a ring homomorphism. This defines the **global section functor** $\Gamma\colon (\mathrm{RingedSpaces}) \to (\mathrm{Ring})$ given on ringed spaces by $\Gamma(X) = \mathcal{O}_X(X)$. For affine schemes, these two functors are "invertible" in the following sense.

Theorem 3.56. *The functors* Spec *and* Γ *form an anti-equivalence of the categories of affine schemes and the category of rings.*

Proof. Trivially, Spec is essentially surjective. Let $\varphi\colon A \to B$ be a ring homomorphism. It is clear from the definitions that $\Gamma(\operatorname{Spec}\varphi) = \varphi$. Now let $(f, f^\sharp)\colon \operatorname{Spec} B \to \operatorname{Spec} A$ be a morphism of affine schemes. TODO □

We use the anti-equivalence of the categories of affine schemes and the category of rings to prove that every principal open subset of an affine scheme is in itself an affine scheme with the inherited locally ringed space structure.

Proposition 3.57. *Let A be a ring and $X = \operatorname{Spec} A$. For each $f \in A$, the open set $(D(f))$ with the restricted sheaf $\mathcal{O}_{X|D(f)}$ is a locally ringed space canonically isomorphic to the affine scheme* $\operatorname{Spec} A_f$.

In this situation we also say that $(D(f), \mathcal{O}_{X|U})$ is an **affine open subscheme** of A.

Proof. Clearly $(D(f), \mathcal{O}_{X|U})$ is a locally ringed space. Let $\varphi\colon A \to A_f$ denote the canonical localisation map and let $j\colon \operatorname{Spec} A_f \to \operatorname{Spec} A$ be the induced continuous map. We proved

in Proposition 2.34 that j is a homeomorphism from $\operatorname{Spec} A_f$ onto the open subset $D(f) \subseteq \operatorname{Spec} A$. Hence there is a morphism of locally ringed spaces

$$(j, j^\sharp) \colon (\operatorname{Spec} A_f, \mathcal{O}_{\operatorname{Spec} A_f}) \to (D(f), \mathcal{O}_{X|U})$$

where $j^\sharp = i^\sharp \circ \varphi^\sharp$ is the composition of morphisms of sheaves induced from φ and the inclusion $i \colon D(f) \hookrightarrow X$. For each open subset $D(g) \subseteq D(f)$, we obtain

$$\mathcal{O}_{X|U}(D(g)) = \mathcal{O}_X(D(g)) = A_g \cong (A_f)_{\varphi(g)} = \mathcal{O}_{\operatorname{Spec} A_f}(D(\varphi(g))) = (j_* \mathcal{O}_{\operatorname{Spec} f})(D(g)).$$

(Use Problem 2.7.) Hence $j^\sharp$ is an isomorphism of sheaves when checked on the basis consisting of principal open subsets, and so $j^\sharp$ is indeed an isomorphism. □

Examples of Affine Schemes and Their Morphisms

Example 3.58. Let k be a field. Then the scheme $\operatorname{Spec} k$ consists of a single point. The structure sheaf of $\operatorname{Spec} k$ is just k.

Example 3.59. Let R be a discrete valuation ring. Then $\operatorname{Spec} R$ consists of two points.

Example 3.60 (The Affine Space)**.** Let R be a ring. Let $\mathbb{A}^n_R$ denote the affine scheme $\operatorname{Spec} R[X_1, \dots, X_n]$. This is called the **affine space of relative dimension** n. The inclusion $R \to R[X_1, \dots, X_n]$ define a morphism of affine schemes $\mathbb{A}^n_R \to \operatorname{Spec} R$.

Let's consider the affine line $\mathbb{A}^1_k$ when k is a field. Since $k[X]$ is a PID, every proper open subset of $\mathbb{A}^1_k$ is of the form $D(f)$ for some non-zero polynomial $f \in k[x]$. Then $\mathcal{O}_{\mathbb{A}^1_k}(D(f)) = k[X]_f = k[X, 1/f]$.

3.7 Problems

3.1 In this problem we extend the notion of exactness to the category of sets. A diagram of the form

$$A \xrightarrow{f} B \underset{h}{\overset{g}{\rightrightarrows}} C$$

is said to be **exact** if f is injective and the image $\operatorname{im} f$ is equal to the set

$$\ker(g, h) := \{b \in B : g(b) = h(b)\}.$$

Show that a presheaf $\mathcal{F}$ of sets on a topological space X is a sheaf if and only if for all open set U and open covering $\{U_i\}_{i \in I}$ of U, the sequence

$$\mathcal{F}(U) \xrightarrow{f} \prod_{i \in I} \mathcal{F}(U_i) \underset{h}{\overset{g}{\rightrightarrows}} \prod_{i,j \in I} \mathcal{F}(U_i \cap U_j)$$

is exact, where the functions f, g, h are given by

$$\begin{aligned} f &: s \mapsto (s|_{U_i})_{i\in I} \\ g &: (s_i)_{i\in I} \mapsto (s_i|_{U_i\cap U_j})_{i,j\in I} \\ h &: (s_j)_{j\in I} \mapsto (s_j|_{U_i\cap U_j})_{i,j\in I}. \end{aligned}$$

3.2 *Kernel Sheaf.* Prove that if $\varphi\colon \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves on a topological space X, then the **kernel sheaf** $\mathcal{K}$ defined on each open subset $U \subseteq X$ by the formula

$$\mathcal{K}(U) = \ker(\varphi_U\colon \mathcal{F}(U) \to \mathcal{G}(U))$$

is a subsheaf of $\mathcal{F}$. The restriction maps of $\mathcal{K}$ is inherited from those of $\mathcal{F}$. The kernel sheaf of φ is also denoted by $\ker\varphi$. Furthermore, show that for all $x \in X$, $\ker(\varphi_x) = \mathcal{K}_x = (\ker\varphi)_x$.

3.3 *Quotient Presheaf.* Let X be a topological space, $\mathcal{F}$ a sheaf of Abelian groups and $\mathcal{F}'$ a sheaf such that $\mathcal{F}'(U)$ is an ideal of $\mathcal{F}(U)$ for all open subset $U \subseteq X$. Show that the assignment

$$U \mapsto \mathcal{F}(U)/\mathcal{F}'(U)$$

is a presheaf on X, called the **quotient presheaf**. Show that the stalk at $x \in X$ of the quotient presheaf is equal to $\mathcal{F}_x/\mathcal{F}'_x$.

3.4 *Product Sheaf.* Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on a topological space X. Show that the presheaf $U \mapsto \mathcal{F}(U) \times \mathcal{G}(U)$ is already sheaf. It is called the **product sheaf** of $\mathcal{F}$ and $\mathcal{G}$.

3.5 Let X be a topological space. Show that if $\mathcal{G}$ is a sheaf and $\mathcal{F}$ a sub-presheaf of $\mathcal{G}$, then there is a smallest subsheaf $\mathcal{H}$ of $\mathcal{G}$ such that $\mathcal{F}$ is a sub-presheaf of $\mathcal{H}$. Furthermore, $\mathcal{F}_x = \mathcal{H}_x$ for all $x \in X$.

3.6 Let X be an irreducible topological space and let A be an Abelian group. Define a presheaf $\mathcal{A}$ on X by

(1.) $\mathcal{A}(\emptyset) = 0$, and,

(2.) for all open subsets $\emptyset \neq U \subseteq X$, $\mathcal{A}(U) = A$.

If $\emptyset \neq V \subseteq U$ are open subsets, then the restriction map $\mathcal{A}(U) \to \mathcal{A}(V)$ is taken to be the identity map. Prove that $\mathcal{A}$ is the sheafification of the constant presheaf with value A.

3.7 Let $(\mathcal{F}_i)_{i\in I}$ be an inductive system of sheaves on a topological space X. Show that the assignment $U \mapsto \varinjlim_{i\in I} \mathcal{F}_i(U)$ for each open subset $U \subseteq X$ define a presheaf on X. Show that the sheafification of this presheaf is the inductive limit of $(\mathcal{F}_i)_{i\in I}$ in the category of sheaves.

3.8 Let A be a ring and $\mathfrak{p}$ a prime ideal. Show that the rings $(A_f)_{f\notin\mathfrak{p}}$ together with homomorphisms $A_f \to A_g$ whenever $g \in \sqrt{fA}$ form a direct system. Show that the limit of this direct system is $A_\mathfrak{p}$.

3.9 Let $\varphi\colon A \to B$ be a homomorphism of rings, and let $f\colon \operatorname{Spec} B \to \operatorname{Spec} A$ be the corresponding morphism of affine schemes.

(a) Show that φ is injective if and only if the morphism of sheaves $f^\sharp\colon \mathcal{O}_{\operatorname{Spec} A} \to f_*\mathcal{O}_{\operatorname{Spec} B}$ is injective.

(b) Show that if φ is surjective, then the morphism of sheaves $f^\sharp\colon \mathcal{O}_{\operatorname{Spec} A} \to f_*\mathcal{O}_{\operatorname{Spec} B}$ is surjective.

3.10 Suppose that $\varphi\colon \mathcal{F} \to \mathcal{G}$ is a morphism of presheaves on a topological space X such that $\varphi_U\colon \mathcal{F}(U) \to \mathcal{G}(U)$ is injective for all $U \subseteq X$. Show that for all $x \in X$, the induced morphism on stalks $\varphi_x\colon \mathcal{F}_x \to \mathcal{G}_x$ is injective. Hence deduce that the associated morphism of associated sheaves $\varphi^\dagger\colon \mathcal{F}^\dagger \to \mathcal{G}^\dagger$ is also injective.

3.11 Let $\mathcal{F}$ be a presheaf and $\mathcal{G}$ be a sheaf on a topological space X. Let $\mathcal{B}$ be a basis of X. Suppose that $\mathcal{F}|_\mathcal{B} = \mathcal{G}|_\mathcal{B}$ are equal as presheaves on $\mathcal{B}$ (hence as sheaves on $\mathcal{B}$ because $F|_\mathcal{B}$ is). Prove that $\mathcal{G}$ is the sheafification of $\mathcal{F}$.

3.12 *Stalks of Pushforward Sheaves from a Subspace.* Let X be a topological space and let $i\colon Z \to X$ denote the inclusion from a subspace Z into X. Let $\mathcal{F}$ be a sheaf (of Abelian groups) on X.

(a) Show that if $x \notin \bar{Z}$, then $(i_*\mathcal{F})_x = \{0\}$. (More generally, it is the final object in the target category of $\mathcal{F}$)

(b) Show that if $x \in Z$, then $(i_*\mathcal{F})_x = \mathcal{F}_x$.

Chapter 4

Schemes

4.1 Schemes

After defining affine schemes, we can now define schemes as locally ringed spaces that are affine schemes glued together.

Definition 4.1. A **scheme** is a locally ringed space $(X, \mathcal{O}_X)$ which admits a covering by open subsets $\{U_i\}_{i \in I}$ such that $(U_i, \mathcal{O}_{X|U_i})$ is isomorphic to an affine scheme for all $i \in I$.

A **morphism of schemes** is a morphism of the underlying locally ringed spaces. Schemes and morphisms of schemes together form the category of schemes, which is denoted by (Sch). The category of affine schemes is then a full subcategory of (Sch). We shall often write X for the scheme $(X, \mathcal{O}_X)$; its structure sheaf is left implicit.

Open Subschemes

Open subsets of a scheme inherit the scheme structure of X, as the following lemma shows.

Lemma 4.2. *Let $(X, \mathcal{O}_X)$ be a scheme and $U \subseteq X$ an open subset. The locally ringed space $(U, \mathcal{O}_{X|U})$ is itself a scheme.*

Proof. It is clear that $(U, \mathcal{O}_{X|U})$ is a locally ringed space. Let $\{U_i\}_{i \in I}$ be an open covering of X such that each $(U_i, \mathcal{O}_{X|U_i})$ is isomorphic to an affine scheme. For each $i \in I$, $U \cap U_i$ is an open subset of U_i. Using the fact that U_i is affine, we see that there is an open covering of $U \cap U_i$ by principal open subsets $D(f_{i,j}), j \in J$. By Proposition 3.57, $D(f_{i,j}) \cong \operatorname{Spec} \Gamma(U_i, \mathcal{O}_X)_{f_{i,j}}$ as affine schemes, and so $U \cap U_i$ can be covered by affine schemes. It then follows that $U = \bigcup_i (U \cap U_i)$ can also be covered by affine schemes. □

Definition 4.3. Let X be a scheme and $U \subseteq X$ an open subset. The scheme $(U, \mathcal{O}_{X|U})$ is called an **open subscheme** of X. If, in addition, U is affine, we say that U is an **affine open subscheme** of X.

The proof of the lemma shows that every scheme has a basis consisting of affine open subschemes. We also record another useful lemma regarding sections of schemes. It says that non-empty open subsets of a scheme cannot have zero sections.

Lemma 4.4. *Let $(X, \mathcal{O}_X)$ be a scheme and $U \subseteq X$ an open subset. Then $\Gamma(U, \mathcal{O}_X) = 0$ if and only if $U = \emptyset$.*

Proof. If $U = \emptyset$, then $\Gamma(U, \mathcal{O}_X) = 0$ which can be seen from the sheaf conditions with respect to the empty covering.

Now suppose $\Gamma(U, \mathcal{O}_X) = 0$. Let $V \subseteq U$ be an affine open subscheme of X, which exists because affine open subschemes form a basis of the topology of a scheme. Then the restriction map is a ring homomorphism

$$0 = \Gamma(U, \mathcal{O}_X) \to \Gamma(V, \mathcal{O}_X),$$

which implies that $\Gamma(V, \mathcal{O}_X) = 0$. Therefore $V = \operatorname{Spec} 0 = \emptyset$ by Theorem 3.56. Taking a covering of U by affine open subschemes, each of which is then the empty set, we conclude that $U = \emptyset$. □

Let X be a scheme and $U \subseteq X$ an open subset. Then the inclusion $j\colon U \hookrightarrow X$ gives rise to a morphism of schemes $(U, \mathcal{O}_{X|U}) \to (X, \mathcal{O}_X)$ whose morphism of sheaves $\mathcal{O}_X \mapsto j_*\mathcal{O}_{X|U}$ we shall shortly define. Let $V \subseteq U$ be an open set. The restriction maps of the structure sheaf $\mathcal{O}_X$ give us the ring homomorphism

$$\Gamma(V, \mathcal{O}_X) \to \Gamma(V \cap U, \mathcal{O}_X) = \Gamma(j^{-1}(V), \mathcal{O}_{X|U}) = \Gamma(V, j_*\mathcal{O}_{X|U}).$$

This is compatible with restriction $W \subseteq V \subseteq U$, and thus define a morphism of sheaves $j^\sharp\colon \mathcal{O}_X \to j_*\mathcal{O}_{X|U}$.

The following lemma will be useful for working with affine subschemes.

Lemma 4.5. *Let X be a scheme, and let U, V be affine open subschemes of X. For each $x \in U \cap V$, there is an open set $W \subseteq U \cap V$ containing x such that W is a principal open subset of U and V.*

Proof. Write $U = \operatorname{Spec} R$ and $V = \operatorname{Spec} S$. There is a principal open subset $D(r) \subseteq \operatorname{Spec} R$ containing x for suitable $r \in R$. Then $D(r) \cap \operatorname{Spec} S$ is open in $\operatorname{Spec} S$, and so there is a principal open subset $D(s) \subseteq D(r) \cap \operatorname{Spec} S$ containing x for suitable $s \in S$. Consider the image of s under the restriction $\varphi\colon \Gamma(\operatorname{Spec} S, \mathcal{O}_X) \to \Gamma(D(r), \mathcal{O}_X) = R_r$. Hence $\varphi(s) = a/r^n$ for suitable $a \in R$ and $n \geq 0$.

Let $y \in D(r) \cap \operatorname{Spec} S$. Then y corresponds to a prime ideal $\mathfrak{q}_y$ of S and a prime ideal $\mathfrak{p}_y R_r$ of R_r, where $\mathfrak{p}_y$ is a prime ideal of R disjoint from the set $\{1, r, r^2, \ldots\}$. Using the

anti-equivalence of the categories (Ring) and (Aff) (Theorem 3.56), they are related by the formula $\varphi^{-1}(\mathfrak{p}_y R_r) = \mathfrak{q}_y$. Therefore

$$s \notin \mathfrak{q}_y \quad \text{if and only if} \quad \frac{a}{r^n} \notin \mathfrak{p}_y R_r.$$

Note that $1/r^n$ is a unit of R_r, and so $s \notin \mathfrak{q}_y$ if and only if $a/1 \notin \mathfrak{p}_y R_r$, if and only if $a \notin \mathfrak{p}_y$. Hence we have the equalities

$$D(s) = D(a) \cap D(r) = D(ar).$$

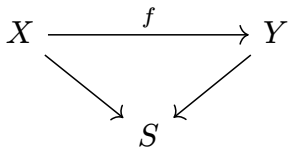
Therefore $W := D(s)$ is equal to both principal open subsets $\operatorname{Spec} S_s$ and $\operatorname{Spec} R_{ar}$. □

Another proof using the notation $s(x) \in \kappa(x)$.

By doing this for all $x \in U \cap V$, we see that $U \cap V$ can be covered by open subsets which is principal open in both U and V.

Schemes Over a Base Scheme

Generalising the notion of varieties defined over a field k, there is a corresponding notion of schemes over a base scheme. Let S be a scheme. An **S-scheme** is a scheme X with a morphism of schemes $X \to S$, called the **structural morphism** of X. Given two S-schemes $X \to S$ and $Y \to S$, a **morphism of S-schemes** $f\colon X \to Y$ is a morphism of schemes such that the diagram



commutes. The scheme S is called the **base scheme**. It is itself an S-scheme with the morphism $\mathrm{id}_S\colon S \to S$. In this manner, we have the **category of S-schemes** (Sch/S). Another way of viewing (Sch/S) is that it is the slice category of (Sch) over S. The set of morphisms between X and Y in the category of S-schemes will be denoted by $\mathrm{Hom}_S(X, Y)$, to avoid confusion with the hom-set $\mathrm{Hom}(X, Y)$ when X, Y are just regarded as schemes.

In many cases when the base scheme is affine, we shall employ a common shorthand notation. If S is affine, say $S = \operatorname{Spec} R$, a scheme over S is also called an **R-scheme** or a **scheme over R**. The hom-set of R-schemes X and Y is written by $\mathrm{Hom}_R(X, Y)$.

4.2 Gluing Lemmas

Lemma 4.6 (Morphism Gluing Lemma)**.** *Let X and Y be locally ringed spaces, and let $\{U_i\}_{i\in I}$ be an open covering of X. Then a family of morphisms $f_i\colon U_i \to Y$ of locally ringed spaces together form a morphism $f\colon X \to Y$ if and only if the morphisms $f_i\colon U_i \to Y$ and $f_j\colon U_j \to Y$ coincide on $U_i \cap U_j$ for all $i, j \in I$. Furthermore, the resulting morphism $X \to Y$ is unique.*

Proof. The only non-trivial part is constructing a morphism $f\colon X \to Y$ from the morphisms $f_i\colon U_i \to Y$. At the level of the underlying topological spaces this is always possible, so we obtain a continuous map $f\colon X \to Y$ such that $f|_{U_i} = f_i$ for all $i \in I$.

We seek to define a morphism of sheaves $f^\sharp\colon \mathcal{O}_Y \to f_*\mathcal{O}_X$. Let $V \subseteq Y$ be an open subset, so that $f^{-1}(V)$ is an open subset of X. For each $i \in I$, we are given a map

$$\mathcal{O}_Y(V) \to (f_i)_*\mathcal{O}_{X|U_i}(V) = \mathcal{O}_X(U_i \cap f^{-1}(V)).$$

For each $s \in \mathcal{O}_Y(V)$, let s_i denote the image of s in $\mathcal{O}_X(U_i \cap f^{-1}(V))$ under the above map. The idea is to glue the sections s_i together and take the resulting section to be $f^\sharp_V(s)$. To make sure that the sections can be glued together, we need to show that

$$s_i|_{U_i \cap U_j \cap f^{-1}(V)} = s_j|_{U_i \cap U_j \cap f^{-1}(V)}$$

for all $i, j \in I$. This is implied by our assumption that f_i and f_j coincide on $U_i \cap U_j$. Furthermore, $f^\sharp_V(s)$ is well-defined by the uniqueness condition of sheaf axioms.

The only thing left to check is that the maps $f^\sharp_V$ are compatible with restriction maps. Let $W \subseteq V$ be open subsets of Y. Then for all $s \in \mathcal{O}_Y(V)$ and $i \in I$,

$$\begin{aligned} f^\sharp_V(s)|_{f^{-1}(W)\cap U_i} &= f^\sharp_V(s)|_{f^{-1}(V)\cap U_i}|_{f^{-1}(W)\cap U_i} \\ &= s_i|_{f^{-1}(W)\cap U_i} \\ &= (s|_W)_i. \end{aligned}$$

By uniqueness, $f^\sharp_V(s)|_{f^{-1}(W)} = f^\sharp_W(s|_W)$. This proves that $f^\sharp$ is compatible with restriction maps and so define a morphism of sheaves $\mathcal{O}_Y \to f_*\mathcal{O}_X$. For each $x \in X$, the induced map on stalks $f^\sharp_x$ is plainly a local ring homomorphism because taking stalks is local. □

Next we show that schemes can be glued together if they are compatible in a sense. We make the following definition.

Definition 4.7. A **gluing datum of schemes** consists of the following data:

(1.) an index set I;

(2.) for all $i \in I$, a scheme U_i;

(3.) for all $i, j \in I$, an open subset $U_{ij} \subseteq U_i$.

(4.) for all $i, j \in I$, an isomorphism $\varphi_{ji}\colon U_{ij} \to U_{ji}$ of schemes such that

  (a) $U_{ii} = U_i$ for all $i \in I$,

  (b) the identity $\varphi_{kj} \circ \varphi_{ji} = \varphi_{ki}$ on $U_{ij} \cap U_{ik}$ for all $i, j, k \in I$. (This is called the **cocycle condition**.)

Lemma 4.8 (Scheme Gluing Lemma)**.** *content...*

4.3 More on Morphisms

Morphisms into an Affine Scheme

Theorem 4.9. *Let X be a scheme and R a ring. Then there is a bijection*

$$\begin{aligned} \mathrm{Hom}_{(\mathrm{Sch})}(X, \operatorname{Spec} R) &\longrightarrow \mathrm{Hom}_{(\mathrm{Ring})}(R, \Gamma(X, \mathcal{O}_X)) \\ f &\longmapsto f^\sharp_{\operatorname{Spec} R} \end{aligned}$$

which is functorial in X and R.

Proof. Since X is a scheme, there is an affine open cover U_i of X. □

Remark 4.10. The theorem still holds if X is a locally ringed space which is not necessarily a ring.

Morphisms from Spectrum of a Field

4.4 Immersions

Open Immersions

Definition 4.11. A morphism of schemes $f\colon Y \to X$ is an **open immersion** if the underlying continuous map is a homeomorphism of Y with an open subset of X, and the sheaf homomorphism $f^\sharp\colon \mathcal{O}_X \to f_*\mathcal{O}_Y$ induces an isomorphism $\mathcal{O}_{X|U} \cong (f_*\mathcal{O}_Y)|_U$.

A continuous map $f\colon Y \to X$ such that f is a homeomorphism of Y onto an open subset of X (that is, Y is homeomorphic to the image $f(Y)$ which is open in X) is called a **topological open immersion**. It is clear that if U is an open subscheme of X, then the inclusion $i\colon U \hookrightarrow X$ is an open immersion.

Closed Immersions

For closed immersions, the definition is more involved. This is because there can be different closed subscheme of a scheme X which has the same underlying topological space. Thus the definition of a closed immersion should related to the structure sheaf of X.

Let $(X, \mathcal{O}_X)$ be a ringed space. A **sheaf of ideal** (or an **ideal sheaf**) is a subsheaf $\mathcal{I} \subseteq \mathcal{O}_X$ such that $\mathcal{I}(U)$ is an ideal of $\mathcal{O}_X(U)$ for all open subset $U \subseteq X$. Define the **quotient sheaf** $\mathcal{O}_X/\mathcal{I}$ to be the sheaf associated to the presheaf

$$U \mapsto \mathcal{O}_X(U)/\mathcal{I}(U).$$

We remark that there is a canonical projection $\mathcal{O}_X \to \mathcal{O}_X/\mathcal{I}$, which is surjective because one may check surjectivity on stalks.

Definition 4.12. Let $j\colon Z \to X$ be a morphism of schemes. We say that j is a **closed immersion** if j is a homeomorphism from Z to a closed subset of X, and the sheaf homomorphism $j^\sharp\colon \mathcal{O}_X \to j_*\mathcal{O}_Z$ is surjective.

In Corollary 2.33, we see that, given a ring A and an ideal $\mathfrak{a}$ of A, the spectrum $\operatorname{Spec}(A/\mathfrak{a})$ is homeomorphic to the closed subset $V(\mathfrak{a})$ of $\operatorname{Spec} A$ via the continuous map induced from the quotient map $\varphi\colon A \to A/\mathfrak{a}$. In fact, the morphism of schemes induced from φ is a closed immersion of $\operatorname{Spec}(A/\mathfrak{a})$ into $\operatorname{Spec} A$, as the proposition below shows.

Proposition 4.13. *Let A be a ring, $\mathfrak{a} \subseteq A$ an ideal, and let $\varphi\colon A \to A/\mathfrak{a}$ be the quotient homomorphism. Define $X = \operatorname{Spec} A$ and $Z = \operatorname{Spec}(A/\mathfrak{a})$. Then the morphism of affine schemes $(j, j^\sharp)\colon Z \to X$ induced from φ is a closed immersion.*

Proof. It is shown in Corollary 2.33 that j is a homeomorphism from Z to the closed subset $V(\mathfrak{a})$ of X. We check that $j^\sharp$ is surjective by checking it on stalks. Let $x \in Z$, and so $\mathfrak{q}_x$ is a prime ideal of $A/\mathfrak{a}$. Hence $\mathfrak{q}_x = \mathfrak{p}/\mathfrak{a}$ for some prime ideal $\mathfrak{p}$ of A containing $\mathfrak{a}$. The morphism $j^\sharp_x$ is then equal to the ring homomorphism between localisations

$$A_\mathfrak{p} \xrightarrow{\varphi_\mathfrak{p}} (A/\mathfrak{a})_{\mathfrak{q}_x}$$

which is induced from $\varphi\colon A \to A/\mathfrak{a}$. This is easily seen to be surjective. □

Corollary 4.14. *If $\varphi\colon A \to B$ is a surjective ring homomorphism, then the induced morphism $\operatorname{Spec}\varphi\colon \operatorname{Spec} B \to \operatorname{Spec} A$ of affine schemes is a closed immersion.*

Proof. By the first isomorphism theorem, $B \cong A/\ker\varphi$. □

In the above situation, we would like to think that the closed subset $V(\mathfrak{a})$ is a closed subscheme of $\operatorname{Spec} A$ with the scheme structure inherited from X. Thus we make the following definition.

Definition 4.15. Let X be a scheme. A **closed subscheme** of X is given by a closed subset $Z \subseteq X$ and an ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$ such that

$$Z = \{x \in X : (\mathcal{O}_X/\mathcal{I})_x \neq 0\}$$

and $(Z, (\mathcal{O}_X/\mathcal{I})|_Z)$ is a scheme.

We need the following lemma about how inverse images and direct images of sheaves with respect to a closed immersion behave.

Lemma 4.16. *Let X be a topological space and let $j\colon Z \to X$ be an inclusion of a closed subset Z into X.*

(1.) Let $\varphi\colon G \to \mathcal{H}$ be a morphism of presheaves on Z. For each $x \in Z$, there is a commutative square

$$\begin{array}{ccc} (j_*\mathcal{G})_x & \xrightarrow{\sim} & \mathcal{G}_x \\ {\scriptstyle (j_*\varphi)_x}\downarrow & & \downarrow{\scriptstyle \varphi_x} \\ (j_*\mathcal{H})_x & \xrightarrow[\sim]{} & \mathcal{H}_x \end{array}$$

where the horizontal maps are isomorphisms.

(2.) For every sheaf $\mathcal{F}$ on X such that $\mathcal{F}_x = 0$ for all $x \in X \setminus Z$, the canonical morphism

$$\eta(\mathcal{F})\colon \mathcal{F} \to j_*j^{-1}\mathcal{F}$$

is an isomorphism.

(3.) If $\mathcal{G}$ is a sheaf on Z, then the canonical morphism

$$j^{-1}j_*\mathcal{G} \to \mathcal{G}$$

is an isomorphism of sheaves.

Proof. (1.) is straightforward after unravelling all the definitions. (See Problem 3.12).

We now turn to (2.). Note that $j^{-1}(U) = U \cap Z$ for all open set $U \subseteq X$. The canonical morphism is the morphism $\eta(\mathcal{F})$ given by the unit of the adjunction $j^{-1} \dashv j_*$. On an open set $U \subseteq X$, it sends $\mathcal{F}(U)$ to $j^+\mathcal{F}(j^{-1}(U))$ via the map

$$\kappa^U_{U\cap Z}\colon \mathcal{F}(U) \to \varinjlim_{\substack{V \supseteq U\cap Z \\ V \text{ open in } X}} \mathcal{F}(V) = j^+\mathcal{F}(U \cap Z),$$

then to the sheafification $j^{-1}\mathcal{F}(U \cap Z)$. Let's use the notation $\eta^{pre}(\mathcal{F})$ for the morphism $\mathcal{F} \to j_*j^+\mathcal{F}$ above and $i\colon j^+\mathcal{F} \to j^{-1}\mathcal{F}$ for the sheafification. With this, $\eta(\mathcal{F})$ is equal to the composition

$$\eta(\mathcal{F}) = j_*(i) \circ \eta^{pre}(\mathcal{F}).$$

If $x \in Z$, then the stalk of $j_*j^+\mathcal{F}$ at $x \in X$ is, by definition,

$$\varinjlim_{U \ni x} \varinjlim_{\substack{V \supseteq U\cap Z \\ V \text{ open in } X}} \mathcal{F}(V).$$

The morphism on stalks $\eta^{pre}(\mathcal{F})_x\colon \mathcal{F}_x \to (j_*j^+\mathcal{F})_x$ sends $[U, s]$ into $[U, \kappa^U_{U\cap Z}(s)]$. We now show that it is an isomorphism. First suppose that $[U, \kappa^U_{U\cap Z}(s)] = [X, 0]$. Then there is an open neighbourhood $W \subseteq U$ of x such that

$$0 = \mathrm{res}^U_W \circ \kappa^U_{U\cap Z}(s) = \kappa^W_{W\cap Z}(s|_W).$$

By definition of colimit, there is an open set $W' \supseteq W \cap Z$ such that $s|_W|_{W'} = 0$ in $\mathcal{F}(W')$. Since $x \in Z \cap W$, it follows that W' also contains x, and thus

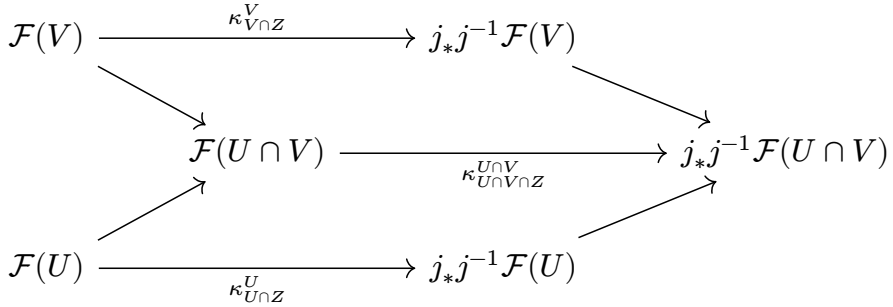
$$[U, s] = [W, s|_{W'}] = [W, 0].$$

Therefore $\eta^{pre}(\mathcal{F})_x$ is injective.

For surjectivity, let $[U, t] \in (j_* j^+ \mathcal{F})_x$. Then U is an open neighbourhood of x, and $t \in j_* j^+ \mathcal{F}(U)$. By definition, there is a section s in $\mathcal{F}(V)$ for some open subset V such that $V \supseteq U \cap Z$ and $\kappa^V_{U\cap Z}(s) = t$. Clearly V is also an open neighbourhood of x which contains $V \cap Z$. We claim that $[U, t] = [V, \kappa^V_{V\cap Z}(s)]$. Indeed, the intersection $U \cap V$ is an open neighbourhood of x, and so

$$t|_{U\cap V} = \mathrm{res}^V_{U\cap V}(\kappa^V_{U\cap Z}(s)) = \kappa^V_{U\cap V\cap Z}(s) = \mathrm{res}^V_{V\cap U}(\kappa^V_{V\cap Z}(s)) = \kappa^V_{V\cap Z}(s)|_{U\cap V},$$

which is obtained by chasing through the diagram



But $[V, \kappa^V_{V\cap Z}(s)]$ is the image of $[V, s]$ under $\eta^{pre}(\mathcal{F})_x$. This shows that $\eta^{pre}(\mathcal{F})_x$ is surjective.

By taking stalks of $\eta(\mathcal{F})$ at $x \in Z$, we get

$$\eta(\mathcal{F})_x = (j_*(i))_x \circ \eta^{pre}(\mathcal{F})_x.$$

We have just shown that $\eta^{pre}(\mathcal{F})_x$ is an isomorphism. Furthermore, Theorem 3.32 says that the morphism $i_x \colon (j^+\mathcal{F})_x \to (j^{-1}\mathcal{F})_x$ is an isomorphism, therefore $(j_*(i))_x$ is also an isomorphism by (1.). It follows that $\eta(\mathcal{F})_x$ is an isomorphism for all $x \in Z$.

Finally, if $x \notin Z$, then Problem 3.12 shows us that the morphism of stalks $\eta(\mathcal{F})_x$ reduces to the zero map $0 \to 0$ which is also an isomorphism. Combining with the above case, we see that $\eta(\mathcal{F})$ is an isomorphism of sheaves.

The proof for (3.) is similar and is omitted. □

Lemma 4.17. *Let $Z \subseteq X$ be a closed subscheme given by an ideal sheaf $\mathcal{I}$. Then the inclusion $j \colon Z \to X$ is a closed immersion.*

Proof. Let us write $\mathcal{O}_Z = j^{-1}(\mathcal{O}_X/\mathcal{I})$. By Lemma 4.16, the canonical morphism given by the unit of the adjunction

$$i^\sharp \colon \mathcal{O}_X/\mathcal{I} \to j_* j^{-1}\mathcal{O}_X/\mathcal{I} = j_*\mathcal{O}_Z.$$

is an isomorphism of sheaves. Now let $j^\sharp$ be the composition of the canonical projection $\mathcal{O}_X \to \mathcal{O}_X/\mathcal{I}$, which is surjective, followed by the isomorphism $i^\sharp$ above. Then $(j, j^\sharp)$ is a closed immersion. □

The converse is also true.

Lemma 4.18. *Let $(j, j^\sharp)\colon Z \to X$ be a closed immersion. Then j induces an isomorphism of schemes between Z and a closed subscheme $j(Z)$ of X.*

Proof. Let's identify Z with the subspace $j(Z)$, so that $j\colon Z \hookrightarrow X$ is the inclusion of a closed subspace. Define $\mathcal{I}$ to be the kernel sheaf of the morphism $j^\sharp\colon \mathcal{O}_X \to j_*\mathcal{O}_Z$. For each $x \in X$, we have

$$\mathcal{I}_x = \ker(j_x^\sharp\colon \mathcal{O}_{X,x} \to (j_*\mathcal{O}_Z)_x).$$

(Problem 3.2.) Since j is a closed immersion, the map of stalks $j_x^\sharp$ is by definition surjective. It then follows that there is an isomorphism

$$\mathcal{O}_{X,x}/\mathcal{I}_x \xrightarrow{\sim} (j_*\mathcal{O}_Z)_x \tag{$*$}$$

induced from $j_x^\sharp$.

If $x \notin Z$, then by Problem 3.12 $(j_*\mathcal{O}_Z)_x = 0$. In this case $(\mathcal{O}_X/\mathcal{I})_x = \mathcal{O}_{X,x}/\mathcal{I}_x = 0$. And if $x \in Z$, then $(\mathcal{O}_X/\mathcal{I})_x = (j_*\mathcal{O}_Z)_x = \mathcal{O}_{Z,x}$ by Problem 3.12 again. Since $(Z, \mathcal{O}_Z)$ is a scheme, the local ring $\mathcal{O}_{Z,x}$ is not zero. Therefore we have

$$Z = \{x \in X : (\mathcal{O}_X/\mathcal{I})_x \neq 0\}.$$

Now notice that the morphism of presheaves

$$\mathcal{O}_X(U)/\mathcal{I}(U) \to j_*\mathcal{O}_Z(U), \quad U \subseteq X \text{ open},$$

gives rise to a morphism of sheaves $g\colon \mathcal{O}_X/\mathcal{I} \to j_*\mathcal{O}_Z$ which is an isomorphism by Eq. ($*$). Therefore the morphism $j^{-1}g\colon j^{-1}\mathcal{O}_X/\mathcal{I} \to j^{-1}j_*\mathcal{O}_Z$ is also an isomorphism. Lemma 4.16 implies that $j^{-1}j_*\mathcal{O}_Z$ is isomorphic as sheaves to $\mathcal{O}_Z$, and so $(Z, j^{-1}(\mathcal{O}_X/\mathcal{I}))$ is a scheme as needed. □

For an affine scheme $X = \operatorname{Spec} A$, closed subschemes of X are in one-to-one correspondence with ideals of A, which is what one expects.

Proposition 4.19. *Let $X = \operatorname{Spec} A$ be an affine scheme and let $\mathfrak{a} \subseteq A$ be an ideal. Then the closed subset $Z = V(\mathfrak{a})$ is a closed subscheme of X with the sheaf structure coming from $V(\mathfrak{a}) \cong \operatorname{Spec} A/\mathfrak{a}$.*

Proposition 4.20. *Let $X = \operatorname{Spec} A$ be an affine scheme.*

(1.) Let Z be a closed subscheme of X, then Z is of the form $V(\mathfrak{a})$ for some ideal $\mathfrak{a}$ of A.

(2.) The map $\mathfrak{a} \mapsto V(\mathfrak{a})$ is a bijection from the set of ideals of A and the set of closed subschemes of X.

Proof. Let $i\colon V(\mathfrak{a}) \to X$ be the inclusion morphism, Note that if $D(f)$ is a principal open subset of X, then the inverse image presheaf of $\mathcal{O}_X$ under i gives

$$(i^+\mathcal{O}_X)(D(f)) = \mathcal{O}_X(D(f) \cap V(\mathfrak{a})) =$$

□

4.5 Functor of Points

4.6 Problems

4.1 *Criterion for Affineness.* Prove that a scheme X is affine if and only if there is a finite set of elements $f_1, \dots, f_n \in \Gamma(X, \mathcal{O}_X)$ such that the open subsets X_{f_i} are affine and the f_i generate the unit ideal of $\Gamma(X, \mathcal{O}_X)$.

Chapter 5

Some Properties of Schemes

5.1 Topological Properties of Schemes

We start with the properties of schemes depending on the underlying topological spaces.

Definition 5.1. Let X be a scheme. We say that X is **connected** (resp. **irreducible**, **quasi-compact**) if the underlying topological space is connected (resp. irreducible, quasi-compact).

▸ **Exercise 5.2.** Show that $\operatorname{Spec} A$ is not connected if and only if A is isomorphic to a direct product of two non-zero rings.

Definition 5.3. A morphism of schemes $f\colon X \to Y$ is said to be **injective** (resp. **surjective**, **bijective**, **open**, **closed**, **dominant** or is a **homeomorphism**) if the continuous map of topological spaces $f\colon X \to Y$ is injective (resp. surjective, bijective, open, closed, dominant or is a homeomorphism).

5.2 Noetherian Schemes

Recall that a ring R is **Noetherian** if it satisfies the *ascending ideal condition*: every chain of ideals

$$I_1 \subseteq I_2 \subseteq \cdots \subseteq I_n \subseteq \cdots$$

is eventually constant, that is, there is $N \in \mathbb{N}$ such that $I_N = I_n$ for all $n \geq N$. Equivalently, R is Noetherian if and only if every ideal of R is finitely generated. The corresponding notion for schemes is as follows.

Definition 5.4. A scheme X is **locally Noetherian** if X admits an open covering $X = \bigcup_i U_i$ by affine open subschemes U_i such that the ring $\mathcal{O}_X(U_i)$ is Noetherian. A scheme is **Noetherian** if it is quasi-compact and locally Noetherian.

The following observations are immediate.

Remark 5.5. Let X be a locally Noetherian scheme. Suppose that $\{U_i\}_{i\in I}$ is an open covering of X by affine open subschemes such that $U_i = \operatorname{Spec} A_i$ and A_i is Noetherian for all $i \in I$.

(1.) Recall that every affine scheme is quasi-compact (Proposition 2.16). Hence if X is both affine and locally Noetherian, then it is automatically Noetherian.

(2.) Localisations of a Noetherian ring are again Noetherian. Consequently, every principal open subset $D(f) = \operatorname{Spec}(A_i)_f$ of U_i is Noetherian. Open sets of this form give a basis of X, and so X has a basis consisting of Noetherian affine open subschemes.

Lemma 5.6. *An affine scheme* $\operatorname{Spec} A$ *is Noetherian if and only if the ring* A *is Noetherian.*

Proof. Clearly the condition is sufficient. For the converse, suppose that $\operatorname{Spec} A$ is Noetherian. As previously discussed, we may cover $\operatorname{Spec} A$ by finitely many principal open subsets $D(f_i)$, $i = 1, \dots, n$, such that each ring A_{f_i} is Noetherian. Let I be an ideal of A. It follows that the ideals $J_i = IA_{f_i}$ of A_{f_i} are finitely generated. The following lemma says that I is finitely generated which concludes the proof. □

Lemma 5.7. *Let* A *be a ring and suppose that there are elements* $f_1, \dots, f_n$ *such that* $\langle f_1, \dots, f_n \rangle = A$. *Let* M *be an* A*-module. Then* M *is finitely generated if and only if for all* $i = 1, \dots, n$ *the localisation* M_{f_i} *is a finitely generated* A_{f_i}*-module.*

Proof. The forward implication is obvious. To prove the converse, suppose that M_{f_i} is finitely generated as an A_{f_i}-module by elements $m_{ij}/(f_i)^{n_{ij}}$, $j = 1, \dots, r_i$, $m_{ij} \in M$. Consider the submodule $N \subseteq M$ generated by all m_{ij}. Then N is finitely generated, and furthermore, $N_{f_i} = M_{f_i}$ for all i.

We claim that the A-module M/N is 0, so that $M = N$. In order to prove this, let $\mathfrak{p}$ be a prime ideal of A. Then one of the f_i is not an element of $\mathfrak{p}$. It then follows that

$$(M/N)_{\mathfrak{p}} = ((M/N)_{f_i})_{\mathfrak{p}} = (M_{f_i}/N_{f_i})_{\mathfrak{p}} = 0.$$

Since $(M/N)_{\mathfrak{p}} = 0$ for all prime ideal $\mathfrak{p}$, $M = N$ as claimed. □

Remark 5.8. As an immediate consequence of Lemma 5.6, we see that if a scheme X can be covered by affine open subschemes U_i such that each U_i is a Noetherian scheme, then X is locally Noetherian.

Proposition 5.9. *The following are equivalent for a scheme X.*

(1.) X is locally Noetherian.

(2.) Each point $x \in X$ has an affine open neighbourhood U such that $\mathcal{O}_X(U)$ is Noetherian.

(3.) For every affine open subset $U \subseteq X$ the ring $\mathcal{O}_X(U)$ is Noetherian.

Proof. Clearly (1) $\iff$ (2) and (3) $\Rightarrow$ (1) because schemes may be covered by affine open subschemes. In turn we only need to show that (1) implies (3).

Assume that X is locally Noetherian and $U \subseteq X$ is an affine open subset. By Remark 5.5, U can be written as a union $U = \bigcup_i U_i$ of Noetherian open subschemes which are affine in X. For each i, Lemma 4.5 states that the intersection $U \cap U_i$ can be covered by open subsets W_{ij}, $j \in J$, such that W_{ij} is principal open in both U and U_i.

Since U_i is Noetherian and W_{ij} is principal open in U_i, it follows that W_{ij} is Noetherian (Remark 5.5). Thus $U = \bigcup_{i,j} W_{ij}$ is a union of affine open subschemes of U which are Noetherian, and so U is locally Noetherian by Remark 5.8. Because U is affine, the ring $\mathcal{O}_X(U)$ is Noetherian by Remark 5.5 and Lemma 5.6. □

Noetherian Topological Spaces

There is a topological version of Noetherianness.

Definition 5.10. A topological space is **Noetherian** if it satisfies the descending chain condition for closed subsets: every chain of closed subsets

$$Z_1 \supseteq Z_2 \supseteq \cdots \supseteq Z_n \supseteq \cdots$$

is eventually constant, that is, there is $N \in \mathbb{N}$ such that $Z_N = Z_n$ for all $n \geq N$.

The following proposition can be proved via Zorn's lemma.

Proposition 5.11. *The following are equivalent for a topological space X.*

(1.) X is Noetherian.

(2.) Every non-empty set of closed subsets of X has a minimal element.

(3.) Every non-empty set of open subsets of X has a maximal element.

(4.) Every open subset is quasi-compact.

Proposition 5.12 (Properties of Noetherian spaces)**.** *Let X be a Noetherian topological space.*

(1.) X has only finitely many irreducible components.

(2.) Every subspace of X is Noetherian.

Proposition 5.13. *If X is a Noetherian scheme, then X is a Noetherian topological space.*

▸ **Exercise 5.14.** Prove the propositions.

Remark 5.15. The converse is not true.

5.3 Generic Points and Irreducible Components

The generic point of a topological space X is a point $\eta \in X$ such that $\overline{\{\eta\}} = X$. Just as for affine schemes, there is a one-to-one correspondence between points of a scheme X and its closed irreducible subsets.

Proposition 5.16. *Let X be a scheme. Then there is a bijection*

$$\begin{aligned} X &\to \{\text{closed irreducible subsets of } X\} \\ x &\mapsto \overline{\{x\}}. \end{aligned}$$

Hence every closed irreducible subset Z of X contains a unique point $\eta \in Z$ such that $\overline{\{\eta\}} = Z$.

Proof. If X is affine, this is Proposition 2.24. For the general case, let X be a scheme and let Z be a closed irreducible subset of X. Take an affine open subset U of X such that $Z \cap U \neq \emptyset$. We claim that the closure of $Z \cap U$ in X is equal to Z. Indeed, let K denote the closure of $Z \cap U$ in X. Then $K \cap Z$ is closed in Z, and thus we may express Z as a union of two closed sets

$$Z = (K \cap Z) \cup ((X \setminus U) \cap Z).$$

But Z is irreducible, so either $(X \setminus U) \cap Z = Z$ or $K \cap Z = Z$. In the former case, we have $X \setminus U \supseteq Z$, contradicting $Z \cap U \neq \emptyset$. Hence we must have $K \cap Z = Z$, i.e. $K \subseteq Z$. Since Z is closed, this shows that $K = Z$.

It then follows from Proposition 2.22 that $Z \cap U$ is irreducible. Therefore $Z \cap U$ is a closed irreducible subset of an affine scheme X. By Corollary 2.28, $Z \cap U$ contains a unique generic point η such that the closure $\overline{\{\eta\}}^U$ in U is $Z \cap U$. We claim that the closure of $\{\eta\}$

in X is Z. By topology, we have

$$\overline{\{\eta\}}^X \cap U = \overline{\{\eta\}}^U = Z \cap U.$$

Using the fact that $\overline{\{\eta\}}^X \supseteq \overline{\{\eta\}}^X \cap U$, we obtain

$$\overline{\{\eta\}}^X = \overline{\overline{\{\eta\}}^X}^X \supseteq \overline{Z \cap U}^X = Z.$$

Since Z is a closed set which also contains η, we conclude that $\overline{\{\eta\}}^X = Z$.

Now let $z \in Z$ be a generic point. Then z is contained in every affine open subset U which meets Z. The uniqueness of generic points in the affine case implies that $z = \eta$. □

5.4 Reduced and Integral Schemes

A ring R is **reduced** if the only nilpotent elements of R is 0. In classical algebraic geometry, the coordinate ring $k[V]$ of an affine algebraic set $V \subseteq \mathbb{A}^n_k$ (k algebraically closed) is always reduced.

Definition 5.17. A scheme X is **reduced** if the local ring $\mathcal{O}_{X,x}$ is reduced for all $x \in X$.

Proposition 5.18. *A scheme X is reduced if and only if for each $U \subseteq X$ the ring $\mathcal{O}_X(U)$ is reduced.*

Proof. First suppose that X is reduced. Let $f \in \mathcal{O}_X(U)$ be such that $f^n = 0$ for some $n \geq 1$. Then for each $x \in U$, $f_x^n = 0$. This implies that $f_x = 0$ because the local ring are reduced. Hence $f = 0$ by the sheaf property.

Conversely, suppose that the ring $\mathcal{O}_X(U)$ are reduced for all open subset U. Let $x \in X$ and let $f_x \in \mathcal{O}_{X,x}$ be a germ satisfying $f_x^n = 0$ for some $n \geq 1$. We may find a section $f \in \mathcal{O}_X(U)$ defined on an open neighbourhood U of x lifting f_x. Then there is an open neighbourhood $V \subseteq U$ of x such that $f^n|_V = 0$. But $\mathcal{O}_X(V)$ is reduced, and thus $f|_V = 0$. This implies that $f_x = 0$. □

Definition 5.19. A scheme X is **integral** if for every non-empty open subset $U \subseteq X$, the ring $\mathcal{O}_X(U)$ is an integral domain.

Proposition 5.20. *A non-empty scheme X is integral if and only if it is reduced and irreducible.*

Proof. Suppose X is integral. For each $U \subseteq X$, the ring $\mathcal{O}_X(U)$ is integral, hence reduced. This shows that X is reduced by Proposition 5.18. Now assume that X is reducible. We may find two non-empty disjoint open subsets U_1, U_2 of X. The sheaf conditions imply

that $\mathcal{O}_X(U_1 \cup U_2) = \mathcal{O}_X(U_1) \times \mathcal{O}_X(U_2)$ is a product of two non-trivial rings (Lemma 4.4) which cannot be an integral domain.

For the converse, suppose that the scheme X is reduced and irreducible; *a fortiori* X cannot be empty. We first show that $\mathcal{O}_X(X)$ is an integral domain. Suppose $f, g \in \mathcal{O}_X(X)$ are such that $fg = 0$. For $s \in \mathcal{O}_X(X)$, define

$$V(s) := \{x \in X : s(x) = 0\} = X \setminus X_s.$$

This is a closed subset of X by Lemma 3.49. We also immediately see from the definitions that $V(st) = V(s) \cup V(t)$.

Since $fg = 0$, we have $X = V(fg) = V(f) \cup V(g)$. But X is irreducible, and so without loss of generality we may assume that $V(f) = X$. We claim that $f = 0$. This can be verified locally, therefore we shall consider the restriction of f to affine open subsets of X.

Set $a := f|_U$ where $U = \operatorname{Spec} A \subseteq X$. Then $a(x) = f(x) = 0$ for all $x \in U$ is equivalent to $a \in \mathfrak{p}_x$ for all prime ideals $\mathfrak{p}_x$ of A. Hence $a \in \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p} = \operatorname{Nil}(A)$. By hypothesis the scheme X is reduced, whence $A = \mathcal{O}_X(U)$ is a reduced ring. This implies $\operatorname{Nil}(A) = 0$ forcing $f|_U = a = 0$. Therefore $f = 0$ as desired.

Now let V be any non-empty open subset of X viewed as a scheme $(V, \mathcal{O}_{X|V})$. It is straightforward from the definition that V is reduced as a scheme. Furthermore, V is irreducible because non-empty open subsets of an irreducible topological space are themselves irreducible. The same argument as above reveals that $\mathcal{O}_{X|V}(V) = \mathcal{O}_X(V)$ is an integral domain. □

Proposition 5.21. *If X is an integral scheme, then the local rings $\mathcal{O}_{X,x}$ are integral domains for all $x \in X$.*

Proof. Follows from the fact that non-zero localisations of an integral domain are again integral. □

Remark 5.22. The converse of Proposition 5.21 is false.

Example 5.23 (Reduced and Integral Affine Schemes)**.** Let $X = \operatorname{Spec} A$ be an affine scheme. Then X is reduced if and only if A is reduced. Similarly, X is integral if and only if A is an integral domain.

Example 5.24. Affine algebraic sets are example of reduced schemes.

Function Field of an Integral Scheme

In classical algebraic geometry, there is the notion of function field $k(V)$ of an algebraic variety $V \subseteq \mathbb{A}^n_k$, where k is algebraically closed. It is defined as the field of fraction of the coordinate ring $k[V]$ which exists because $k[V]$ is an integral domain. Its elements are rational functions defined on an open subset of V. The following is an analogue for integral schemes.

Definition 5.25. Let X be an integral scheme with the generic point $\eta \in X$. The local ring $\mathcal{O}_{X,\eta}$, which is indeed a field, is called the **function field** of X and is denoted by $K(X)$.

Example 5.26. Let k be a field, so that the scheme $\mathbb{A}^1_k = \operatorname{Spec} k[T]$ is integral. The function field $K(\mathbb{A}^1_k)$ of $\mathbb{A}^1_k$ is by definition the localisation of $k[T]$ at the zero prime ideal. Hence we have

$$\mathcal{O}_{X,\eta} = k[T]_{(0)} = k(T)$$

and so the function field of $\mathbb{A}^1_k$ is the field of rational functions.

The following proposition shows that, when X is an integral scheme, the rings $\mathcal{O}_X(U)$ and the local rings $\mathcal{O}_{X,x}$, $x \in X$, can be viewed as subrings of the function field $K(X)$.

Proposition 5.27. *Let X be an integral scheme with η its generic point.*

(1.) If $U = \operatorname{Spec} A$ is a non-empty affine open subscheme of X, then $K(X) = \operatorname{Frac}(A)$. If $x \in X$, then $\operatorname{Frac}(\mathcal{O}_{X,x}) = K(X)$.

(2.) Let $U \subseteq V \subseteq X$ be non-empty open subsets, then the morphisms

$$\Gamma(V, \mathcal{O}_X) \xrightarrow{\operatorname{res}^V_U} \Gamma(U, \mathcal{O}_X) \xrightarrow{f \mapsto f_\eta} K(X)$$

are injective.

(3.) For every non-empty open subset $U \subseteq X$ and for every covering $U = \bigcup_i U_i$ by non-empty open subsets U_i, we have

$$\Gamma(U, \mathcal{O}_X) = \bigcap_i \Gamma(U_i, \mathcal{O}_X) = \bigcap_{x \in U} \mathcal{O}_{X,x},$$

where the intersection takes place in $K(X)$.

Proof. (1.) Since U is non-empty, it contains the generic point η. By definition, the ring A is an integral domain, and so the zero ideal is the unique generic point of U. But the closure of η in U is $\overline{\{\eta\}}^U = U$, and so η is also the generic point of U. Therefore η corresponds to the zero ideal of A. This implies that $\operatorname{Frac}(A) = A_{(0)} = \mathcal{O}_{X,\eta} = K(X)$. Furthermore, since A is an integral domain, the fraction field of every local ring $\mathcal{O}_{X,x} = A_{\mathfrak{p}_x}$ is also $K(X)$.

(2.) It suffices to show that $\Gamma(U, \mathcal{O}_X) \xrightarrow{f \mapsto f_\eta} K(X)$ is injective. (Given this, it is an easy exercise to show that res^V_U is injective for all $U \subseteq V$.) Let $f \in \mathcal{O}_X(U)$ be such that $f_\eta = 0$. Note that we can check that $f = 0$ locally on affine open subschemes of U.

Let W be an open non-empty affine subscheme of $W \subseteq U$. Then W contains the generic point η and so $0 = f_\eta = (f|_W)_\eta$. But by (1.) the stalk map $\mathcal{O}_X(W) \to \mathcal{O}_{X,\eta} = K(X)$ is the inclusion $\mathcal{O}_X(W) \to K(W)$ of $\mathcal{O}_X(W)$ into its field of fractions which is injective.

Hence $f|_W = 0$. Since this holds for every open non-empty affine subscheme of U, the sheaf condition implies that $f = 0$.

(3.) First let's prove that $\Gamma(U, \mathcal{O}_X) = \bigcap_i \Gamma(U_i, \mathcal{O}_X)$. The inclusion $\subseteq$ follows from the injectivity of inclusion maps proved in (2.). For the inverse inclusion $\supseteq$, note that if $s \in \Gamma(U_i, \mathcal{O}_X)$, then it trivially satisfies the condition to glue to a section in $\Gamma(U, \mathcal{O}_X)$.

Finally we prove that $\Gamma(U, \mathcal{O}_X) = \bigcap_{x \in U} \mathcal{O}_{X,x}$. Let $x \in U$ be contained in an open affine subscheme $W \subseteq U$ such that $W = \operatorname{Spec} A$. Then there are injective ring homomorphisms

$$\Gamma(U, \mathcal{O}_X) \to \Gamma(W, \mathcal{O}_X) = A \to A_{\mathfrak{p}_x} = \mathcal{O}_{X,x}.$$

Hence $\Gamma(U, \mathcal{O}_X) \subseteq \mathcal{O}_{X,x}$ for all x, and thus the forward inclusion holds.

For the other inclusion, suppose that $s \in \mathcal{O}_{X,x}$ for all $x \in U$. Let $W \subseteq U$ be a non-empty open affine subscheme and write $W = \operatorname{Spec} A$. Then we have $s \in \operatorname{Frac}(A)$ and $s \in A_{\mathfrak{p}_x}$ for all prime ideal $\mathfrak{p}_x$ of X.

Consider the set $\mathfrak{a} = \{h \in A : sh \in A\}$. It is easy to check that this is an ideal of A. If $\mathfrak{a}$ is proper, then it is contained in a maximal ideal $\mathfrak{m}$. But by assumption $s \in A_{\mathfrak{m}}$ which is equivalent to $s = b/g$ for suitable $b \in A$, $g \notin \mathfrak{m}$. This means that $sg = b \in A$, and so $g \in \mathfrak{a}$, a contradiction. Therefore $\mathfrak{a} = A$, implying that $s = s1 \in A = \Gamma(W, \mathcal{O}_X)$. Since this holds for any non-empty open affine subscheme $W = U$ which form an open cover of U, we conclude that $s \in \bigcap \Gamma(W, \mathcal{O}_X) = \Gamma(U, \mathcal{O}_X)$ by the first part. □

5.5 Underlying Reduced Subscheme

Given a scheme X, there could be distinct closed subschemes with same underlying topological space as X. As an example, given an ideal $\mathfrak{a} \subseteq A$, the subschemes $V(\mathfrak{a})$ and $V(\mathfrak{a}^k)$ agrees as topological subspaces of the scheme $\operatorname{Spec} A$.

However, among closed subschemes of X with the same underlying topological space, there is a smallest one which is characterised by the fact that it is reduced. To prove this result, we define the subsheaf $\mathcal{N}_X$ of $\mathcal{O}_X$ to be the sheaf associated to the presheaf

$$U \mapsto \operatorname{Nil}(\Gamma(U, \mathcal{O}_X)), \quad U \subseteq X \text{ open}.$$

This is called the **nilradical sheaf** of X.

Theorem 5.28. *The ringed space $(X_{\mathrm{red}}, \mathcal{N}_X)$ is a reduced closed subscheme of X with the same underlying topological space as X. Furthermore, if $T \subseteq X$ is a closed subscheme of X with the same underlying topological space as X, then the inclusion $X_{\mathrm{red}} \hookrightarrow X$ factors through a closed immersion $X_{\mathrm{red}} \to T$.*

Proof. We first show that X_{red} is a scheme. It suffice to prove this in the case when $X = \operatorname{Spec} A$ is affine. We claim that the presheaf $U \mapsto \operatorname{Nil}(\Gamma(U, \mathcal{O}_X))$ restricted to the basis of principal open subsets of X is a sheaf. Indeed, suppose $D(f)$ is principal open and $D(f) = \bigcup_{i \in I} D(f_i)$ is an open covering, which we may assume that I is finite because $D(f)$ is quasi-compact. The uniqueness part of the sheaf condition is clearly satisfied, so we only need to show that it satisfies gluing condition.

Let $s_i \in \mathrm{Nil}(\Gamma(D(f_i), \mathcal{O}_X)) = \mathrm{Nil}(A_{f_i})$ be sections such that $s_i|_{D(f_i f_j)} = s_j|_{D(f_i f_j)}$ for all $i, j \in J$. They glue to a section $s \in \Gamma(D(f_i), \mathcal{O}_X)$ such that $s|_{D(f_i)} = s_i$ for all i, and thus we have to check whether s is nilpotent or not. Because I is finite, there is $N \geq 1$ such that $(s|_{D(f_i)})^N = s_i^N = 0$ independent of i, forcing $s^N = 0$. This proves the claim.

TODO. □

Corollary 5.29. *If $X = \operatorname{Spec} A$ is an affine scheme, then $\mathcal{N}_X(U) = \mathrm{Nil}(\mathcal{O}_X(U))$ for all affine open subscheme $U \subseteq X$ and $X_{\mathrm{red}} = \operatorname{Spec}(A/\operatorname{Nil}(A))$.*

Definition 5.30.

5.6 Properties of Morphisms of Schemes

Quasi-compact Morphisms

Definition 5.31. A morphism $f\colon X \to Y$ of schemes is **quasi-compact** if for all quasi-compact open subset $V \subseteq Y$, the preimage $f^{-1}(V)$ is quasi-compact.

Proposition 5.32. *A morphism $f\colon X \to Y$ is quasi-compact if and only if there is an affine open cover $\{V_i\}$ of Y such that for all i, $f^{-1}(V_i)$ is quasi-compact*

Proof. The condition is clearly necessary. TODO. □

Morphisms (Locally) of Finite Type

Definition 5.33. Let X, Y be schemes. A morphism $f\colon X \to Y$ is **locally of finite type** if there is an open cover $\{V_i\}_{i \in I}$ of Y by affine open subschemes $V_i = \operatorname{Spec} B_i$ such that

- for each i, $f^{-1}(V_i)$ can be covered by affine open subschemes $U_{ij} = \operatorname{Spec} A_{ij}$ where A_{ij} is a finitely generated B_i-algebra.

Definition 5.34. A morphism $f\colon X \to Y$ is **of finite type** if it is both locally of finite type and quasi-compact.

5.7 Problems

5.1 Show that every scheme X is a Kolmogorov space, i.e. if $x \neq y$, then there is an open subset of X which contains only one of the points x or y.

5.2 *Properties Preserved by Localisations.* Let R be a ring and $S \subseteq R$ a multiplicative subset.

(a) Show that if R is Noetherian, then $S^{-1}R$ is Noetherian.

(b) Show that if R is reduced, then $S^{-1}R$ is reduced.

(c) Show that if R is an integral domain, then $S^{-1}R$ is an integral domain provided that $S^{-1}R \neq 0$.

Chapter 6

Projective Spaces

In this chapter we define projective spaces in two different ways. One is based on gluing affine spaces together and the other is based from the Proj construction. The two definitions are equivalent.

6.1 Projective Spaces as Gluing of Affine Schemes

In this section we construct the projective space $\mathbb{P}^n_R$ over a ring R via gluing $n+1$ copies of the affine space. Let $X_0, X_1, \dots, X_n$ be variables. Define, for each $i = 0, \dots, n$,

$$U_i = \operatorname{Spec} R\left[\frac{X_0}{X_i}, \dots, \widehat{\frac{X_i}{X_i}}, \dots, \frac{X_n}{X_i}\right],$$

where $\widehat{T}$ means that T is omitted. Therefore U_i is isomorphic to the affine space $\mathbb{A}^n_R$. We like to think that everything is happening inside the ring of Laurent polynomials $S = R[X_0, \dots, X_n, X_0^{-1}, \dots, X_n^{-1}]$ which clearly contains these defining rings of U_i as subrings. The gluing datum for $\mathbb{P}^n_R$ is as follows.

(1.) The index set is $\{0, 1, \dots, n\}$ and the schemes are the U_i.

(2.) For each $0 \leq i, j \leq n$ and $i \neq j$, set

$$U_{ij} = D_{U_i}\left(\frac{X_j}{X_i}\right) \subseteq U_i.$$

Otherwise set $U_{ii} = U_i$.

(3.) Define $\varphi_{ii} = \mathrm{id}_{U_i}$; and if $i \neq j$, define

$$\varphi_{ji} \colon U_{ij} \to U_{ji}$$

to be the isomorphism corresponding to the identity map

$$R\left[\frac{X_0}{X_i}, \dots, \widehat{\frac{X_i}{X_i}}, \dots, \frac{X_n}{X_i}\right]_{X_j/X_i} \leftarrow R\left[\frac{X_0}{X_j}, \dots, \widehat{\frac{X_j}{X_j}}, \dots, \frac{X_n}{X_j}\right]_{X_i/X_j}.$$

Since the morphisms φ_{ji} are defined by equalities the cocycle condition holds trivially for them. The resulting scheme after gluing is called the **projective space of relative dimension n over R** and is denoted by $\mathbb{P}^n_R$. The scheme U_i is regarded as a subscheme of $\mathbb{P}^n_R$ and is denoted by $D_+(X_i)$.

Proposition 6.1. *The canonical ring homomorphism $R \to \Gamma(\mathbb{P}^n_R, \mathcal{O}_{\mathbb{P}^n_R})$ is an isomorphism.*

Proof. If $n = 0$, then $\mathbb{P}^n_R = \operatorname{Spec} R$ and there is nothing to prove. So let $n \geq 1$. For each i, write

$$R_i = R\left[\frac{X_0}{X_i}, \dots, \widehat{\frac{X_i}{X_i}}, \dots, \frac{X_n}{X_i}\right].$$

Firstly, observe that the restriction morphism $\Gamma(U_i, \mathcal{O}_{\mathbb{P}^n_R}) \to \Gamma(U_{ij}, \mathcal{O}_{\mathbb{P}^n_R})$ is injective because it is the localisation map

$$R_i = R\left[\frac{X_0}{X_i}, \dots, \widehat{\frac{X_i}{X_i}}, \dots, \frac{X_n}{X_i}\right] \to R\left[\frac{X_0}{X_i}, \dots, \widehat{\frac{X_i}{X_i}}, \dots, \frac{X_n}{X_i}\right]_{X_j/X_i}.$$

The element X_j/X_i is a non-zerodivisor of the ring R_i. This gives the above observation. Hence we may regard this restriction map as an inclusion of subrings. The sheaf condition with respect to the open cover $\{U_i\}$ of $\mathbb{P}^n_R$ implies that

$$\Gamma(\mathbb{P}^n_R, \mathcal{O}_{\mathbb{P}^n_R}) = \left\{(f_i) \in \prod_{i=1}^n \Gamma(U_i, \mathcal{O}_{\mathbb{P}^n_R}) : f_i|_{U_{ij}} = f_j|_{U_{ji}} \text{ for all } i, j\right\}.$$

By the above observation, $f_i|_{U_{ij}} = f_i$ for all i, j. Therefore an element $f \in S$ defines a global section of $\mathbb{P}^n_R$ if and only if $f \in R_i$ for all i. In other words,

$$\Gamma(\mathbb{P}^n_R, \mathcal{O}_{\mathbb{P}^n_R}) \cong \bigcap_{i=1}^n R_i.$$

An element $f \in S$ belongs to R_i if and only if the total degree is zero and each monomial $rX_0^{\alpha_0} \cdots X_n^{\alpha_n}$ of f with $r \in R$, $\alpha_n \in \mathbb{Z}$ satisfies $\alpha_k \geq 0$ for all $k \neq i$. From this it follows that $\Gamma(\mathbb{P}^n_R, \mathcal{O}_{\mathbb{P}^n_R}) \cong \bigcap_{i=1}^n R_i = R$. □

Morphisms into Projective Spaces

Proposition 6.2. *Let*

Corollary 6.3. *If $n > 0$, then $\mathbb{P}^n_R$ is not affine.*

Proof. Suppose not. Then $\mathbb{P}^n_R \cong \operatorname{Spec} R$ by the anti-equivalence of the category of affine schemes and the category of rings (Theorem 3.56). □

6.2 Vanishing Schemes

6.3 Proj Construction

Chapter 7

Varieties as Schemes

7.1 Schemes Locally of Finite Type over a Field

For schemes over a field k, the definition of morphisms locally of finite type simplifies considerably. This is because there is only one possible affine covering of $\operatorname{Spec} k$, namely $\operatorname{Spec} k$ itself. Hence we arrive at the following proposition.

Proposition 7.1. *A scheme X over a field k is locally of finite type over k if there is an affine open cover U_i of X such that for all i, $U_i = \operatorname{Spec} A_i$ where A_i is a finitely generated k-algebra.*

The Hilbert basis theorem states that for all $n \geq 1$ the polynomial ring $k[x_1, \ldots, x_n]$ is Noetherian. This implies that every finitely generated k-algebra is also Noetherian. Therefore a scheme locally of finite type over k is locally Noetherian.

Proposition 7.2. *Let X be a k-scheme locally of finite type. For any open affine subset $U \subseteq X$, $\mathcal{O}_X(U)$ is a finitely generated k-algebra.*

7.2 Prevarieties as Schemes

7.3 Dimension Theory

Definition 7.3. Let X be a topological space.

(1.) A chain
$$X_0 \supsetneq X_1 \supsetneq \cdots \supsetneq X_l$$
of closed irreducible subsets of X is said to be of **length** l.

(2.) The **dimension** $\dim X$ of X is the supremum of the lengths of all possible of closed irreducible subsets.

A topological space is called **equidimensional** if all of its irreducible components have the same dimension.

Remark 7.4. The dimension of the empty topological space is set to be $-\infty$. In general the dimensions of non-empty topological spaces can be a non-negative integers or ∞.

Lemma 7.5 (Properties of Dimension)**.** *Let X be a topological space.*

(1.) Suppose Y is a subspace of X. Then $\dim Y \leq \dim X$.

(2.) If X is irreducible, $\dim X < \infty$ and $Y \subsetneq X$ is a proper closed subset, then $\dim Y < \dim X$.

(3.) Let $\{U_i\}_{i\in I}$ be an open covering of X. Then $\dim X = \sup_{i\in I} \dim U_i$.

(4.) Let J be the set of irreducible components of X. Then $\dim X = \sup_{Y\in J} \dim Y$.

Proof. (1.) Let $Z_0 \supsetneq Z_1 \supsetneq \cdots \supsetneq Z_l$ be a chain of closed irreducible subsets of Y. Taking closure preserves strict inequalities, as one may check. Therefore we obtain the chain $\overline{Z_0} \supsetneq \overline{Z_1} \supsetneq \cdots \supsetneq \overline{Z_l}$. Since the closure of an irreducible set is irreducible, this proves (1).

(2.) Under the given assumption, any chains of closed irreducible subset of Y can be extended by adding X. This proves (2).

(3.) By (1) we have $\dim U_i \leq \dim X$, and so $\sup_{i\in I} \dim U_i \leq \dim X$.

(4.) Obvious. □

The **dimension of a scheme** X is simply the dimension of the underlying topological space. For affine schemes this coincides with the Krull dimension. Recall that the Krull dimension of a ring A is defined to be the supremum of the lengths of all chains
$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_l$$

of prime ideals in A.

Proposition 7.6. *Let $X = \operatorname{Spec} A$. Then the dimension of X as a scheme is equal to the Krull dimension of A.*

Proof. Follows from Proposition 2.24 where it is proved that prime ideals correspond bijectively to irreducible closed subsets of X, and that this bijection is order-reversing. □

7.4 Bézout's Theorem

Chapter 8

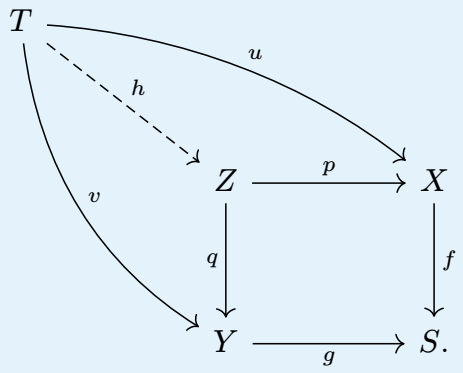
Fibre Products

8.1 Fibre Product in General Categories

We start with the definition of fibre products in any category.

Definition 8.1. Let $\mathcal{C}$ be a category, and let $f\colon X \to S$ and $g\colon Y \to S$ be morphisms in $\mathcal{C}$. A **fibre product** or **pullback** of f and g is an object Z with morphisms $p\colon Z \to X$ and $q\colon Z \to Y$ which is universal with respect to the following property:

The equation $f \circ p = g \circ q$ holds; and furthermore, for any object T with morphisms $u\colon T \to X$ and $v\colon T \to Y$ such that $f \circ u = g \circ v$, there exists a unique morphism $h\colon T \to Z$ such that the following diagram commutes



A fibre product of f and g is also called a **fibre product of X and Y over S**. As with any universal objects, the fibre product of f and g is unique up to unique isomorphism if it exists. Thus we may denote the fibre product Z of f and g by $X \times_S Y$. The morphisms $X \times_S Y \to X$ and $X \times_S Y \to Y$ are called the **projections** to the first and second factor, respectively.

The unique morphism h arising from u and v as in the definition of fibre product will often be denoted by $(u, v)_S$. The notation emphasises the fact that $p \circ (u, v)_S = u$ and $q \circ (u, v)_S = q$, showing that p and q acts like projections.

For familiar categories such as the category of sets and the category of topological

spaces, fibre products always exist and can be explictly described.

Example 8.2. Let $f\colon X \to S$ and $g\colon Y \to S$ be functions. Then the set

$$Z = \{(x, y) \in X \times Y : f(x) = g(y)\},$$

together with projection maps $p\colon (x, y) \mapsto x$ and $q\colon (x, y) \mapsto y$, is the fibre product of f and g in (Set).

For the category of topological spaces, equip the set Z above with the subspace topology induced from the product topology of $X \times Y$. The projection maps p and q are then continuous, and together with Z form the fibre product of f and g in (Top).

Keeping the notation of Example 8.2, the fibre product of f and g (in the category of sets) can also be written as a union of products of fibres

$$Z = X \times_S Y = \bigcup_{s \in S} f^{-1}(s) \times g^{-1}(s),$$

hence the name *fibre product*.

▸ **Exercise 8.3.** For any category $\mathcal{C}$ and object $S \in \mathcal{C}$, prove that the fibre product of $f\colon X \to S$ and $Y \to S$ is the same as the product of f and g taken in the slice category $\mathcal{C}/S$.

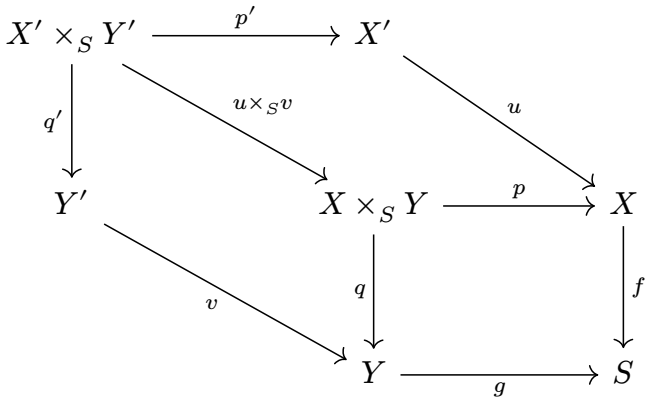
Functoriality

We now turn to functoriality of fibre products. Suppose that in the category $\mathcal{C}$ all fibre products exist. Given morphisms $f \circ X \to S$ and $g \circ Y \to S$ we may form the fibre product $X \times_S Y$:

$$\begin{array}{ccc} X \times_S Y & \xrightarrow{\ p\ } & X \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle f} \\ Y & \xrightarrow[\ g\ ]{} & S. \end{array}$$

Now suppose X' and Y' are S-objects with S-morphisms $u\colon X' \to X$ and $v\colon Y' \to Y$. Again we make form the fibre product $(X' \times_S Y', p', q')$. Then there is a unique morphism

$u \times_S v \colon X' \times_S Y' \to X \times_S Y$ such that the diagram



is commutative. In fact, $u \times_S v = (u \circ p', v \circ q')_S$ is obtained by the universal property of fibre products, as one can easily check.

The following proposition is easy to prove either directly or via the Yoneda lemma.

Proposition 8.4. *Let S be an object in $\mathcal{C}$ and let X, Y, Z be S-objects. Then there are "canonical" isomorphisms*

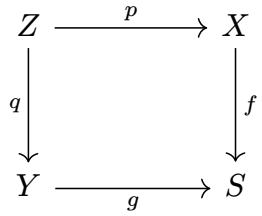
(1.) $X \times_S S \xrightarrow{\sim} X$.

(2.) $X \times_S Y \xrightarrow{\sim} Y \times_S X$.

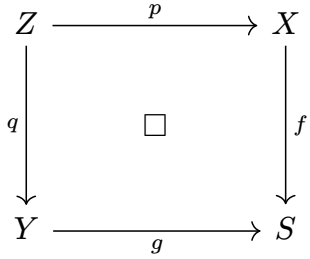
(3.) $X \times_S (Y \times_S Z) \xrightarrow{\sim} (X \times_S Y) \times_S Z$.

All of the isomorphisms are functorial in X, Y, and Z.

Given a commutative diagram

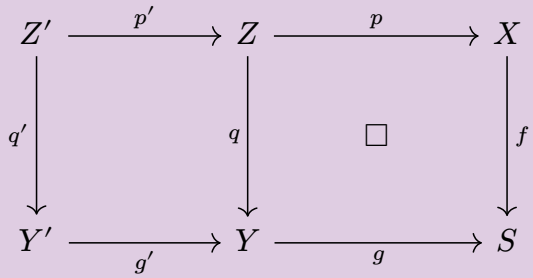


we say that this diagram is **cartesian** if Z and the morphisms p, q is the fibre product of f and g. We denote a cartesian diagram using a square $\square$ in the middle as shown below:



We finish this section with the *pullback lemma.*

Lemma 8.5 (Pullback lemma)**.** *Suppose we are given the following commutative diagram in a category $\mathcal{C}$*



in which the right square is cartesian. Then the left square is cartesian if and only if the outer rectangle is cartesian.

▸ **Exercise 8.6.** Prove the pullback lemma either by diagram chasing or by the Yoneda lemma.

8.2 Fibre Product of Schemes

The goal of this section is to prove that fibre products always exists in the category of schemes. We begin by showing that if all the schemes X, Y, and S involved are affine, then the fibre product of X and Y over S exists.

Theorem 8.7. *Let $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} B$, and $S = \operatorname{Spec} R$ be affine schemes, and let $f\colon X \to S$ and $g\colon Y \to S$ be morphisms of schemes. Then the fibre product of f and g exists and is given by the affine scheme $Z = \operatorname{Spec}(A \otimes_R B)$. The projections $p\colon Z \to X$ and $q\colon Z \to Y$ are the morphisms corresponding to the ring homomorphisms*

$$\alpha\colon A \to A \otimes_R B, \quad a \mapsto a \otimes 1,$$
$$\beta\colon B \to A \otimes_R B, \quad b \mapsto 1 \otimes b.$$

Proof. Recall that a morphism of affine schemes $f\colon X \to S$ corresponds to a ring homomorphism $\varphi\colon R \to A$. Similarly, g corresponds to a ring homomorphism $\psi\colon R \to B$. Hence A and B are R-algebras and so the tensor product $A \otimes_R B$ exists. Let α, β be ring homomorphisms as defined in the theorem statement. Then $\alpha \circ \varphi = \beta \circ \psi$, as can easily be checked. This implies that $f \circ p = g \circ q$.

Let $T \to S$ be a scheme with morphisms $p'\colon T \to X$ and $q'\colon T \to Y$ such that

$f \circ p' = g \circ q'$. By, we have a series of bijections

$$\begin{aligned}\mathrm{Hom}_{(\mathrm{Sch}/S)}(T, Z) &\cong \mathrm{Hom}_{(R-\mathrm{Alg})}(A \otimes_R B, \Gamma(T, \mathcal{O}_T)) \\ &\cong \mathrm{Hom}_{(R-\mathrm{Alg})}(A, \Gamma(T, \mathcal{O}_T)) \times \mathrm{Hom}_{(R-\mathrm{Alg})}(A, \Gamma(T, \mathcal{O}_T)) \\ &\cong \mathrm{Hom}_{(\mathrm{Sch}/S)}(T, A) \times \mathrm{Hom}_{(\mathrm{Sch}/S)}(T, B),\end{aligned}$$

where the second bijection is induced by composition with α and β. This is functorial in T. Therefore, by the Yoneda lemma, Z is the fibre product of X and Y over S. □

Theorem 8.8 (Fibre Product of Schemes)**.** *Let S be a scheme and let X, Y be S-schemes. Then the fibre product $X \times_S Y$ exists in the category of schemes.*

8.3 Fibres

8.4 Base Changes

Definition 8.9. Let S be a scheme and let X be an S-scheme. Suppose $u\colon S' \to S$ is a morphism of schemes. The **base change of X along u** is defined to be the fibre product $X \times_S S'$ of X and S' fitting in the following cartesian diagram

$$\begin{array}{ccc} X \times_S S' & \xrightarrow{\ p\ } & X \\ {\scriptstyle q}\downarrow & \square & \downarrow \\ S' & \xrightarrow[\ u\ ]{} & S. \end{array}$$

The scheme $X \times_S S'$ is an S'-scheme via the projection $q\colon X \times_S S' \to S$.

We also write $u^*(X)$ and $X_{S'}$ for the base change $X \times_S S'$. If $f\colon X \to Y$ is a morphism of S-schemes, we can also base change the morphism f by defining $f_{S'} := f \times_S \mathrm{id}_{S'}\colon X \times_S S' \to Y \times_S S'$. Furthermore, if $S' = \operatorname{Spec} B$ is affine, we also write X_B for $X_{S'}$ and f_B for $f_{S'}$. Hence base change defines a covariant functor

$$u^*\colon (\mathrm{Sch}/S) \to (\mathrm{Sch}/S')$$

from the category of S-schemes to the category of S'-schemes.

Example 8.10 (Base Change of Finitely Generated Algebras)**.** Let R be a ring and set $S = \operatorname{Spec} R$. Consider the S-scheme $X = \operatorname{Spec} R[X_1, \dots, X_n]/(f_1, \dots, f_m)$ where $X_1, \dots, X_n$ are variables and $f_1, \dots, f_m$ are polynomials in those variables. If $\varphi\colon R \to T$ is a ring homomorphism, then it induces a morphism of affine schemes $u\colon S' \coloneqq \operatorname{Spec} T \to S$. The base change of X along u is then the S'-scheme

$$\begin{aligned} X \times_S S' &= \operatorname{Spec}\left(R[X_1, \dots, X_n]/(f_1, \dots, f_m) \otimes_R T\right) \\ &= \operatorname{Spec}(T[X_1, \dots, X_n]/(f_1, \dots, f_m)), \end{aligned}$$

where in the last equality we viewed $f_1, \dots, f_m$ as polynomials in $T[X_1, \dots, X_n]$ via the morphism φ.

8.5 Problems

Chapter 9

Separated Morphisms

Recall that a topological space X is Hausdorff if for every two distinct points $x \neq y$, there are open neighbourhoods U of x and V of y such that $U \cap V = \emptyset$. The following conditions are equivalent to X being Hausdorff:

(1.) The diagonal set $\Delta := \{(x, x) : x \in X\}$ is closed in $X \times X$, where $X \times X$ is equipped with the product topology.

(2.) For every topological space Y and continuous map $f\colon Y \to X$, the graph $\{(y, f(y)) : y \in Y\}$ is closed in $Y \times X$.

(3.) For every topological space Y and continuous maps $f, g\colon Y \to X$, the equaliser $\{y \in Y : f(y) = g(y)\}$ is closed in Y.

▸ **Exercise 9.1.** Prove the asserted equivalence.

Schemes are in general not Hausdorff. Nevertheless, the above characterisation serves as a motivation for an analogue of Hausdorff property of schemes.

9.1 Diagonals, Graphs, and Equalisers

We first need to define diagonals, graphs and equalisers for arbitrary schemes. In more generality we shall define them for any category which has fibre products.

In this section, let $\mathcal{C}$ be a category admitting all fibre products. Let $S \in \mathcal{C}$ and $u\colon X \to S$ be an S-object. If $T \to S$ is another S-object, the set of all S-morphisms $T \to X$ is denoted by $X_S(T)$.

Definition 9.2. Situation as above.

(1.) The **diagonal** of $u\colon X \to S$ is the morphism

$$\Delta_{X/S} := \Delta_u := (\mathrm{id}_X, \mathrm{id}_X)_S\colon X \to X \times_S X$$

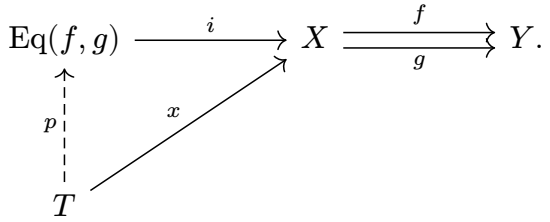
obtained from the universal property of fibre products. Sometimes we write Δ for $\Delta_{X/S}$ if no confusion can arise

(2.) Let $f\colon X \to Y$ be an S-morphism. The **graph** of f is the morphism

$$\Gamma_f := (\mathrm{id}_X, f)_S\colon X \to X \times_S Y.$$

(3.) Let $f, g\colon X \to Y$ be two S-morphisms. The **equaliser** $\mathrm{Eq}(f,g)$ of f, g is the categorical equaliser of f and g in the slice category over S. The universal S-morphism $i\colon \mathrm{Eq}(f,g) \to X$ is called the **canonical** morphism.

Unravelling the definition of equalisers, we can give another definition using universal mapping property as follows. The canonical morphism $i\colon \mathrm{Eq}(f,g) \to X$ satisfies $f \circ i = g \circ i$ and for all S-morphism $x\colon T \to X$ such that $f \circ x = g \circ x$, there is a unique morphism $p\colon T \to \mathrm{Eq}(f,g)$ such that $i \circ p = x$. This is indicated in the following commutative diagram:



Alternatively, $i\colon K \to X$ is an equaliser of f, g if and only if it represents the contravariant functor sending an S-object T to the set $\{x \in X_S(T) : f(T)(x) = g(T)(x)\}$.

▸ **Exercise 9.3.** Show that the canonical morphism $i\colon \mathrm{Eq}(f,g) \to X$ is a monomorphism in the category of S-objects.

The following lemma shows that the equaliser of two S-morphisms can be expressed in terms of fibre products, and so all equalisers exist in $\mathcal{C}$.

Lemma 9.4.

Example 9.5. In the category of sets, the diagonal of an S-object $u\colon X \to S$ is the usual diagonal function:

$$\begin{aligned} \Delta_u\colon X &\to X \times_S X \\ x &\mapsto (x, x). \end{aligned}$$

Similarly, if $f\colon X \to Y$ is a S-morphism, then the graph of f is the function

$$\begin{aligned} \Gamma_f\colon X &\to X \times_S Y \\ x &\mapsto (x, f(x)). \end{aligned}$$

The set of pairs $\{(x, f(x)) : x \in X\}$ is of course the set-theoretic graph of f.

Diagonals, Graphs, and Equalisers in the Category of Schemes

The diagonal morphism and the graph of any morphism in the category of affine schemes have simple descriptions. The proof is simple and is omitted.

Proposition 9.6. *Let $S = \operatorname{Spec} R$, $X = \operatorname{Spec} A$, and $Y = \operatorname{Spec} B$ be affine schemes.*

(1.) The diagonal morphism $\Delta_{X/S}\colon X \to X \times_S X$ is the morphism corresponding to the ring homomorphism

$$\begin{aligned} \Delta_{A/R}\colon A \otimes_R A &\to A \\ a \otimes a' &\mapsto aa'. \end{aligned}$$

(2.) If $f\colon X \to Y$ is a morphism of schemes corresponding to a ring homomorphism $\varphi\colon B \to A$, then the graph of f is the morphism corresponding to the ring homomorphism

$$\begin{aligned} \Gamma_\varphi\colon A \otimes_R B &\to A \\ a \otimes b &\mapsto a\varphi(b). \end{aligned}$$

Corollary 9.7. *Let $S = \operatorname{Spec} R$, $X = \operatorname{Spec} A$, and $Y = \operatorname{Spec} B$ be affine schemes. Let $f\colon X \to Y$ be a morphism of affine schemes. Then the diagonal morphism and the graph of f are closed immersion.*

Proof. It is easily checked that the diagonal homomorphism $\Delta_{A/R}$ and the graph homomorphism Γ_φ are surjective. The conclusion follows from Corollary 4.14. □

In the general case of schemes, the diagonal homomorphism and the graph need not be closed immersions.

9.2 Separated Morphisms

Definition 9.8. A morphism of scheme $f : Y \to S$ is said to be **separated** if the diagonal $\Delta_{Y/S}$ is a closed immersion. We then say that Y is **separated over** S. If $Y \to \operatorname{Spec} \mathbb{Z}$ is separated, then Y is said to be a **separated scheme**.

Example 9.9. The affine line with double origin is not separated. Why?

Proposition 9.10. *Let $f : X \to Y$ be a morphism of affine schemes, then f is separated.*

Chapter 10

Proper Morphisms

Definition 10.1 (Closed and Universally Closed Morphisms)**.** Let $f\colon X \to Y$ be a morphism of schemes.

(1.) We say that f is **closed** if for every closed subset $Z \subseteq X$, its image $f(Z)$ is closed in Y. In other words, f is closed as a topological map.

(2.) The morphism f is **universally closed** if for every morphism $Y' \to Y$, the base change $X \times_Y Y' \to Y'$ is closed.

Definition 10.2. A morphism $f\colon X \to Y$ is **proper** if it is separated, of finite type, and universally closed.

Lemma 10.3. *Let* $\mathbf{P}$ *be a property of morphisms such that*

(1.) $\mathbf{P}$ *is stable under composition;*

(2.) $\mathbf{P}$ *is stable under base change;*

(3.) Every closed immersion has $\mathbf{P}$*.*

Proposition 10.4. *Closed immersions are proper.*

Chapter 11

Sheaves of Modules

11.1 Sheaves of Modules

Definition 11.1. Let $(X, \mathcal{O}_X)$ be a ringed space. A **sheaf of modules** on X or an **$\mathcal{O}_X$-module** is an Abelian sheaf $\mathcal{F}$ on X satisfying the following properties:

(1.) For each open subset U of X, $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$-module.

(2.) The module structures are compatible with restriction maps: if $V \subseteq U$ are open subsets, $a \in \mathcal{O}_X(U)$, and $s \in \mathcal{F}(U)$, then

$$(as)|_V = a|_V s|_V.$$

We may rephrase the definition of $\mathcal{O}_X$-modules in terms of product sheaves. Recall that if $\mathcal{F}$ and $\mathcal{G}$ are sheaves on a topological space X, the presheaf

$$U \mapsto \mathcal{F}(U) \times \mathcal{G}(U)$$

is a sheaf (Problem 3.4). Then an $\mathcal{O}_X$-module consists of a sheaf $\mathcal{F}$ on X, the *addition morphism* $\mathcal{F} \times \mathcal{F} \to \mathcal{F}$ defined for each open subset $U \subseteq X$ by

$$\begin{aligned} \mathcal{F}(U) \times \mathcal{F}(U) &\to \mathcal{F}(U) \\ (s, s') &\mapsto s + s', \end{aligned}$$

and the *multiplication morphism* $\mathcal{O}_X \times \mathcal{F} \to \mathcal{F}$ defined for each open subset $U \subseteq X$ by

$$\begin{aligned} \mathcal{O}_X(U) \times \mathcal{F}(U) &\to \mathcal{F}(U) \\ (a, s) &\mapsto as', \end{aligned}$$

such that $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$-module.

Example 11.2. The sheaf $\mathcal{O}_X$ is trivially an $\mathcal{O}_X$-module. More generally, if $i \in \mathbb{N}$, then the sheaf

$$U \mapsto \underbrace{\mathcal{O}_X(U) \oplus \cdots \oplus \mathcal{O}_X(U)}_{i \text{ summands}}$$

is an $\mathcal{O}_X$-module. We will generalise this construction later.

Let $\mathcal{F}$ and $\mathcal{G}$ be $\mathcal{O}_X$-modules. A **morphism of $\mathcal{O}_X$-modules** $\varphi\colon \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves such that for each open subset $U \subseteq X$, the map on

$$\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$$

is also an $\mathcal{O}_X(U)$-module homomorphism. The identity morphism $\mathrm{id}\colon \mathcal{F} \to \mathcal{F}$ is immediately seen to be a morphism of $\mathcal{O}_X$-modules. Moreover, the composition of two morphisms of $\mathcal{O}_X$-modules is again a morphism of $\mathcal{O}_X$-modules. We thus obtain the category $(\mathcal{O}_X\text{—Mod})$ of $\mathcal{O}_X$-modules. The set of all morphisms of $\mathcal{O}_X$-modules from $\mathcal{F}$ to $\mathcal{G}$ is denoted by $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$.

▸ **Exercise 11.3.** Show that $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ is naturally an $\mathcal{O}_X(X)$-module. In particular, the hom-set $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ is an Abelian group.

▸ **Exercise 11.4.** Show that, for each $\mathcal{O}_X$-module $\mathcal{F}$, the stalk $\mathcal{F}_x$ at $x \in X$ has the induced $\mathcal{O}_{X,x}$-module structure. The multiplication of $[U, a]$ and $[V, s]$ where U, V are open neighbourhood of x, $a \in \mathcal{O}_X(U)$ and $s \in \mathcal{F}(V)$, is defined to be

$$[U, a][V, s] = [U \cap V, a|_{U\cap V} s|_{U\cap V}].$$

Show that if $\varphi\colon \mathcal{F} \to \mathcal{G}$ is a morphism of $\mathcal{O}_X$-modules, then the map of stalks $\varphi_x\colon \mathcal{F}_x \to \mathcal{G}_x$ is a homomorphism of $\mathcal{O}_{X,x}$-modules.

The result of Exercise 11.4 implies that taking stalks defines a functor from $(\mathcal{O}_X\text{-Mod})$ to $(\mathcal{O}_{X,x}\text{-Mod})$.

Definition 11.5. Let $(X, \mathcal{O}_X)$ be a locally ringed space, $\mathcal{F}$ an $\mathcal{O}_X$-module, and $x \in X$. By Exercise 11.4, the stalk $\mathcal{F}_x$ is an $\mathcal{O}_{X,x}$-module. The tensor product

$$\mathcal{F}(x) := \mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x),$$

where $\kappa(x)$ is the residue class field of x, is an $\kappa(x)$-vector space. This is called the **stalk of $\mathcal{F}$ at x.**

By definition the stalk at x of $\mathcal{O}_X$, viewed as an $\mathcal{O}_X$-module, is the local ring $\mathcal{O}_{X,x}$ itself.

We now turn to more examples of $\mathcal{O}_X$-modules. For the first example, recall that a *zero object* in a category $\mathcal{C}$ is an object which is both initial and final.

Example 11.6 (The Zero Sheaf)**.** The sheaf 0 defined by $0(U) = \{0\}$ for all open subset $U \subseteq X$ is an $\mathcal{O}_X$-module. For every $\mathcal{O}_X$-module $\mathcal{F}$, there are unique morphisms $\mathcal{F} \to 0$ and $0 \to \mathcal{F}$ making 0 the zero object in $(\mathcal{O}_X\text{—Mod})$.

Example 11.7. Let A be a ring. Let X be a space that consists of a single point and let $\mathcal{O}_X$ be the sheaf of rings with $\mathcal{O}_X(X) = A$. Then an $\mathcal{O}_X$-module $\mathcal{F}$ is just an A-module M.

Example 11.8. Here is an example from differential geometry. Let X be a smooth real manifold, and let $C^\infty(X)$ be the sheaf of rings of smooth functions on X. For each $i \geq 0$, let $\Omega^i(X)$ be the sheaf of smooth differential forms of degree i on X. Then $\Omega^i(X)$ is a $C^\infty(X)$-module.

Proposition 11.9 (Sheaves of Abelian Groups as $\underline{\mathbb{Z}}$-modules)**.** *Let X be a topological space and let $\underline{\mathbb{Z}}$ be the constant sheaf of rings on X with value $\mathbb{Z}$. Then sheaves of Abelian groups are the same thing as $\underline{\mathbb{Z}}$-modules.*

Proof. Plainly, every $\underline{\mathbb{Z}}$-module is a sheaf of Abelian groups. On the other hand, suppose $\mathcal{F}$ is a sheaf of Abelian groups. For each open set U of X, a section $a \in \underline{\mathbb{Z}}(U)$ is a locally constant function $U \to \mathbb{Z}$. Therefore the collection $\{a^{-1}(n)\}_{n\in\mathbb{Z}}$ is an open cover of U by disjoint open sets. We define the scalar multiplication by $a \in \underline{\mathbb{Z}}$ on $s \in \mathcal{F}(U)$ by

$$as := \text{ the section obtained from gluing } n \cdot (s|_{a^{-1}(n)}) \text{ over the covering } \{a^{-1}(n)\}_{n\in\mathbb{Z}}.$$

Because our covering is disjoint, the sections $n \cdot (s|_{a^{-1}(n)})$ automatically satisfy the gluing condition, hence as is well-defined.

Finally we check that this gives rise to a morphism of sheaves $\underline{\mathbb{Z}} \times \mathcal{F} \to \mathcal{F}$. This is equivalent to checking that, if $U \subseteq V$ are open subsets of X and $(a, s) \in \underline{\mathbb{Z}} \times \mathcal{F}(V)$, then

$$\operatorname{res}^V_U(as) = a|_U s|_U. \tag{11.1}$$

This can be locally checked on the covering $\{a^{-1}(n) \cap U\}_{n\in\mathbb{Z}}$ of U. On $a^{-1}(n) \cap U$, we have

$$\operatorname{res}^U_{a^{-1}(n)\cap U} \circ \operatorname{res}^V_U(as) = \operatorname{res}^V_{a^{-1}(n)\cap U}(as) = n \cdot (s|_{a^{-1}(n)\cap U}).$$

This follows because $as|_{a^{-1}(n)} = n \cdot s|_{a^{-1}(n)}$ by definition, hence the result after restricting this section further to $a^{-1}(n) \cap U$.

For the right hand side of the equation, we see that $a|_U$ is obtained by restricting the domain of a from V to U. The resulting function $a|_U$ is therefore constant on each open set in the covering $\{a^{-1}(n) \cap U\}_{n\in\mathbb{Z}}$. Hence $a|_U s|_U$ is, by definition, the section obtained from gluing $n \cdot (s|_U)|_{a^{-1}(n)\cap U}$. It then follows immediately that

$$(a|_U s|_U)|_{a^{-1}(n)\cap U} = n \cdot (s|_{a^{-1}\cap U}).$$

As a consequence, the restrictions of the sections on each side of 11.1 are equal on $a^{-1}(n) \cap U$ for all n. Thus 11.1 holds by the sheaf condition.

We left the reader to verify that this multiplication satisfies the module axioms. The proof involves checking the axioms on good covering then gluing the sections back just like the one above. This establishes the fact that a sheaf of Abelian groups is the same as a $\underline{\mathbb{Z}}$-module. □

Definition 11.10. Let X be a scheme and $\mathcal{F}$ an $\mathcal{O}_X$-module. An $\mathcal{O}_X$-**submodule** of $\mathcal{F}$ is an $\mathcal{O}_X$-module $\mathcal{G}$ such that $\mathcal{G}(U)$ is an $\mathcal{O}_X(U)$-submodule of $\mathcal{F}(U)$ for all open $U \subseteq X$.

The inclusion $i\colon \mathcal{G} \to \mathcal{F}$ clearly defines a morphism of $\mathcal{O}_X$-modules. In the case where $\mathcal{F} = \mathcal{O}_X$, an $\mathcal{O}_X$-submodule of $\mathcal{O}_X$ is called an **ideal** of $\mathcal{O}_X$.

11.2 Constructions on $\mathcal{O}_X$-modules

In this section we discuss the construction of various $\mathcal{O}_X$-modules. This is in parallel to the theory of modules over a ring.

Remark 11.11. Clearly there is also the notion of a **presheaf of modules**, i.e. a sheaf $\mathcal{F}$ such that for all open subsets $U \subseteq X$, $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$-module and the module strictures are compatible with restriction maps. The sheafification of a presheaf of modules inherits the $\mathcal{O}_X$-module structure, so that the sheafification is then a sheaf of modules.

In general, some constructions, such as the image presheaf or the cokernel presheaf of a morphism of $\mathcal{O}_X$-modules, fail to be sheaves. In this case we shall sheafify to obtain the sheaves.

Kernel, Images and Cokernels

Let X be a scheme and $w\colon \mathcal{F} \to \mathcal{F}'$ be a morphism of $\mathcal{O}_X$-modules.

(1.) The **kernel** of w is the sheaf $U \mapsto \ker(w_u\colon \mathcal{F}(U) \to \mathcal{F}'(U))$ denoted by $\ker(w)$.

(2.) The **image** of w is the sheaf $\operatorname{im}(w)$ associated to the presheaf $U \mapsto \operatorname{im}(w_u\colon \mathcal{F}(U) \to \mathcal{F}'(U))$.

(3.) The **cokernel** of w is the sheaf $\operatorname{coker}(w)$ associated to the presheaf $U \mapsto \operatorname{coker}(w_u\colon \mathcal{F}(U) \to \mathcal{F}'(U))$.

Since the stalk at $x \in X$ of an associated sheaf agrees with the stalk at x of the original presheaf, we deduce the following proposition.

Proposition 11.12. *Let X be a scheme and $w\colon \mathcal{F} \to \mathcal{F}'$ be a morphism of $\mathcal{O}_X$-modules. Then*

$$\ker(w)_x = \ker(w_x), \quad \operatorname{im}(w)_x = \operatorname{im}(w_x), \quad \textit{and } \operatorname{coker}(w)_x = \operatorname{coker}(w_x).$$

Quotient Modules

Let X be a scheme, and let $\mathcal{G}$ be an $\mathcal{O}_X$-submodule of $\mathcal{F}$. We define the **quotient module** of $\mathcal{F}$ by $\mathcal{G}$ to be the sheaf $\mathcal{F}/\mathcal{G}$ associated to the presheaf

$$U \mapsto \mathcal{F}(U)/\mathcal{G}(U).$$

It is an $\mathcal{O}_X$-module via the addition and scalar multiplication induced from $\mathcal{F}$.

Let's determine the stalk at $x \in X$ of the quotient module $\mathcal{F}/\mathcal{G}$. Again, we may instead compute the stalk of the presheaf $U \mapsto \mathcal{F}(U)/\mathcal{G}(U)$. For all $x \in X$, we have

$$\varinjlim_{D(f)\ni x} \mathcal{F}(U)/\mathcal{G}(U) = \mathcal{F}_x/\mathcal{G}_x.$$

Hence $(\mathcal{F}/\mathcal{G})_x = \mathcal{F}_x/\mathcal{G}_x$ for all $x \in X$.

Proposition 11.13. *Every morphism $w\colon \mathcal{F} \to \mathcal{F}'$ of $\mathcal{O}_X$-modules give rise to canonical isomorphisms*

$$\mathcal{F}/\ker(w) \xrightarrow{\sim} \operatorname{im}(w) \quad \textit{and} \quad \mathcal{F}'/\operatorname{im}(w) \xrightarrow{\sim} \operatorname{coker}(w).$$

Exact Sequences

A sequence

$$\mathcal{F} \xrightarrow{w} \mathcal{F}' \xrightarrow{w'} \mathcal{F}''$$

of $\mathcal{O}_X$-modules is said to be **exact** if $\operatorname{im}(w) = \ker(w')$. Equivalently, the sequence is exact if and only if the induced sequence of $\mathcal{O}_{X,x}$-modules

$$\mathcal{F}_x \xrightarrow{w_x} \mathcal{F}'_x \xrightarrow{w'_x} \mathcal{F}''_x$$

is exact for all $x \in X$.

An infinite sequence of $\mathcal{O}_X$-modules indexed by $\mathbb{N}$

$$\cdots \to \mathcal{F}_{i-2} \to \mathcal{F}_{i-1} \xrightarrow{w_{i-1}} \mathcal{F}_i \xrightarrow{w_i} \mathcal{F}_{i+1} \to \mathcal{F}_{i+2} \to \cdots$$

is exact if $\mathcal{F}_{i-1} \xrightarrow{w_{i-1}} \mathcal{F}_i \xrightarrow{w_i} \mathcal{F}_{i+1}$ is exact for all $i \in \mathbb{N}$. A finite sequence of $\mathcal{O}_X$-modules can be viewed as an infinite sequence by padding zero modules.

Module of Homomorphisms

Given two homomorphisms of $\mathcal{O}_X$-modules $w, w'\colon \mathcal{F} \to \mathcal{F}'$, their sum $w+w'$ is the obvious morphism defined by pointwise addition on each open set:

$$(w + w')_U := w_U + w'_U\colon \mathcal{F}(U) \to \mathcal{F}'(U).$$

Now suppose $a \in \Gamma(X, \mathcal{O}_X)$. Then the multiplication aw is defined by

$$(aw)_U : (a|_U)w|_U : \mathcal{F}(U) \to \mathcal{F}'(U).$$

In this way, the set of $\mathcal{O}_X$-modules homomorphism $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{F}')$ has the structure of an $\Gamma(X, \mathcal{O}_X)$-module.

Proposition 11.14 (Left Exactness of Hom)**.** *A sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}''$ of $\mathcal{O}_X$-modules is exact if and only if for all open subsets $U \subseteq X$ and for all $\mathcal{O}_U$-modules $\mathcal{G}$, the sequence*

$$0 \to \mathrm{Hom}(\mathcal{G}, \mathcal{F}'|_U) \to \mathrm{Hom}(\mathcal{G}, \mathcal{F}|_U) \to \mathrm{Hom}(\mathcal{G}, \mathcal{F}''|_U)$$

is an exact sequence of $\Gamma(U, \mathcal{O}_X)$-modules. Dually, $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}''$ is exact if and only if for all open subsets $U \subseteq X$ and for all $\mathcal{O}_U$-modules $\mathcal{G}$, the sequence

$$0 \to \mathrm{Hom}(\mathcal{F}''|_U, \mathcal{G}) \to \mathrm{Hom}(\mathcal{F}|_U, \mathcal{G}) \to \mathrm{Hom}(\mathcal{F}'|_U, \mathcal{G})$$

is an exact sequence of $\Gamma(U, \mathcal{O}_X)$-modules.

This result also follows from the category of sheaves of modules being Abelian (proven below) combined with the general fact that the Hom functor on an Abelian category is left exact.

Proof. Should I prove this? □

Direct Sums and Products

Let $(\mathcal{F}_i)_{i \in I}$ be a family of $\mathcal{O}_X$-modules. Then we define the product presheaf by

$$U \mapsto \prod_{i \in I} \mathcal{F}_i(U).$$

It is actually a sheaf, and so we obtain the **product sheaf** of $(\mathcal{F}_i)_{i \in I}$, which we shall denote by $\prod_i \mathcal{F}_i$. Similarly there is the direct sum presheaf defined by

$$U \mapsto \bigoplus_{i \in I} \mathcal{F}_i(U).$$

However this is in general not a sheaf. Therefore the **direct sum sheaf** of $(\mathcal{F}_i)_{i \in I}$ is defined to be the associated sheaf of this presheaf, denoted by $\oplus_i \mathcal{F}_i$. The component-wise addition and scular multiplication endow those sheaves with the structure of $\mathcal{O}_X$-modules. There is a natural injective homomorphism

$$\bigoplus_{i \in I} \mathcal{F}_i \to \prod_{i \in I} \mathcal{F}_i.$$

This homomorphism is an isomorphism if I is finite.

In the important case when $\mathcal{F}_i = \mathcal{F}$ for all i, we shall use the notation $\mathcal{F}^{(I)}$ for the direct sum $\oplus_i \mathcal{F}_i$, and $\mathcal{F}^I$ for the product sheaf $\prod_i \mathcal{F}_i$.

Stalks of direct sums and product sheaves.

Theorem 11.15. *The category of $\mathcal{O}_X$-modules on a scheme X is Abelian.*

Tensor Products

Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}_X$-modules. We have the presheaf

$$U \mapsto \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$$

whose restriction maps are tensor products of the restriction maps of $\mathcal{F}$ and $\mathcal{G}$. This presheaf has the obvious addition and scalar multiplication making it into a presheaf of modules. However, it is in general not a sheaf, and so we sheafify it to get the **tensor product sheaf**, denoted $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$.

Remark 11.16. The tensor product sheaf $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ can be characterised by a universal property similar to the one for tensor products of R-modules, although we shall not spell it out here.

For each $s \in \mathcal{F}(U)$ and $t \in \mathcal{G}(U)$ we denote again by $s \otimes t$ the image of $s \otimes t \in \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$ in $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})(U)$.

Proposition 11.17 (Stalks of the Tensor Product Sheaf)**.** *For all $x \in X$, the canonical homomorphism of $\mathcal{O}_{X,x}$-modules*

$$(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})_x \to \mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{G}_x$$

is an isomorphism.

This follows from the fact that inductive limits commute with tensor products. (Recall that the tensor product functor is left exact, hence it preserves colimits.)

We define the n-times tensor product of $\mathcal{F}$ ($n \geq 0$) by the formula

$$\begin{aligned} \mathcal{F}^{\otimes 0} &:= \mathcal{O}_X, \\ \mathcal{F}^{\otimes n} &:= \underbrace{\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{F} \otimes_{\mathcal{O}_X} \cdots \otimes_{\mathcal{O}_X} \mathcal{F}}_{n \text{ terms}}. \end{aligned}$$

11.3 Direct Image and Inverse Image

11.4 Modules Generated by Sections

For a module M over a ring A, an A-module homomorphism $\varphi\colon A \to M$ is completely determined by the image of 1 under φ, because $\varphi(r) = r\varphi(1)$ for all $r \in A$. There is an analogous statement for $\mathcal{O}_X$-modules.

Proposition 11.18. *Let X be a scheme and let $\mathcal{F}$ be an $\mathcal{O}_X$-module. There is an isomorphism of $\Gamma(X, \mathcal{O}_X)$-modules*

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{F}) &\xrightarrow{\sim} \Gamma(X, \mathcal{F}) \\ w &\mapsto w_X(1) \end{aligned}$$

which is functorial in $\mathcal{F}$.

Proof. It is clear that this map is $\Gamma(X, \mathcal{O}_X)$-linear and functorial in $\mathcal{F}$. Now let $s \in \Gamma(X, \mathcal{F})$. Define a morphism $\psi_s \colon \mathcal{O}_X \to \mathcal{F}$ so that each morphism $\psi_{s,U} \colon \mathcal{O}_X(U) \to \mathcal{F}(U)$ is the unique homomorphism of $\mathcal{O}_X(U)$-modules determined by $\psi_{s,U}(1) = s|_U$. It is easy to see that ψ_s is in fact a morphism of $\mathcal{O}_X$-modules, and that this gives the requisite bijection of the proposition. □

Let I be any index set. Then we have the isomorphism of $\mathcal{O}_X$-modules

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X^{(I)}, \mathcal{F}) = \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{F})^I \xrightarrow{\sim} \Gamma(X, \mathcal{F})^I$$

which is functorial in $\mathcal{F}$. Note that this says that if we have some global sections $(s_i)_{i \in I}$ of $\mathcal{F}(X)$, then we can define a morphism $\mathcal{O}_X^{(I)} \to \mathcal{F}$.

Definition 11.19. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module, and $(s_i)_{i \in I}$ a family of sections in $\Gamma(X, \mathcal{F})$.

(1.) $\mathcal{F}$ is **generated by the sections** (s_i) if the corresponding morphism $\mathcal{O}_X^{(I)} \to \mathcal{F}$ is an epimorphism.

(2.) $\mathcal{F}$ is **generated by global sections** if there exists an epimorphism $\mathcal{O}_X^{(I)} \to \mathcal{F}$ for some index set I.

(3.) $\mathcal{F}$ is **locally generated by sections** if for each $x \in X$, there is an open neighbourhood U of x such that $\mathcal{F}|_U$ is generated by global sections.

Lemma 11.20. *An $\mathcal{O}_X$-module $\mathcal{F}$ is generated by global sections if and only if for all $x \in X$, the canonical homomorphism $\mathcal{F}(X) \otimes_{\mathcal{O}_X(X)} \mathcal{O}_{X,x} \to \mathcal{F}_x$ is surjective.*

Proof. content... □

Module of Homomorphisms and Dual Modules

Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}_X$-modules. The presheaf

$$U \mapsto \mathrm{Hom}_{\mathcal{O}_{X|U}}(\mathcal{F}|_U, \mathcal{G}|_U)$$

is a sheaf (This is the Sheaf Gluing Lemma, Lemma 3.45). Clearly $\mathrm{Hom}_{\mathcal{O}_{X|U}}(\mathcal{F}|_U, \mathcal{G}|_U)$ has the structure of an $\Gamma(U, \mathcal{O}_X)$-module. Thus this is an $\mathcal{O}_X$-module which we shall promptly denote it by $\mathscr{H}\mathrm{om}_{\mathcal{O}_{X|U}}(\mathcal{F}, \mathcal{G})$.

Definition 11.21. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Its **dual $\mathcal{O}_X$-module** $\mathcal{F}^\vee$ is defined to be the $\mathcal{O}_X$-module $\mathscr{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{O}_X)$.

11.5 Finite Locally Free Modules

The most important class of $\mathcal{O}_X$-modules are the locally free modules. As the name suggests, these are $\mathcal{O}_X$-modules that are locally isomorphic to $\mathcal{O}_X^{(I)}$. Here is the more precise definition.

Definition 11.22. Let X be a ringed space. An $\mathcal{O}_X$-module $\mathcal{F}$ is **locally free** if for all $x \in X$ there exists an open neighbourhood U of x such that $\mathcal{F}|_U$ is isomorphic to an $\mathcal{O}_{X|U}$-module of the form $\mathcal{O}_{X|U}^{(I)}$ for some index set I (I may depend on x).

If, in the above definition, the index sets I are all finite, then we say that $\mathcal{F}$ is **finite locally free** or **locally free of finite type**.

If $\mathcal{F} \cong \mathcal{O}_X^{(I)}$, then we say that $\mathcal{F}$ is **free**. The **rank** of $\mathcal{F}$ is equal to the (finite or infinite) cardinality of I, and so the rank of a free $\mathcal{O}_X$-module is either a non-negative integer or ∞.

In general, if $\mathcal{F}$ is only locally free, then the cardinality of I might vary on X. However, if $\mathcal{F}|_U$ is isomorphic to $\mathcal{O}_{X|U}^{(I)}$ on an open subset U, then for each $x \in U$, the stalk $\mathcal{F}_x$ is isomorphic to $\mathcal{O}_{X,x}^{(I)}$ as an $\mathcal{O}_{X,x}$-module. Recall that the rank of a free module is well-defined. In particular, this implies that the cardinality of I depends only on $\mathcal{F}$ and x. Thus we define the **rank** of $\mathcal{F}$ to be the function $\mathrm{rk}(\mathcal{F}) \colon X \to \mathbb{N} \cup \{\infty\}$ sending x to the rank of the module $\mathcal{F}_x$. This function is clearly locally constant. Furthermore, if $\mathcal{F}$ is a free $\mathcal{O}_X$-module such that $\mathcal{F} \cong \mathcal{O}_X^{(I)}$, then its rank is the constant function $x \mapsto |I|$.

We are mostly interested in $\mathcal{O}_X$-modules of constant finite rank, i.e. $\mathrm{rk}(\mathcal{F})$ is a constant function $X \to \mathbb{N}$.

Definition 11.23. An $\mathcal{O}_X$-module $\mathcal{F}$ is said to be **locally free of rank n** (also called a **vector bundle** of rank n) if $\mathrm{rk}(\mathcal{F})$ is the constant function $x \mapsto n$.

A few remarks is in order. Firstly, the definition of vector bundles is different, but it turns out that locally free $\mathcal{O}_X$-module of rank n is the same thing as a vector bundle of rank n. It is for this reason that a locally free $\mathcal{O}_X$-module of rank 1 is also called a **line bundle**.

Secondly, we may rephrase the definition above slightly by saying that $\mathcal{F}$ is locally free of rank n if for all $x \in X$, $\mathcal{F}_x$ is isomorphic to $\mathcal{O}_{X,x}^{(n)}$ as an $\mathcal{O}_{X,x}$-module.

Proposition 11.24. *Let $\mathcal{F}$ be an $\mathcal{O}_X$-module locally free of rank n and $\mathcal{G}$ an $\mathcal{O}_X$-module locally free of rank m. Then the $\mathcal{O}_X$-module $\mathscr{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{F},\mathcal{G})$ is locally free of rank mn.*

Proof. Let $x \in X$. Then there exists an open neighbourhood U of x such that $\mathcal{F}|_U \cong \mathcal{O}_{X|U}^{(n)}$ and $\mathcal{G}|_U \cong \mathcal{O}_{X|U}^{(m)}$. It follows that

$$\begin{aligned}\mathscr{H}\mathrm{om}_{\mathcal{O}_{X|U}}(\mathcal{F}|_U,\mathcal{G}|_U) &\cong \mathscr{H}\mathrm{om}_{\mathcal{O}_{X|U}}(\mathcal{O}_{X|U}^{(n)},\mathcal{O}_{X|U}^{(m)})\\ &\cong (\mathcal{O}_{X|U}^{(m)})^{(n)}\\ &\cong \mathcal{O}_{X|U}^{(mn)}.\end{aligned}$$

Hence the rank of $\mathscr{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{F},\mathcal{G})$ at x is mn. Since this holds for all x, this proves the proposition. □

In particular, if $\mathcal{F}$ is locally free of rank n, then its dual $\mathcal{F}^\vee = \mathscr{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{F},\mathcal{O}_X)$ is also locally free of rank n.

Line Bundles as Invertible $\mathcal{O}_X$-Modules

TODO.

11.6 Sheaves of Modules Attached to Modules

In this section we show that modules over a ring give rise to sheaves of modules over the spectrum of that ring. Let A be a ring and M an A-module. Set $X = \operatorname{Spec} A$. Recall that the collection

$$\mathcal{B} = \{D(f) : f \in A\}$$

of principal open subsets of X is a basis of X. Define the sheaf $\tilde{M}$ on each principal open set $D(f) \subseteq X$ ($f \in A$) by the formula

$$\tilde{M}(D(f)) = M_f,$$

where M_f is the localisation of M with respect to the multiplicative set $\{f^i : i \geq 0\}$. If $D(f) \subseteq D(g)$, then the restriction map $\tilde{M}(D(g)) \to \tilde{M}(D(f))$ is taken to be the homomorphism of A_g-modules $M_g \to M_f$ obtained from the universal property of localisation. The same argument as in the definition of affine schemes show that $\tilde{M}$ is indeed a sheaf on the basis $\mathcal{B}$.

▸ **Exercise 11.25.** The assignment $\tilde{M}$ is a sheaf on the basis $\mathcal{B}$.

Hence $\tilde{M}$ extends to a sheaf on X which we again denote by $\tilde{M}$. As is readily seen from the construction, $\tilde{M}(D(f)) = M_f$ is a module over $\mathcal{O}_X(D(f)) = A_f$. In fact, the module structure extends to the whole of $\tilde{M}$, rendering $\tilde{M}$ into an $\mathcal{O}_X$-module.

Lemma 11.26. *Let $U \subseteq \operatorname{Spec} A$ be an open subset. Then $\tilde{M}(U)$ carries an $\mathcal{O}_X(U)$-module structure making $\tilde{M}$ into an $\mathcal{O}_X$-module.*

Proof. Let $a \in \mathcal{O}_X(U)$ and $s \in \tilde{M}(U)$. We define the multiplication as locally on each principal open subset of U and then show that these local sections can be glued together. Let $D(f) \subseteq U$, $f \in A$ be a principal open subset. Then $a|_{D(f)} \in \mathcal{O}_X(D(f)) = A_f$ and $s|_{D(f)} \in \tilde{M}(D(f)) = M_f$. In particular, the multiplication $a|_{D(f)}s|_{D(f)}$ is defined plainly as the scalar multiplication of M_f.

In order to glue these sections together, we have to check that for two principal open subsets $D(f)$ and $D(g)$ of U,

$$(a|_{D(f)}s|_{D(f)})|_{D(f)\cap D(g)} = (a|_{D(g)}s|_{D(g)})|_{D(f)\cap D(g)}.$$

Recall that $D(f) \cap D(g) = D(fg)$, and so the restriction map $\tilde{M}(D(f)) \to \tilde{M}(D(fg))$ is simply the A_f-module homomorphism $M_f \to M_{fg}$. Similarly the restriction map $\tilde{M}(D(f)) \to \tilde{M}(D(fg))$ is the A_g-module homomorphism $M_g \to M_{fg}$. Using the fact that the restriction maps are module homomorphisms, we get

$$\begin{aligned}(a|_{D(f)}s|_{D(f)})|_{D(f)\cap D(g)} &= (a|_{D(f)})|_{D(f)\cap D(g)}(s|_{D(f)})|_{D(f)\cap D(g)} \\ &= a|_{D(fg)}s|_{D(fg)} \\ &= (a|_{D(g)})|_{D(f)\cap D(g)}(s|_{D(g)})|_{D(f)\cap D(g)} \\ &= (a|_{D(g)}s|_{D(g)})|_{D(f)\cap D(g)}.\end{aligned}$$

Hence the sections $a|_{D(f)}s|_{D(f)}$, where $D(f)$ runs over all principal open subsets of U, glue to a unique section $as \in \tilde{M}(U)$. The module axioms for $\tilde{M}(U)$ and the fact that the module structures are compatible with restriction maps can be checked locally in the similar manner as the above. □

We remark that, by viewing A as an A-module, $\tilde{A} = \mathcal{O}_X$. Another easy consequence of the above construction is the stalks of $\tilde{M}$ are simply localisation of M at prime ideals.

Lemma 11.27. *The stalk of $\tilde{M}$ at $x \in X$ is equal to $M_{\mathfrak{p}_x}$.*

Proof. Follows from

$$\tilde{M}_x = \varinjlim_{D(f)\ni x} \Gamma(D(f), \tilde{M}) = \varinjlim_{f\notin \mathfrak{p}_x} M_f = M_{\mathfrak{p}_x}.$$

□

Suppose $\varphi\colon M \to N$ is a homomorphism of A-modules. Then we have the induced maps $\varphi_f\colon M_f \to N_f$ of A_f-modules for all $f \in A$. These morphism together define a

morphism of sheaves $\tilde{\varphi}\colon \tilde{M} \to \tilde{N}$. Indeed, if $D(f) \subseteq D(g)$, then we have a commutative diagram

$$\begin{array}{ccc} M_f & \xrightarrow{\varphi_f} & N_f \\ \downarrow & & \downarrow \\ M_g & \xrightarrow[\varphi_g]{} & N_g \end{array}$$

which proves the claim. Moreover, since each φ_f is an A_f-module homomorphism, $\tilde{\varphi}$ is in fact a morphism of $\mathcal{O}_X$-modules. It is then clear that $M \mapsto \tilde{M}$ defines a functor from the category of A-modules to the category of $\mathcal{O}_X$-modules.

Theorem 11.28. *Let A be a ring and $X = \operatorname{Spec} A$. Denote by Γ the global section functor. Then there is a bijection*

$$\operatorname{Hom}_{A-\mathrm{Mod}}(M, N) \underset{\Gamma(-)}{\overset{M \mapsto \tilde{M}}{\rightleftarrows}} \operatorname{Hom}_{\mathcal{O}_X}(\tilde{M}, \tilde{N})$$

Consequently $M \mapsto \tilde{M}$ is fully faithful.

Proof. It follows form the definitions that $\Gamma(\tilde{\varphi}) = \varphi$. Now let $w\colon \tilde{M} \to \tilde{N}$ be a morphism of $\mathcal{O}_X$-modules, and set $u := \Gamma(w)\colon M \to N$. For each $f \in A$, consider the morphism $w_{D(f)}\colon \tilde{M}(D(f)) \to \tilde{N}(D(f))$ which gives us the commutative diagram

$$\begin{array}{ccc} M & \xrightarrow{\qquad u \qquad} & N \\ \downarrow & & \downarrow \\ \tilde{M}(D(f)) = M_f & \xrightarrow{w_{D(f)}} & \tilde{N}(D(f)) = N_f. \end{array}$$

But the lower horizontal homomorphism is also equal to u_f by the universal property of localisation. Hence $u_f = w_{D(f)}$ for all f, and so $\tilde{u} = w$. □

Theorem 11.29 (The Functor Associating $\mathcal{O}_X$-modules is Exact)**.** *Let A be a ring and $X = \operatorname{Spec} A$.*

(1.) A sequence of A-modules $M \to N \to P$ is exact if and only if the sequence of $\mathcal{O}_X$-modules $\tilde{M} \to \tilde{N} \to \tilde{P}$ is exact.

(2.) Let $\varphi\colon M \to N$ be a homomorphism of A-modules. Then $\ker(\varphi)^\sim \cong \ker(\tilde{\varphi})$, $\operatorname{im}(\varphi)^\sim \cong \operatorname{im}(\tilde{\varphi})$ and $\operatorname{coker}(\varphi)^\sim \cong \operatorname{coker}(\tilde{\varphi})$.

(3.) If $\varphi\colon M \to N$ be a homomorphism of A-modules, then u is injective (resp. surjective, resp. bijective) if and only if $\tilde{u}$ is injective (resp. surjective, resp. bijective).

(4.) Let $(M_i)_{i\in I}$ be a family of A-modules. Then

$$\bigoplus_{i\in I} \tilde{M}_i = \left(\bigoplus_{i\in I} M_i\right)^\sim.$$

(5.) Let $(M_\lambda)_{\lambda\in\Lambda}$ be a direct system of A-modules. Then the direct limit of the direct system $(\tilde{M}_\lambda)_{\lambda\in\Lambda}$ exists and is given by

$$\lim_{\lambda\in\Lambda} \tilde{M}_i = \left(\lim_{\lambda\in\Lambda} M_i\right)^\sim.$$

Proof. By commutative algebra, the sequence $M \to N \to P$ is exact if and only if for all $x \in X$ the sequence $M_{\mathfrak{p}_x} \to N_{\mathfrak{p}_x} \to P_{\mathfrak{p}_x}$ is exact. The latter statement is equivalent to $\tilde{M} \to \tilde{N} \to \tilde{P}$ being exact by Lemma 11.27. This proves (1.). The assertions (2.) and (3.) are immediate consequences of (1.).

(4.) TODO. □

Remark 11.30. Let A be a ring. Every A-module M gives rise to an exact sequence

$$A^{(J)} \to A^{(I)} \to M \to 0.$$

Hence by taking the associated $\mathcal{O}_{\operatorname{Spec} A}$-modules we obtain the exact sequence

$$\mathcal{O}_X^{(J)} \to \mathcal{O}_X^{(I)} \to \tilde{M} \to 0.$$

The exactnesss is the consequence of Theorem 11.29.

11.7 Quasi-coherent Modules

In the previous section, we saw that the functor $M \mapsto \tilde{M}$ is fully faithful. The next goal is to identify the essential image of this functor and obtain the subcategory of the category of $\mathcal{O}_{\operatorname{Spec} A}$-modules which is equivalent to the category of A-modules. Remark 11.30 gives

a necessary condition for an $\mathcal{O}_X$-module $\mathcal{F}$ to be isomorphic to $\tilde{M}$ for some A-module M, that is, it must fit in an exact sequence of the form

$$\mathcal{O}_X^{(J)} \to \mathcal{O}_X^{(I)} \to \mathcal{F} \to 0.$$

This leads to the following definition.

Definition 11.31. Let $(X, \mathcal{O}_X)$ be a ringed space. An $\mathcal{O}_X$-module $\mathcal{F}$ is **quasi-coherent** if for all $x \in X$, there exists an open neighbourhood U of x and an exact sequence of $\mathcal{O}_{X|U}$-modules

$$\mathcal{O}_{X|U}^{(J)} \longrightarrow \mathcal{O}_{X|U}^{(I)} \longrightarrow \mathcal{F}|_U \to 0,$$

where I and J are arbitrary index sets depending on x.

Example 11.32. The structure sheaf $\mathcal{O}_X$ is quasi-coherent because the sequence

$$0 \to \mathcal{O}_X \xrightarrow{\text{id}} \mathcal{O}_X \to 0$$

is exact.

It turns out that this condition is already sufficient for $X = \operatorname{Spec} A$. An $\mathcal{O}_X$-module $\mathcal{F}$ is isomorphic to $\tilde{M}$ for some A-module M if and only if $\mathcal{F}$ is quasi-coherent. We need the following remark.

Remark 11.33. Let X be any locally ringed space, and $f \in \Gamma(X, \mathcal{O}_X)$. Recall the definition of X_f as given in Lemma 3.49:

$$X_f = \{x \in X : f(x) \neq 0\}$$

and that X_f is an open subset of X. We claim that the image of f under the restriction map $\Gamma(X, \mathcal{O}_X) \to \Gamma(X_f, \mathcal{O}_X)$ is invertible. As show in the proof of Lemma 3.49, at each point $x \in X$, there is an open neighbourhood V of x and a section $g \in \mathcal{O}_X(V)$ such that $f|_V g = 1$, so we have glue these sections g together. Suppose that we have $y \in X_f$, W an open neighbourhood of y and $h \in \mathcal{O}_X(W)$ such that $f|_W h = 1$. Then on $W \cap V$,

$$f|_{W\cap V} g|_{W\cap V} = 1|_{W\cap V} = f|_{W\cap V} h|_{W\cap V},$$

and so the sections indeed glue to a section $g \in \mathcal{O}_X(X_f)$ such that $fg = 1$.

It follows from the above remark that the restriction $\Gamma(X, \mathcal{O}_X) \to \Gamma(X_f, \mathcal{O}_X)$ induces a homomorphism of $\Gamma(X, \mathcal{O}_X)$-modules

$$\Gamma(X, \mathcal{O}_X)_f \to \Gamma(X_f, \mathcal{O}_X). \tag{11.2}$$

In the case where $X = \operatorname{Spec} A$, the open set X_f is simply the principal open set $D(f)$. For an A-module M and the $\mathcal{O}_X$-module $\mathcal{F} = \tilde{M}$, the morphism of Eq. (11.2) is simply the identity map $M_f \to M_f$.

Theorem 11.34. *Let X be a scheme and let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Then the following are equivalent*

(1.) $\mathcal{F}$ is quasi-coherent.

(2.) For every open affine subset $U = \operatorname{Spec} A$ of X, there exists an A-module M such that $\mathcal{F}|_U \cong \tilde{M}$.

(3.) There exists an open covering by subschemes $U_i = \operatorname{Spec} A_i$, $i \in I$, of X and a family of A_i-modules M_i such that $\mathcal{F}|_{U_i} \cong \tilde{M}_i$ for all i.

(4.) For every open affine subset $U = \operatorname{Spec} A$ of X and every $f \in A$, the homomorphism in Eq. (11.2)

$$\Gamma(U, \mathcal{F})_f \to \Gamma(D(f), \mathcal{F})$$

is an isomorphism.

Proof. It is easy to see that (2.) $\Rightarrow$ (3.) and (3.) $\Rightarrow$ (1.).

We show that (4.) $\Rightarrow$ (2.). We may assume that $X = \operatorname{Spec} A$ so that $\Gamma(X, \mathcal{O}_X) = A$. Take $M = \Gamma(X, \mathcal{F})$ which is an A-module. The statement says that $\Gamma(U, \mathcal{F})_f \cong \tilde{M}$. Write it out.

To show that (1.) □

Corollary 11.35. *Let A be a ring and $X = \operatorname{Spec} A$. Then there is an equivalence between the category of A-modules and the category of quasi-coherent $\mathcal{O}_X$-modules.*

11.8 Coherent Modules

11.9 Problems

11.1 Let A be a ring, $X = \operatorname{Spec} A$, and let M be an A-module. Prove that the $\mathcal{O}_X$-module $\tilde{M}$ is the sheafification of the presheaf $U \mapsto M \otimes_R \mathcal{O}_X(U)$.

Chapter 12

Line Bundles and Divisors

12.1 Cartier Divisors on Integral Schemes

Let X be an integral scheme with the generic point η and let $K(X) = \mathcal{O}_{X,\eta}$ be the function field of X. Denote by $\mathcal{K}_X$ the constant sheaf with value $K(X)$. Since X is irreducible, $\mathcal{K}_X$ is simply the presheaf $U \mapsto K(X)$ for all non-empty open set $U \subseteq X$, and for $\emptyset \neq V \subseteq U$, the restriction map $\mathcal{K}_X(U) \to \mathcal{K}_X(V)$ is the identity map.

Definition 12.1. A **Cartier divisor** on an integral scheme X is a tuple $(U_i, f_i)_{i\in I}$ satisfying

(1.) the sets U_i form an open cover of X, and,

(2.) $f_i \in K(X)^\times$ are elements such that $f_i f_j^{-1} \in \Gamma(U_i \cap U_j, \mathcal{O}_X^\times)$ for all i, j.

Two tuples $(U_i, f_i)_{i\in I}$ and $(V_j, g_j)_{j\in J}$ define the same Cartier divisor if $f_i g_j^{-1} \in \Gamma(U_i \cap V_j, \mathcal{O}_X^\times)$ for all $i \in I$ and $j \in J$.

The set of Cartier divisors on X is denoted by $\mathrm{Div}(X)$. We shall equip a structure of an abelian group on $\mathrm{Div}(X)$ as follows. Given Cartier divisors $(U_i, f_i)_{i\in I}$ and $(V_j, g_j)_{j\in J}$, define their sum to be the Cartier divisor $(U_i \cap V_j, f_i g_j)$. Note that if $i_1, i_2 \in I$ and $j_1, j_2 \in J$, then $f_{i_1} f_{i_2}^{-1} \in \Gamma(U_{i_1} \cap U_{i_2}, \mathcal{O}_X^\times)$ and $g_{j_1} g_{j_2}^{-1} \in \Gamma(V_{j_1} \cap V_{j_2}, \mathcal{O}_X^\times)$. Hence

$$f_{i_1} g_{j_1} (f_{i_2} g_{j_2})^{-1} = f_{i_1} f_{i_2}^{-1} g_{j_1} g_{j_2}^{-1}$$

is an element of $\Gamma(U_{i_1} \cap U_{i_2} \cap V_{j_1} \cap V_{j_2}, \mathcal{O}_X^\times)$. Furthermore, this is independent of the choice of the tuples. Indeed, assume that $(U_i, f_i)_{i\in I}$ defines the same divisor as $(U'_k, h_k)_{k\in K}$, and that $(V_j, g_j)_{j\in J}$ and $(V'_m, a_m)_{m\in M}$ define the same Cartier divisor. We then have two tuples

$$(U_i \cap V_j, f_i g_j) \quad \text{and} \quad (U'_k \cap V'_m, h_k a_m).$$

It is easily checked that $f_i g_j (h_k a_m)^{-1} \in \Gamma(U_i \cap V_j \cap U'_k \cap V'_m, \mathcal{O}_X^\times)$, so the two tuples are equivalent Cartier divisors. The defined operation is clearly commutative and associative.

The identity of $\mathrm{Div}(X)$ is the Cartier divisor $(X, 1)$ and the inverse of the Cartier divisor $(U_i, f_i)_{i\in I}$ is then $(U_i, f_i^{-1})_{i\in I}$. We will use the additive notation for this Abelian group structure on $\mathrm{Div}(X)$.

Remark 12.2. If X be an integral scheme, then $\mathrm{Div}(X) = \Gamma(X, \mathcal{K}_X^\times/\mathcal{O}_X^\times)$.

Definition 12.3. A Cartier divisor is **principal** if it is equal to the divisor given by (X, f) for some $f \in K(X)^\times$. Two Cartier divisors D and E are **linearly equivalent** if their difference $D - E$ is principal.

It is easily seen that the set of principal Cartier divisors is a subgroup of $\mathrm{Div}(X)$ which is isomorphic to $K(X)^\times$. The quotient of $\mathrm{Div}(X)$ by the subgroup of principal divisors is called the **group of divisor classes** on X, and is denoted by $\mathrm{DivCl}(X)$.

Proposition 12.4. *Let X be an integral scheme. We have an exact sequence*

$$1 \to \Gamma(X, \mathcal{O}_X)^\times \to K(X)^\times \to \mathrm{Div}(X) \to \mathrm{DivCl}(X) \to 0$$

of Abelian groups.

Proof. The sequence is exact at $\Gamma(X, \mathcal{O}_X)^\times$ because the map $\Gamma(X, \mathcal{O}_X)^\times \to K(X)^\times$ is injective. Next, if $a \in \Gamma(X, \mathcal{O}_X)^\times$, then the Cartier divisor (X, a) is equal to $(X, 1)$ because $a = a1 \in \Gamma(X, \mathcal{O}_X^\times)$, and so the sequence is exact at $K(X)^\times$. Lastly the exactness at $\mathrm{Div}(X)$ and at $\mathrm{DivCl}(X)$ are simple consequences of the definition of $\mathrm{DivCl}(X)$. □

Line Bundle attached to a Cartier Divisor

Let $D = (U_i, f_i)_{i\in I}$ be a Cartier divisor on X. We define a line bundle $\mathcal{O}_X(D)$ attached to D by the assignment

$$\Gamma(U, \mathcal{O}_X(D)) = \{f \in K(X) : \text{for all } i,\ f_i f \in \Gamma(U_i \cap U, \mathcal{O}_X)\}$$

for every non-empty open subset $U \subseteq X$. It is easily checked that $\mathcal{O}_X(D)$ is an $\mathcal{O}_X$-module.

Remark 12.5. For each U_i, consider the restriction $\mathcal{O}_X(D)|_{U_i}$. For each open subset $V \subseteq U_i$, $\Gamma(V, \mathcal{O}_X(D)|_{U_i})$ is the free submodule of $\Gamma(V, \mathcal{O}_X)$ generated by f_i^{-1}.

Let us temporarily denote this submodule by M. Clearly, M is free. For all rf_i^{-1}, $r \in \mathcal{O}_X(V)$, and for all $j \in I$, we have

$$f_j(rf_i^{-1}) = rf_jf_i^{-1} \in K(X).$$

By the definition of Cartier divisor, $f_jf_i^{-1} \in \Gamma(U_i \cap U_j, \mathcal{O}_X^\times)$. Since $U_i \cap U_j \cap V = U_j \cap V$, we may view $\Gamma(U_i \cap U_j, \mathcal{O}_X^\times)$ as a subset of $\Gamma(U_j \cap V, \mathcal{O}_X^\times) \subseteq M$, and so we see that $f_j(rf_i^{-1}) \in \Gamma(V, \mathcal{O}_X(D_i)|_{U_i})$. Therefore $M \subseteq \Gamma(V, \mathcal{O}_X(D_i)|_{U_i})$.

Conversely, let $f \in \Gamma(V, \mathcal{O}_X(D_i)|_{U_i})$. Then $f \in K(X)$ and $f_i f \in \Gamma(U_i \cap V, \mathcal{O}_X) = \Gamma(V, \mathcal{O}_X)$. Thus $f = (f_i f) f_i^{-1} \in M$. This proves the remark.

From the remark we see that $\mathcal{O}_X(D)|_{U_i} \cong \mathcal{O}_{X|U_i}$, and so $\mathcal{O}_X(D)$ is indeed a line bundle on X.

12.2 Cartier Divisors

12.3 Weil Divisors

Chapter 13

Smoothness and Differentials

Chapter 14

Projective Schemes

Chapter 15

Čech Cohomology

Let X be a ringed space and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open covering of X.

Chapter 16

Cohomology of Sheaves of Modules

Chapter 17

Serre Duality

Chapter 18

Cohomology and Base Change

Chapter 19

Grassmannians, Flag Varieties, Schubert Varieties

Chapter 20

Torsors

20.1 Torsors

Let X be a topological space and G a sheaf of groups on X. If T is a sheaf of sets on X, then we say that G **acts on** T or that T is a **G-sheaf** if there is a morphism of sheaves $G \times T \to T$ such that for every open subset $U \subseteq X$, the map $G(U) \times T(U) \to T(U)$ is a left group action of $G(U)$ on $T(U)$.

A **morphism of G-sheaves** is a morphism of sheaves $\varphi \colon T \to T'$ such that $\varphi_U \colon T(U) \to T'(U)$ is $G(U)$-equivariant for all open subset $U \subseteq X$. By definition, this means that $\varphi_U(g \cdot t) = g \cdot \varphi_U(t)$ for all open subset $U \subseteq X$, $g \in G(U)$, and $t \in T(U)$.

Definition 20.1. A G-sheaf T is called a **G-torsor** if it satisfies the following two properties:

(1.) The group $G(U)$ acts simply transitively on $T(U)$ for every open subset $U \subseteq X$. (A group G acts *simply transitively* on X if for all $t, s \in X$, there is a unique $g \in G$ such that $g \cdot t = s$.)

(2.) There exists an open covering $\{U_i\}_{i \in I}$ of X such that $T(U_i) \neq \emptyset$ for all $i \in I$.

An example of a G-torsor is G itself under left multiplication action. It is called the **trivial G-torsor**.

Remark 20.2. A G-torsor T is isomorphic to the trivial G-torsor if and only if $T(X) \neq \emptyset$. To see this, take any $t \in T(X)$ and define a morphism $f_U \colon G(U) \to T(U)$ by $g \mapsto gt|_U$ for all open subset $U \subseteq X$ and $g \in G(U)$. The fact that $G(U)$ acts simply transitively implies that each f_U is an isomorphism. The morphisms f_U are compatible with restriction maps and hence together form an isomorphism $f \colon G \to T$ of sheaves.

Chapter 21

Curves

Chapter 22

Category Theory